

TRAINING FOR TRAIL RUNNING

In this article you can read my training philosophy. It's not a plan to follow nor a comprehensive summary of what the current sport science or my training journal that can be found in other posts in this website. In here you will find a training philosophy and methodology based on my personal experience, the experience of other athletes and the science I've read and found relevant for endurance mountain activities. This article is also alive. What's written here today can be wrong, because science and experience evolve, and so when I learn something relevant, I will update it.

First, we often reduce training to the physical exercise, but training, in my opinion, is much more:

- a) Training is the act of transformation to perform in a task.*
- b) Training is the acquisition of skills, knowledge and experience for a task.*
- c) Training is a way of living in connection with our body and our mind.*
- d) Training is changing the biology of the different physiological systems and cognitive functions.*

General Training Principles

Don't Whine

I was running on the Tahoe Rim Trail, a 165 miles trail between Nevada and California, when after more than 20h of non stop running trying to break a FKT I realized I had been running for the last hour in the wrong direction. I turned back and told my pacer, "well, more kilometers, more fun". I don't know even today if I believed truly in that or if it was a euphemism of "that really sucks" but anyway, it's been a big part of my training philosophy. To try to always look on the optimistic side among realism. We can't change the present but to accept it and work from there to change it.

Have Fun

Adaptations in training come from repetition of stimuli over time. Adaptations can be acute, those will be achieved in a short amount of time, but will disappear fast when we stop the stimuli, or long term. Long term adaptations are the ones that happen on a biological level, so we change our biology to make us more adapted to certain efforts. For that is necessary a long term succession of stimuli. To be able to train for that long I believe the easiest way is to enjoy (almost) every session. To have a passion where you see training not as an obligation or something that is good for you, but as a part of yourself. To have fun training, to LOVE training.

How >>> What

Society prioritizes results / outcome above all, that's not different in sport, but the contrary. To excel in sport we use a hierarchization of the results, ranks, marks taking away the context. Despite that, since the origin of sport activities there have been rules for two fundamental purposes: preserving the essence of the sport and ensuring fair play among participants. Even beyond the boundaries of the regulations of the sport, I believe the context always matters more than the outcome and in the greatness of a performance the how has been done weighs more than what's been achieved.

Think plural

We think of our body as an engine, with its parts working separately for a unique purpose. But our body is made by different organisms, we have millions of living organisms inside and nearly around us that are necessary for our life and our functions, and multiple systems that work with different connections. For some we have more knowledge on how it happens, for some we still don't, but not for that we should ignore them.

The Why

Before even planning a training program (a season, a year, a multiple year program) we need to answer some questions.

Why? Why do I run, ski or climb? What is the ultimate reason that makes me wake up, endure the difficulties and commit to it?

Who am I as an athlete? What are my capacities in metabolism, neuromuscular, adaptations to temperature, altitude, different terrains, conditions... What are my strengths and weaknesses? How do I adapt or react to different stimuli? How do I cope with different situations?

What are the conditions required for the event I want to do? Then check what I should work on, what I shouldn't, when and how.

Vitamin N. The commitment

In an interview, Eliud Kipchogue said that the most important vitamin is the vitamin N. N from No. To say no. Our life, a day is limited in time and in the energy we have. We can't do everything. We need to prioritize first on what is most essential and focus on that. Then compromise the things that are less essential, that can mean to do it in a lower amount, to do it at another time in the year or to not doing them at all. To accept this commitment can be hard but it's the key for a long term career.

Low of the lowest effort

At the end – at the very beginning – we don't train our muscles, lungs or heart. We stress our body with the intention – if we do it in the right amount – by making adaptations so our body is capable of tolerating this stress better. The way we create these adaptations can be different from athlete to athlete, because the adaptation pathways will always follow the law of the lowest effort. Our system will find the easiest way for them to adapt to those stimuli, and that “easiest way” might be different depending on individual genetics or epigenetic. At a biological level an adaptation will make changes in our epigenetic. Those adaptations are made by the repetition of stimuli over long periods of time. Stress (physical load – metabolic, neuromuscular + cognitive load) + Rest = Adaptation. With no stimuli repetition there's no adaptation, with no recovery of the stimuli, there's no adaptation.

KISS

“Keep It Simple, Stupid” I stole this quote from the great report by Nils van der Poel “How to skate a 10k”. We find more fancy to focus on that highlighted 1% that it's told to us that makes the difference. That's bullshit. The 99% vs the 1%. Focus on building a great methodology that you find sustainable to do every day, week, month, year. Life as an athlete is to be able to follow a simple life, what is not easy at all, to isolate oneself from distractions. When you have achieved that, the biggest part of the work is done.

Eagle vision, sniper execution

The summit of Everest is very similar to the summit of the local hill behind your home. The difference it's in its base. Before pretending to improve our top game – winning capacities - we need to develop a big base to handle that top gear and to be able to start building it from higher up. Work on building a great base (aerobic base in metabolism, general and diverse strength and neuromuscular connections, mental resilience...) and then surgically execute the interventions to get specifically ready for the done event, stimulating the situations we will be facing during the event from all the aspects.

T-A=0

Talk minus action equals zero. At the end, you need to get out and do the work. Is not about fancy and complicated workouts, planning and so on, but about the continuous stimulation. To be consistent day after day, week after week and year after year.

The Task

Before starting to talk about *how* to train, we need to understand what is the activity we want to perform. Our training is done to change our capacities to perform a task better, so the deeper our understanding of the task, the better we can plan our training. That's pretty obvious but it's astonishing how many people are training without any knowledge on what are the characteristics of what they train for and therefore what are the adaptations they are willing to get.

The task can be from simple to very complex. That doesn't mean that the difficulty to perform at a simpler task is easier, but that the task requires less complexity to be performed. For example running a 10km is more simple - a repetition of a mechanic movement over time, with a metabolic charge easy to calculate, a range of conditions pretty stable - compared to open a new technical route in a virgin summit at high altitude, where the technical requirements will demand multiple biomechanical capacities, different metabolic capacities, cognitive skills, etc.

Basically that means that the more complex a task is, we would need to develop different – and often opposite – capacities, so the absolute performance in any of those capacities will be lower. The simpler a task is, the capacities to develop would be less so we could reach a higher specialisation and therefore a more elevated performance.

Depending on the activity / sport we want to perform we need to become more of a specialist or a generalist, and that will have deep implications in the training structure.

Some of the factors affecting the complexity of the task are:

1. Stability of conditions (terrain, temperature, wind...)
2. Stability of mechanics (repetition of same movement vs multiple techniques to be used without a correlation)
3. Stability of the metabolic ways used to perform the task (always the same duration of efforts vs different durations and intensities)
4. Stability of decision making (Simple decision making, constant analyze of multiple factors to take a decision)

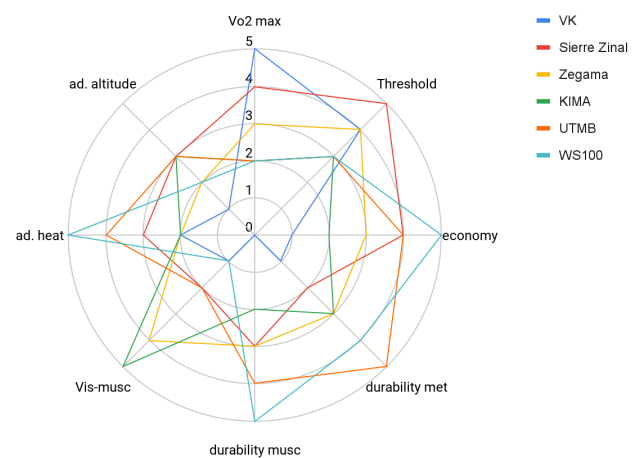
Making a profile of the task:

We will do a profile of the task requirements, to determine what should be the athlete's characteristics to succeed in the task. If the task is complex we should do different profiles and determine which ones are transversal in the different inner-tasks and whose diverge.

To make this profile we can estimate the requirements in the different systems:

- Metabolism
- Neuromuscular
- Oxygen
- Temperature
- Cognitive

Here an example of different general profiles in different trail running races:



The Athlete

Every person is different. It has a different genetic, a different morphology and a different history (physiological and psychological history). So the training will need to be different for each person even if the final performance is the same.

Why do we get tired?

If we were able to not get tired, we could just sprint at our maximum capacity to the end of a competition, or eventually to death. But we can't, something makes us slow down before that happens, and we don't know why.

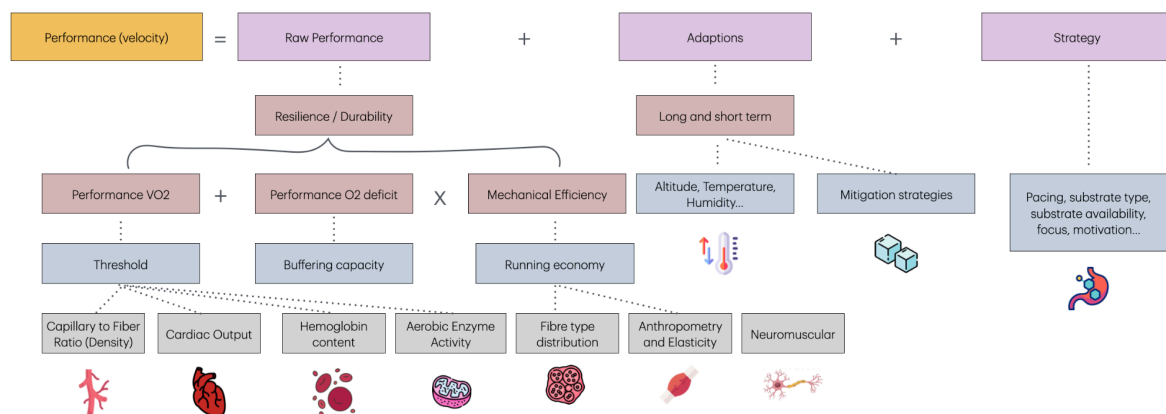
The most common theory says that the central governor- via perceived exertion, starts to "shut down" or slow down functions to prevent death or serious condition before that happens.

What limits performance:

1. Pain tolerance: We can't support the pain or perceived exertion and so we slow down to a more comfortable effort.
2. Metabolism: We can't transform energy in the metabolism to move our muscles.
3. Substrate availability: We cannot get more energy.
4. Temperature: Our body gets hot
5. Oxygen: We can't get enough oxygen to our muscles.
6. Neuromuscular: Peripheral fatigue, brain-muscle connections and coordination
7. Acidity: Our body acid levels are too high.
8. Cognitive: capacity to focus, to tolerate pain, etc.

In general, for endurance sports we can determine the individual performance capacity (velocity to do the task) looking at the possible raw performance with the performance of the $\dot{V}O_2$ and $\dot{V}O_2$ deficit metabolism and the mechanical efficiency and the capacity of reproduce that performance repeatedly and/or for a long time or/or after a previous effort. Also we need to take into account the environmental and specific (equipment, route...) adaptations as well as the strategy to execute the task.

Here a diagram based on Michael J. Joyner running performance:



But, why do we need to train? We see other animals that *don't train* and perform the tasks they are required to do very well. Cheetahs, for example, can sprint very fast and they're not going to the gym and doing HIIT. As humans, we need to train for 2 main reasons, first to perform tasks that we are not "programmed" to do, and secondly because our lifestyle based on the non necessity of exercising for survival makes that we do need training to get to our maximum genetic expression.

Training therefore is the way we have to reach our maximal genetic expression. With training we can change a lot of the physiology and biology of our organism, but not everything.

Not modifiable:

- Genetics
- Anthropometry

Different pathways to create adaptations:

- Epigenetics
- Neuromuscular
- Knowledge – experience
- Environmental adaptations
- Cognitive
- Metabolism

Making the Athlete profile:

Before starting to draw a training plan we need to understand the requirements of the task (the competition, project) and the capacities of the athlete. To see what we need to train, what we don't need to train, and what we should prioritize.

1. Metabolism: We will see how we are able to perform at different intensities and durations, the power or speed profile.
2. Substrate: We will look into our energy budget, our maximum energy expenditure, our oxidation capacities for carbs and fat.
3. Temperature: We will look into our decrease of performance related to core temperature, our sweat rate and sweat composition.
4. Oxygen: We will look into our capacity to obtain oxygen (rarely a limiter), our capacity to transport it (Hb mass) and our capillary. Also what is our preferred way to adapt to altitude.
5. Neuromuscular: We will look into our technique in different terrains and in our efficiency in different paces and different durations.
6. Acidity: We will look into our buffering capacity and the immune response.
7. Cognitive: We will look into how we get “up” or “down” before and during races, our pain tolerance and comfort zones, our capacity to focus in and out, etc.

Here is a very simplified example of a requirements-capacities table to access What we already have, What we miss, What I can adapt, What I can't adapt, to then build the training plan.

System	Race requirements	Athlete capacities
Metabolism	<i>Pace at 10% under threshold. Some uphill (20') above threshold.</i>	<i>I have a high Vo2 max speed but my threshold is low.</i>
Substrate	<i>The race consumption will be 8300cal. At 10% under T (60°F,40°C), at T (95°C)</i>	<i>I can oxydate 85gr/cho/h and 150 gr/fat/h</i>
Temperature	<i>The race will be at 35° and 60% h</i>	<i>My sweat rate at 30° is 1,3L/h and sodium loss is 1200 mg/h</i>
Oxygen	<i>The race is at 3500m of altitude</i>	<i>I struggle to acclimatize, my Hb is low and I'm a capillary adapter.</i>
Neuromuscular	<i>The inclines of the race are ... and paces to run ... there are technical downhill</i>	<i>My eye-muscle connection is great. My NM performance decreases at under 5% incline and pace X</i>
Acidity	<i>There's not much water</i>	<i>I have a very good buffering cap.</i>

Cognitive	<i>There is a fast start, pressure, a dangerous part.</i>	<i>I overperform with pressure. I get scared and freeze in technical sections.</i>
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Other considerations when it comes to the athlete's capacities:

- Vo2 max has a big genetic part, in young people can work Vo2 max. Pregnant women can develop Vo2 max during the pregnancy.
- We're more efficient (metabolic and muscular) at paces / intensities we're used to train.
- Technical skills are easier to acquire at young ages
- Economy comes with time, years and volume of practice.

Formation period:

- Start as young as possible in a non competitive multisport practice of endurance. Keep it as a game.
- Practice of multiple sports is important to develop different qualities, keep the interest and motivation.
- Entering competitions in young categories is not to take serious goals but as a formation to learn technique, strategy, and get used to the competition environment.
- Start training structured as early as possible (13-14 years old) to develop the metabolic ways well.
- Prioritize the long term, focus on building a aerobic base (z2 training) and stimuli on intensity less important, mostly to be done on races (what is also important to learn to race)
- Practice of technical sports is key to perform in these sports (ski, climbing...)
- When entering specific training the volume should increase by 10% year until reaching the elite volume.
- First years practicing a sport is about creating adaptations and getting used to the stressors of the sport. To race a lot can be the most interesting way to get used to racing (pressure, tactics, fueling, the mental aspect, the change of routines...) Without really giving importance to the result itself. Later on a career this is not as important since it is already acquired.

Useful links:



What limits performance: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4606539>



Extreme energy expenditure: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6551185>

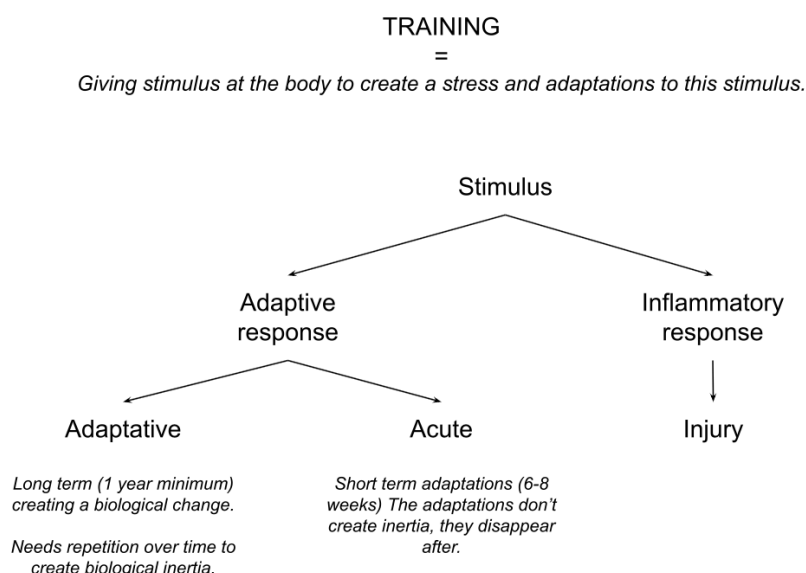
Training Stimuli

To change our capacities (via change in epigenetics, biology, neuromuscular connections, knowledge, etc) to perform the task we stimulate the different systems of our body. But the body systems always seek Homeostasis, the body's tendency to maintain a stable internal environment (things like temperature, pH, hydration, oxygen levels, glucose levels, etc.) So when we do a stimulus we disrupt this homeostasis. If the stimulus is enough to disrupt the homeostasis, not too high that the system fight it, and constantly so the system see that it is not a one time thing but something that needs to adapt to it, then it will change the baseline of homeostasis to fit those stimuli.

If the stimuli is too high, it will bring an inflammatory response, which can lead to a chronic inflammatory process and injuries or disease.

If the stimuli is good, it will bring an adaptive response (there is inflammation as a response of the stimuli that disappears quickly after to create the adaptation). This response can be adaptive in the long term, which will change our biology. That requires constant stimulation over a long period of time (a minimum of 1 year, often several years needed). But once the adaptation has been happening, changing our epigenetics and biology, they will last for a very long time. That is what explains when a retired athlete that has been doing nothing for a year, comes back and performs well since the first day.

The adaptive response can also be acute. Those adaptations don't require a long period of constant stimulation, but disappear quickly after we stop the stimuli.



Physiological and cognitive systems for endurance performance:

First of all it's important to understand that the different systems of the body don't work separately. We are not made as a machine where each piece does its function by itself but as a network where all the systems and other organisms living on us (bacterias, microbiota, etc) are working together and influencing each other as a complex system.

1. Cardiovascular System

- Role: Delivers oxygen and nutrients to muscles and removes waste.
- Trainable aspects:
 - Stroke volume (more blood per heartbeat)
 - Cardiac output (more total blood per minute)
 - Capillarization (more capillaries in muscle = better exchange)

2. Respiratory System

- Role: Supplies oxygen to the blood and removes CO₂.
- Trainable aspects:
 - Respiratory muscle strength and endurance
 - Ventilation efficiency (e.g., breathing patterns)
 - Oxygen diffusion capacity

3. Muscular System

- Role: Executes movement and uses energy.
- Trainable aspects:
 - Mitochondrial density and efficiency
 - Muscle fiber recruitment (especially Type I and fatigue-resistant Type IIa)
 - Capillary density
 - Resistance to muscle damage (eccentric loading)

4. Metabolic System

- Role: Produces energy (ATP) through different fuel pathways.
- Trainable aspects:
 - Aerobic metabolism: Efficiency in using oxygen to burn fats and carbs
 - Fat oxidation: Increased ability to use fat at higher intensities
 - Lactate threshold: Delaying the point where lactate accumulates
 - Glycogen storage and sparing: Using energy more efficiently

5. Thermoregulatory System

- Role: Regulates body temperature during exercise.
- Trainable aspects:
 - Sweating rate and onset
 - Plasma volume expansion
 - Skin blood flow
 - Heat tolerance (via heat acclimation)

6. Neuromuscular System

- Role: Coordinates muscle activation and movement economy.
- Trainable aspects:
 - Motor unit recruitment efficiency
 - Movement economy (running form, muscle stiffness)
 - Proprioception and balance

7. Hormonal System

- Role: Regulates energy usage, recovery, and adaptation.
- Trainable aspects:
 - Sensitivity to training stress
 - Hormonal responses (e.g., cortisol, adrenaline)
 - Adaptation to fuel use (e.g., insulin sensitivity, glucagon)

8. Cognitive, Psychological and Perceptual Systems

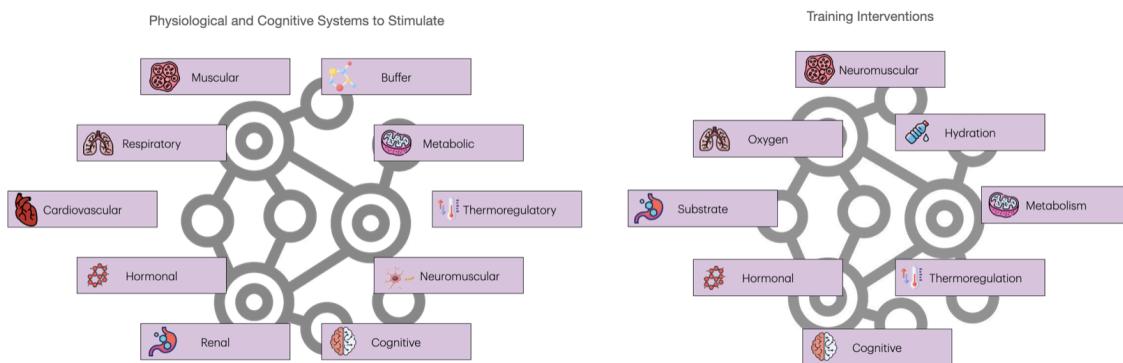
- Role: Governs motivation, pain tolerance, pacing, and decision-making.
- Trainable aspects:
 - Mental resilience
 - Focus and flow state
 - Pacing strategy and internal feedback use
 - Coping mechanisms for fatigue

9. Chemical Buffer Systems

- Role: Resists changes in blood and muscle pH via bicarbonate, phosphate, protein buffers and carnosine.
- Trainable aspects:
 - Increase bicarbonate availability
 - Increase muscle carnosine levels
 - Improve intracellular buffer capacity

10. Renal System

- Role: Acid-base balance, electrolyte balance, fluid balance and blood pressure, EPO production
- Trainable aspects:
 - Acid excretion and HCO_3^- reabsorption
 - Electrolyte handling
 - Fluid retention
 - EPO release



When it comes to practical interventions to stimulate those systems we can divide interventions into 8 types of stimuli or interventions (metabolic, neuromuscular, thermoregulation, oxygen, cognitive, substrate, hydration and hormonal). As the systems are interconnected, when we do a training intervention we will work all of the systems at the same time, in different measures or intensities.

Stimuli	Systems	Interventions	measurements	devices
Metabolism	Glucolysis, Lipolysis, PC, mitochondria	Volume, intensity distribution	Duration, HR, breathing frequency, feelings, lactate	HR band, tymewear, lactate meter, RPE, metabolic chart
Neuromuscular	Muscle fibers type, muscle recruitment, neural connections	Cadence, technical work, "strength" work, plyometrics	cadence, time of contact, weight, power	
Thermoregulation	thermoregulation,	Train in hot conditions, dehydration,	Temperature, sweat rate, electrolytes on sweat	Thermometer, CORE, fluid loss, sweat test
Oxygen	Mitochondrial oxidative capacity, O2 intake, SatO2 limitCentral (lungs volume,	Train in altitude, train breathing muscles, train with oxygen restriction, Apnea,	SatO2, SmO2, Hemoglobin, breathin	Pulsoxymeter, NIRS, tymewear

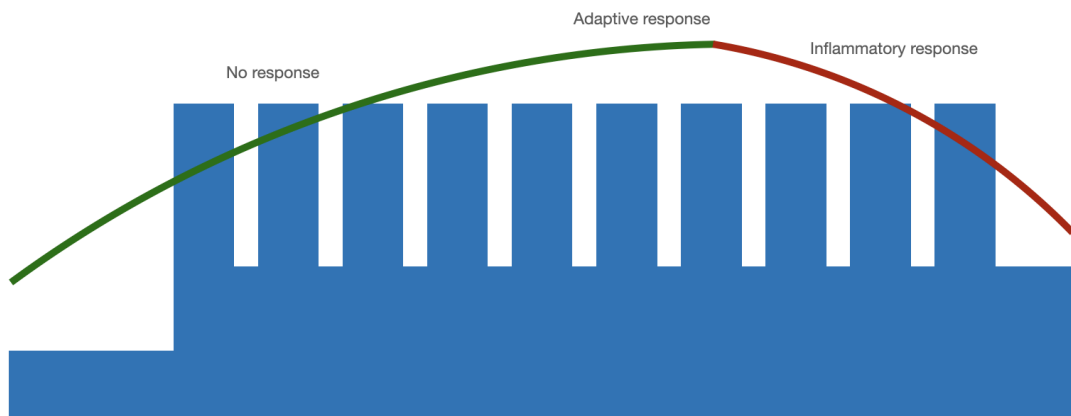
	breathing frequency, transport Hb...) peripheral (capillarization muscles)			
Cognitive	neural, hormonal,	Sleep deprivation, stress, multi task during training, cutting senses,	Brain activity, eyes activity,	EEG, eyes tracker, pupillometry
Substrate	glucolysis, lipolysis, hormonal	Fasting, different substrate and quantities, diet, supplementation	cal intake and consumption, lactate, ketones, glucose, triglycerides, cholesterol, microbiota	lactate, glucose, ketone trackers, diet monitoring
Hydration	renal	hydration deprivation, hydration regulation, electrolytes	pH, RAA, Osmosis	Osmose, pH bands
Hormonal	hormonal	Sleep deprivation, exposure sun, diet, circadian rhythms, lifestyle	hormone and protein blood levels	blood test and protein test

Same stimuli $\neq$ same adaptation. When we do an intervention we might expect similar results in terms of adaptations for different individuals, in general that is the case, but not always. Some athletes might produce an adaptive response to a stimulus and others an inflammatory response to the same stimuli, that can be explained by the genetics and the history of the athletes. Also, the pathway to adapt from the same stimuli or exposition can differ between persons. Since biology will always search for the easiest way to adapt, that way can be different to individuals. For example in the exposition of altitude, some individuals will adapt increasing their hemoglobin concentration, while others will stay in low hemoglobin and have adaptations in a hormonal level or in the capillarization. (see study: [10.1146/annurev-anthro-102313-030000](https://doi.org/10.1146/annurev-anthro-102313-030000)) That is important to take into account when we are trying to monitor adaptations, to be sure that we look at final adoption and not only a single possible pathway to adapt.

It's important to take into account that adaptation will happen if the system's functions are healthy. If the body is facing inflammation or is not in balance the stimuli will probably aggravate the problem. That's why before doing a training intervention we must be aware of the status of the system we want to stimulate so we are sure we are stressing a healthy system that can give an adaptive response. For example if we want to stimulate the renal function with some dehydration stimuli we need to be sure that renal function is healthy, otherwise the stimuli can bring damage. So it's primordial to first work on having systemic health before thinking about specific training stimuli.

Training is a game of optimisation, not maximization.

Even if the total volume of training is linked with performance, more is not always better. Training is about optimizing the stimuli required to create adaptations. The charge of the stimuli should be enough to challenge homeostasis and signal to create new adaptations but not too much to create a fight response and an inflammatory state. Since our system is always changing and affected by multiple factors, it's hard to *measure* the optimal stimulation, but I believe that with experience we are able to detect that. The session should therefore not be dictated by a set charge (duration / number of repetitions) but be adaptable to stop a bit earlier if we feel we are reaching an overstimulation or to prolong if there is not stimulation yet.



Metabolism

The goal when we train our metabolism is to be able to convert more energy we consume or have stored into mechanical energy (muscle movement). For that we will improve the capacity of absorbing substrate, the capacity to store substrate and to produce/recycle substrate from other substrates, to transport the substrate and the oxygen to the muscles, to absorb in the muscles that substrate and oxygen and then to convert it to muscle movement.

Cardiovascular System:

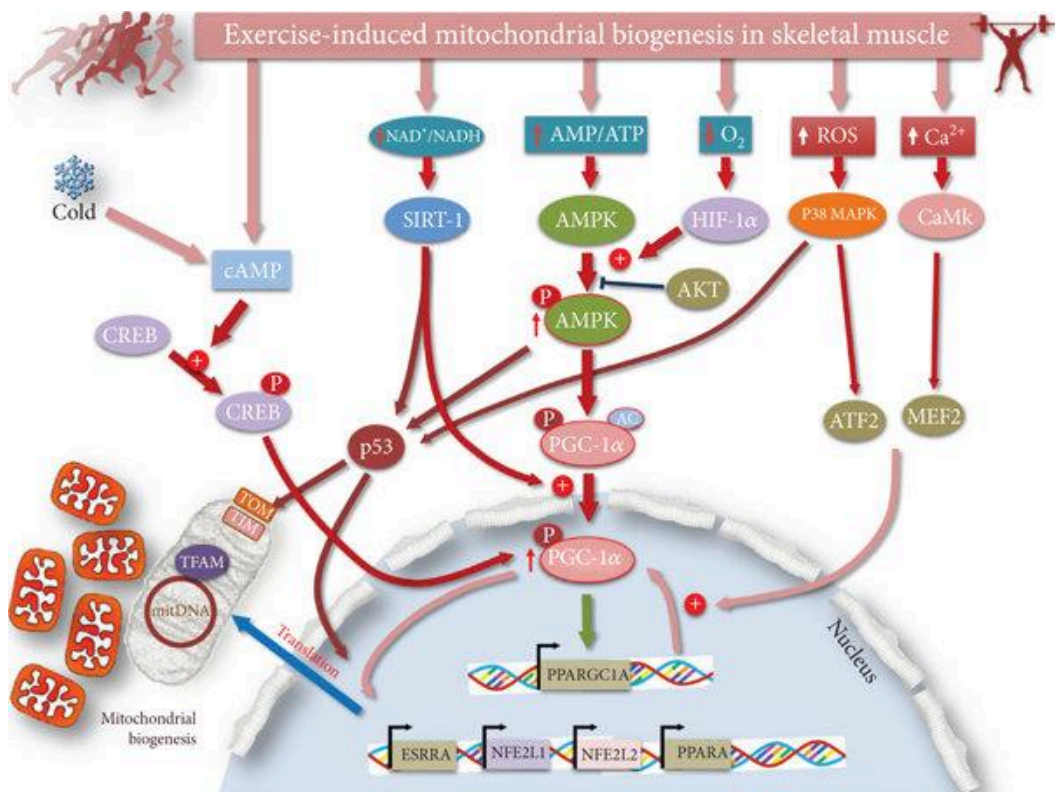
- ↑ Stroke volume (more blood transported per beat)
- ↑ Ventricular wall thickness
- ↑ Contractile functions
- ↑ Size of individual car
- ↑ Capillary density (better O₂ delivery)
- ↑ Blood plasma volume

Hepatic system:

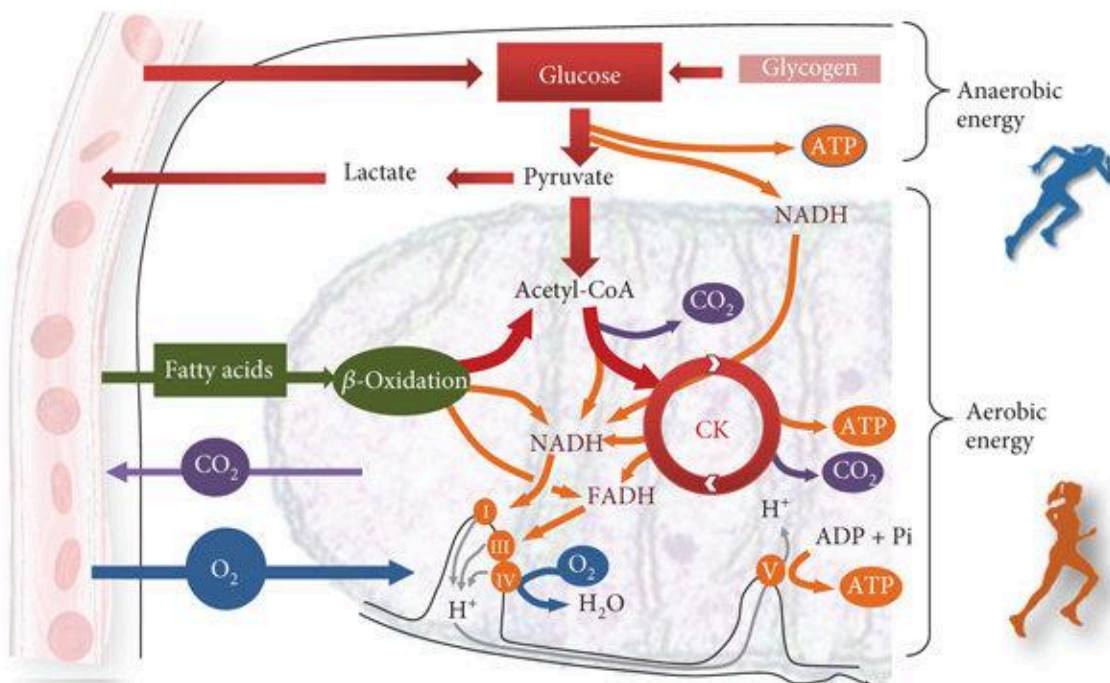
- ↑ Insulin sensitivity
- ↑ Gluconeogenesis and glucose uptake
- ↑ Fatty acid oxidation

Muscular Adaptations:

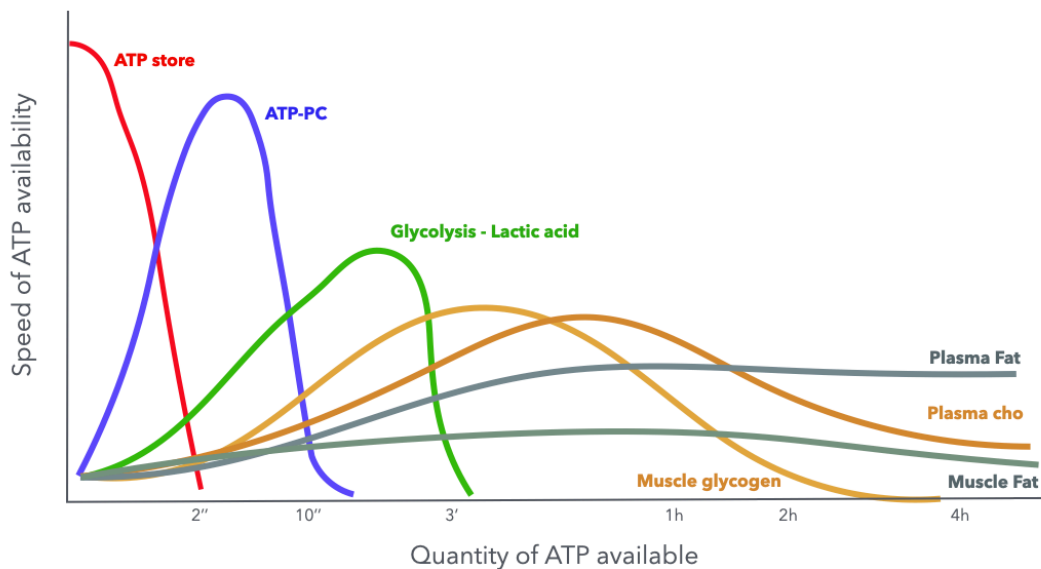
- ↑ Mitochondrial biogenesis - density and function
- ↑ Oxidative enzymes
- ↑ Myoglobin content
- ↑ Fat oxidation capacity
- ↑ Efficiency ATP production



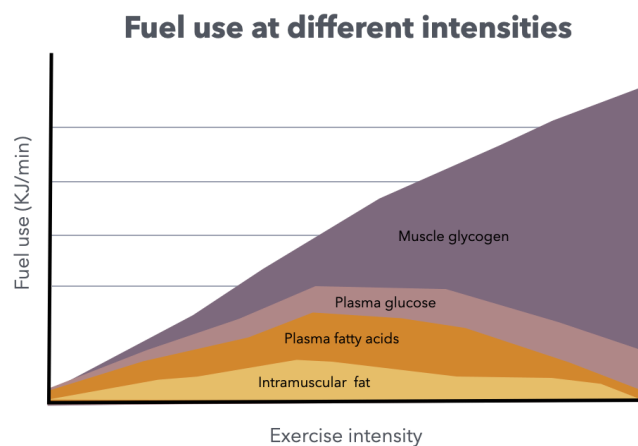
Use of different metabolic pathways to obtain energy, different substrates, different chemical reactions with oxygen (fat or cho oxidation) or without oxygen, and recycling [Lactate shuttle](#)



Use of metabolic pathways depending of the energy requirements of the task:



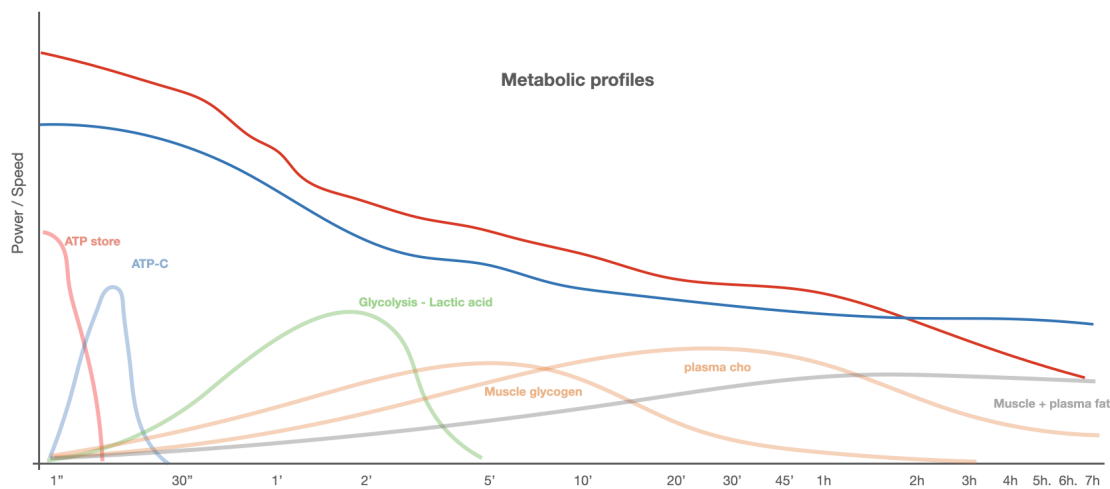
Metabolic flexibility is not to switch from glycolysis to fat oxidation, is not like the body goes from one to the other but that depending of the requirements (more intensity) and the availability (what we have, what we are consuming), it will activate more or less pathways and in different amounts. So when we have a good flexibility it means that we are able to activate at the same time different metabolic pathways. So at the end of a race, if we are doing a sprint, if we are adapted we will be using at its maximum the fat oxidation, the glycolysis, the PC, the lactate at the same time. That's why it is important in training to not only stimulate (via metabolic stimuli and substrate availability - using different substrates at training) to train the different pathways.



Athlete Metabolic Profile:

The metabolic profile will be the maximum energy we can produce at different durations. It can be expressed on physiological intensity markers ($\text{VO}_2\ldots$) or with external intensity

markers (Power, speed, pace, grade adjusted pace). And we can see what metabolic pathway we are more efficient at. During training we will try to modify that profile to get most efficient to the requirements of the race.



Thresholds and stable states:

Stable state refers to a condition where variables (like heart rate, oxygen uptake, lactate concentration, etc.) remain constant over time, but in exercise that is never achieved. Even if we exercise at a low intensity, after some duration the physiological demands will change, the internal temperature will increase, the muscle fiber fatigue will accumulate, the energy stores will deplete, etc. The body will adjust to those changes through feedback mechanisms (hormonal control, nervous system signaling...) but over time the compensation will not be enough to maintain the state stable, so we will see decoupling of heart rate, increase of lactate levels, increase of breathing frequency, changes in fuel use, decrease of heart rate or other changes.

With that into mind then we can think about what are the thresholds and how to use them in training. A threshold refers to an intensity zone where the energy use and production shifts. As we see that a real stable state doesn't exist we need to take into account when talking about thresholds, since those will also change with the duration of the activity.

Aerobic Threshold:

Depending if we measure it with air exchanges it will be called "ventilatory threshold / VT1" or if we measure with the lactate "LT1". It is the highest intensity at which our body can sustain energy production mainly through aerobic metabolism. Is also called "fatmax" since it is where the fat metabolism (lipolysis) is higher.

Anaerobic Threshold:

Also called “lactate threshold / LT2” or “ventilatory threshold 2 /VT2” or “critical power” depending on how we measure it.

Is the highest intensity at which lactate production and clearance are balanced. Beyond this, lactate starts to accumulate quickly.

Vo2 Max:

Is the upper limit of aerobic energy production, above that point the anaerobic metabolism dominates.

Know your training zones:

Training zones are not physiological zones, they are a useful way to prescribe or monitor our training intensity during sessions. There's not abrupt changes in physiology or metabolism pathways from one zone to another.

Z1 regenerate, your basal levels are better than before the session - you can talk easily, breathe by nose without effort. slow muscle fibers, fat ox mostly

Z2 fat ox, type 2 muscle fibers, lactate clearance - you can say few sentences. fat max, max stretch of slow muscle fibers, changing to fast - improvement from 2-3 months

Z3 transition from fat max, glucose increases. - you can say a couple of words, breathe hard, fatox go down, glycogen up, lactate curve break 1

Z4 threshold, glycolysis (where races are won) no fat oxidation. Fast muscle fibers. - no talk, type 2 fibers, inflection point, lactate threshold glycogen system - takes 8-10h on 3-6weeks

Z5 Vo2 max.

Z6 Anaerobic speed capacity - no need to breathe (max 30" efforts)



Aerobic base,
“Boring” slow and middle training
Takes long time to improve it

It's key to have a BIG one there to improve the rest

80% of the days

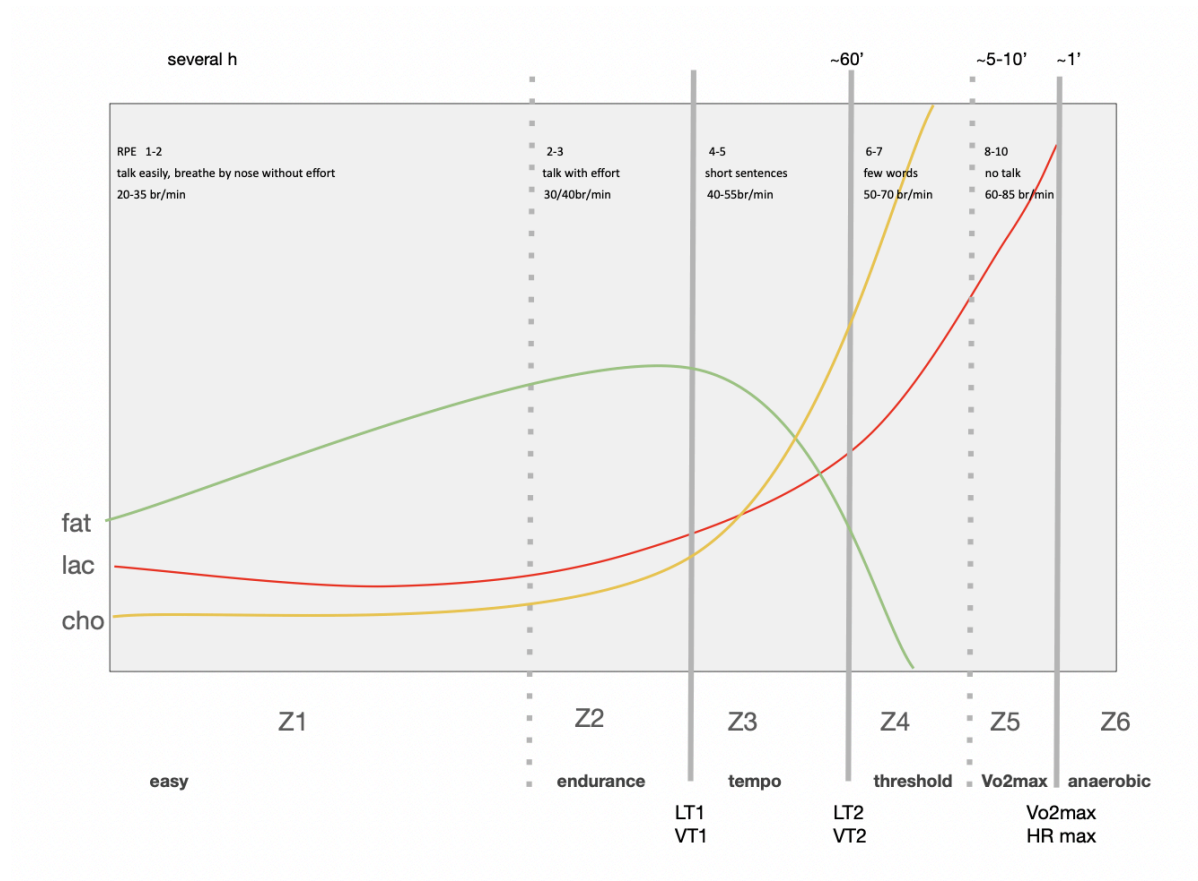


Takes short time to create adaptations

It's key to win races

If doing too much easy to get injured, overtrained

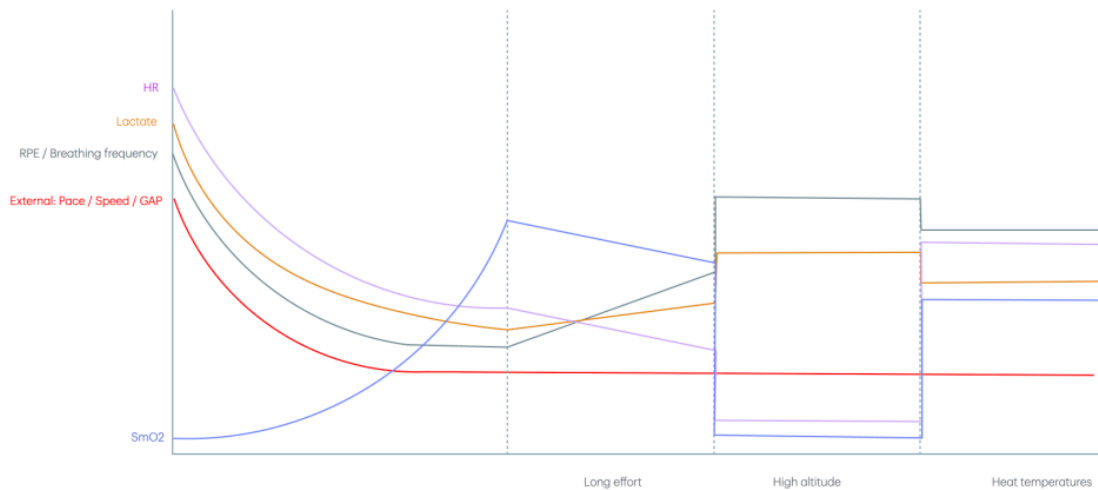
20% of the days



Measuring intensity:

When we measure something we are looking at a single data point from a single part or reaction of a single system. That data point can be impacted by many factors. Feeling / RPE, even if it is not measurable with a device, it takes into account all the different systems that are going on. So as athletes, it is key to develop sensibility to how we feel, specially the senses of nociception, interoception, thermoception and mitoception. That requires time and self awareness, but in the long term it is the best way we can have to measure intensity. There's why it is important in the formation stages to not only rely on external measurements of intensity (heart rate, lactate...) but to use those measurements to develop a connection between those markers and the feeling.

We can measure intensity in external metrics (Power, pace, speed, GAP...) or internal data (heart rate, lactate, temperature, msO2, breathing frequency...) but it is important to acknowledge what are the factors that influence those measurements (eg: higher lactate if you eat more carbs, higher HR if caffeine, lower HR in altitude, lower power in altitude or when hot temperatures...)



Metric	How	+	-
RPE	Our feeling	Takes into account all systems	Takes time to develop accurate perception
HR	Heart rate band	Easy to measure and interpret. Great for low and medium intensities.	Affected by environment, caffeine. Not reliable in high intensities (anaerobic efforts).
Lactate	Lactate meters	Direct insight into glycolytic metabolism	Invasive, delayed, influenced by nutrition/substrate availability
Power / Speed / GAP	GPS, power meters, stopwatch	Objective, easy to measure and interpret	Affected by external factors (e.g. terrain, altitude, wind, temperature)
mSatO₂	NIRS device	Immediate feedback on local muscle oxygen saturation	Hard to interpret. Sensitive to environment and device placement
breathing frequency	breathing bands, metabolic charts	Easy to measure and interpret. The most related to RPE.	Availability of devices, learning process.

Intervals vs continuous effort:

At the end what matters the most is the time spent at the targeted intensity more than the structure of the workout.

Doing intervals we might get a higher total volume at the target intensity, especially on days where motivation is not high or in build up moments of the season when the perceived effort is high at intensities we are not yet used to. Also it might have a better quality per effort, better technique and neuromuscular quality for the same reasons. There is also a reduced fatigue and injury risk with intervals, since there is less muscle and tendon strain per bout. Doing a continuous effort at targeted intensity has more specificity, so we will adapt more to the race demands (metabolic and neuromuscular) as well as getting used to perception of effort. Those efforts have a stronger signaling and can better stimulate mitochondrial biogenesis and capillarisation. In general I would recommend to start the season with more interval training at targeted intensity and evolve versus continuous effort at targeted intensity when we get closer to the goals.

Examples of sessions:

I believe that the structure of the session has less importance than the amount of time at the zone and the stimuli we are reaching. The structure of the session would be a way to make it more possible to reach the time at zone or the stimuli.

Easy and short – Z1
Goal: To regenerate.
The lactate should go down under 1mmol/l during the session and muscle soreness decreases.

Base – Z1
Goal: To create an aerobic base, to grow mitochondria.
Long continuous easy session, it should feel possible to breathe with the nose all time.

Base – Z2

Goal: To create aerobic base, to increase lactate clearance, to maximize fat oxidation, to increase mitochondria density.

Long continuous session.

Long Progressive. Warm up, then long reps (every uphill for example) increase from lower Z2 to higher Z2 or Z3. Here it also works the glycogen if taking Z3 and lactate clearance.

Mountain long day: Some technical terrain where working more neuromuscularly, uphills to push into higher Z2 and downhills to recover.

TEMPO – Z3

Goal: To get used to race pace (biomechanics, neuromuscular) and work max fat ox and glycogen metabolism.

Steady tempo (early season can be a 30' and increase to race distance)

Long intervals (repetitions up to 30')

Up & Flat: steep uphill, long reps (20-10' or short 3'-5' depending motivation) downhill recovery + flat: some repetitions of 1k, 5k or 10k

Race simulation: A session simulating the race you will do. There it can be complete if training for shorter races or counting on the fatigue from the previous days (for example doing the long progressive or the up and flat or steady long runs the days before this session) if training for longer distances.

THRESHOLD – Z4

Goal: To improve glycogen metabolism

Long intervals: 5000m, 3000m, 2000m, 1000m, or in time 20', 10', 5'

Short intervals: 30''-30'' / 400m with short recovery / 60''-60''

Fartlek: a bit with the terrain, changes of speed of around 1 to 3' at Z4 while recovering 1-3' at Z2.

Double threshold: Long intervals in the morning and short intervals in the evening. Shorter intervals in the evening would go slightly higher in intensity, usually right at or right above the anaerobic threshold, while the morning was lower.

Recovery can be short, just to recover breathing, it can be active or it can be a long separation of the intervals with long tempo (Z2-Z3) in between to stimulate more the durability capacity.

SPEED – Z5

Goal: To improve anaerobic pathways

repetitions of 15'' to 1 minute

hill repetitions of 20''-1'

Multi-jumps

100-200m sprints

Long recoveries in between reps (total recovery, several minutes in between repetitions)

Useful links:

 Stephen Seiler about the training zones: <https://www.youtube.com/watch?v=NPwyk9B0j-s>

 Lactate measuring: <https://upsidestrength.com/lactate-testing-for-athletes/>

 Plan well: <https://podcasts.apple.com/es/podcast/fast-talk/id1490521721?i=1000559692393>

 Lactate and long term planning:
<https://podcasts.apple.com/es/podcast/trainingpeaks-coachcast/id1435395422?i=1000535184548>

 Lactate threshold training: <https://www.mdpi.com/1660-4601/20/5/3782>

 Lactate threshold training Marius Bakken:
<http://www.mariusbakken.com/the-norwegian-model.html>

Neuromuscular

The Neuromuscular capacity refers to the ability of the nervous system and muscles to work together efficiently to produce force, speed, coordination, and endurance. It plays a crucial role in stabilizing the body, absorbing and producing force during climbs and descents, and maintaining running form when fatigue.

1. **Motor unit recruitment** – how many and which muscle fibers are activated.
2. **Rate of force development** – how quickly you can produce force.
3. **Intermuscular coordination** – how different muscles work together.
4. **Proprioception & reflexes** – awareness of body position and fast adjustments on uneven terrain.
5. **Elasticity** – how tendons/muscles store and release energy .

How it applies in:

- **Uphill:** Strong hip/knee/ankle extensors and fast coordination.
- **Downhill:** Eccentric strength, quick and precise foot placement.
- **Technical terrain:** Quick reflexes, joint stiffness modulation, and dynamic balance.
- **Fatigue resistance:** Better neural control delays muscle breakdown and keeps technique efficient.
- **Speed and stride frequency**
- **Stride length efficiency**
- **Running economy** lower cost of effort at the same speed

We are the most efficient (running economy) at the pace we are used to training, so it is important to spend time at the pace we will do the race and in different physiological states (muscularly tired, with energy depletion, in different temperatures...)

In technical downhills, there is more a question of the eye-brain-muscle connection than strength. So adapting the feet to go where we want and how we want, and adapting with what the feet feel in the terrain without the need to look at it is key. Anticipation with the vision 2-3 steps ahead and memorization of the terrain so we can look further and go faster without the need to look at the feet.

Developing the feelings of the different terrains and surfaces is key to anticipate muscle reactions (let glide in soft terrain, block in hard terrain, angle incline, low pressure in slippery terrain...)

Examples of sessions:

Plyometrics
Goal: Stimulates stretch-shortening cycle and improves elastic force.
Bounding, hopping, box jumps, single-leg hops.

Cadence + Hill Sprints or hill jumps
Goal: Improves RFD and motor recruitment, especially uphill.
Do 20'' to 1'(until before neuromuscular failure) of max cadence spinning then: 8-15'' at max speed with full recovery, or 3-5 steps with a exaggerated length in uphill, or 8-10'' sprint rolling.

Downhill and flat adaptations
Goal: Enhances shock absorption and leg control. Improves running economy
Long downhills, long runs at race pace cadence.

Strides
Goal: Keeps the nervous system sharp without fatigue.
Short, fast strides after easy runs (e.g., 4×20 sec).

Strength session

Goal: Develops max force and joint stability.

Squats, lunges, deadlifts (low reps 5-7, high weight)

Uphill Strength + Cadence

Goal: Neuromuscular connections, uphill strength and cadence

In uphill (can be done in light uphill or very steep uphill) alternate 1-2' of maximum cadence (short steps) with very long steps ("jumped"). That can also be done in the bike with Torque and high cadence.

Technical Visio-muscular connection

Goal: real-time coordination and proprioception.

Run on technical terrain, change cadence (exaggerated fast-exaggerated slow) with close eyes for some seconds to work on visual memory and feet perception. Other activities that involve speed and decision making with body placement also help (ski, mountain bike)

Kick off

Goal: Strength in fatigue

. At the end of a training (threshold or in a alone session) do kick-off (change the biomechanics – longer steps, more cadence) For example: 200m running steady, then 100m bouncing (big explosive jumps) then 200m sprint max at high cadence.

Thermoregulation

80% of the energy we transform is in the form of heat. That means that if we eat 1000 cal, 800 of them will be expended to produce heat and only 200 will be available for our muscles.

Loss of performance on high temperature environments:

Performance can drop by 1-2% for every 1°C above ~20°C, especially in long-duration events.

At 30–35°C, endurance performance can decline by 10–15% or more, depending on acclimation, humidity, and individual physiology.

That's because of different factors, first the cardiovascular strain. More blood is diverted to the skin for cooling, reducing the amount available to working muscles → decreased stroke volume → higher heart rate at the same effort.

Secondly, an increased core temperature: When it rises above ~39°C, the central nervous system begins to inhibit motor drive to prevent overheating (central fatigue).

Sweating and dehydration: Fluid and electrolyte loss leads to lower plasma volume → reduced thermoregulation and cardiovascular efficiency.

And also a higher RPE (Rate of Perceived Effort): Same power/pace feels harder in heat.

Gastrointestinal stress: Reduced gut blood flow + heat = higher risk of GI issues, especially with high-carb intake.

Sun radiation:

Direct sunlight heats the skin, increasing skin temperature by 5–10°C.

This reduces the skin-to-air temperature gradient, making heat loss by convection and radiation less effective.

Core temperature rises faster because skin acts like a solar panel absorbing energy and the heat absorbed must be dissipated via sweating, which is less efficient at high humidity or low airflow.

Heat training:

Training in the heat induces a series of adaptive responses that improve both performance in the heat and can also boost performance in cooler conditions.

Main Adaptations:

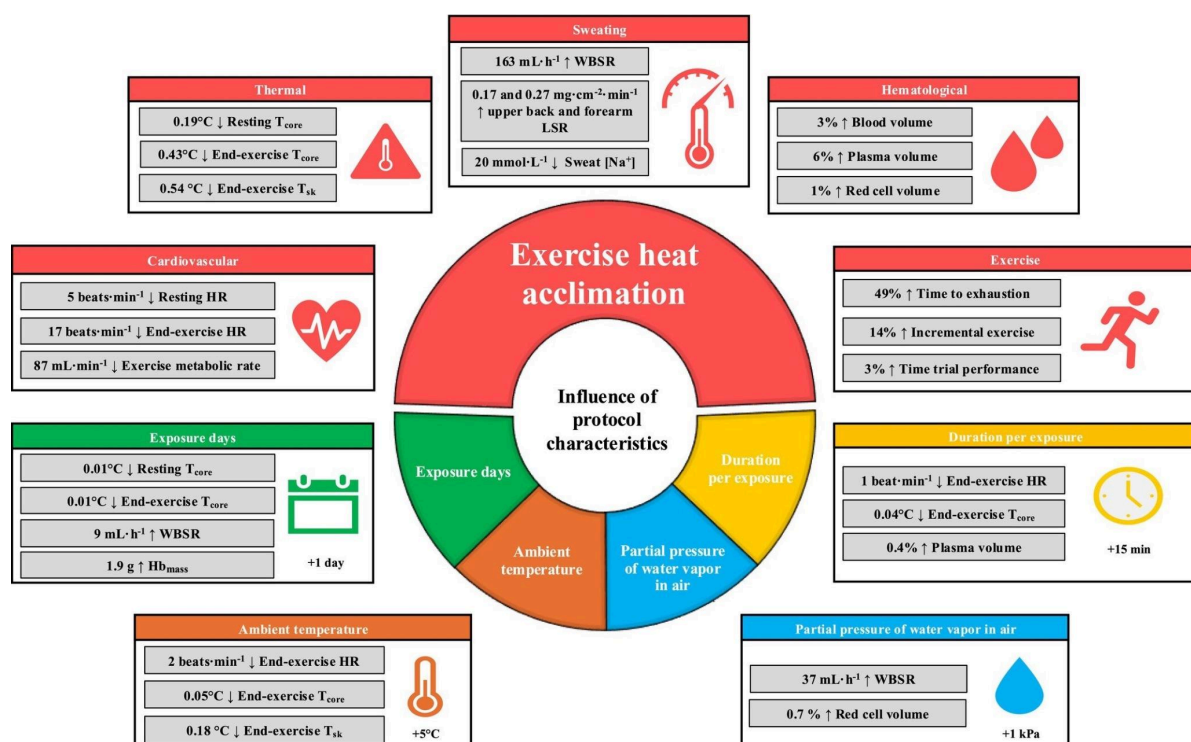
- Lower core temperature at rest and during exercise
- Plasma volume expansion ($\uparrow 10\text{-}20\%$) $\rightarrow$ better stroke volume and cardiac output
- Earlier onset and increased sweat rate $\rightarrow$ better evaporative cooling
- Reduced electrolyte concentration in sweat $\rightarrow$ more efficient hydration
- Improved thermal comfort and reduced RPE
- Stabilized gut function under heat stress

Benefits when racing in Heat:

- Lower heart rate and core temp at given pace
- Better pacing and cognitive function
- Improved ability to ingest and absorb carbs

Benefits when racing in cool temperatures:

- Plasma volume expansion $\rightarrow$ higher blood volume = better cardiac efficiency
- Improved thermoregulation $\rightarrow$ slightly lower core temp rise in long races
- Potentially increased mitochondrial enzyme activity and heat shock proteins that improve muscle function

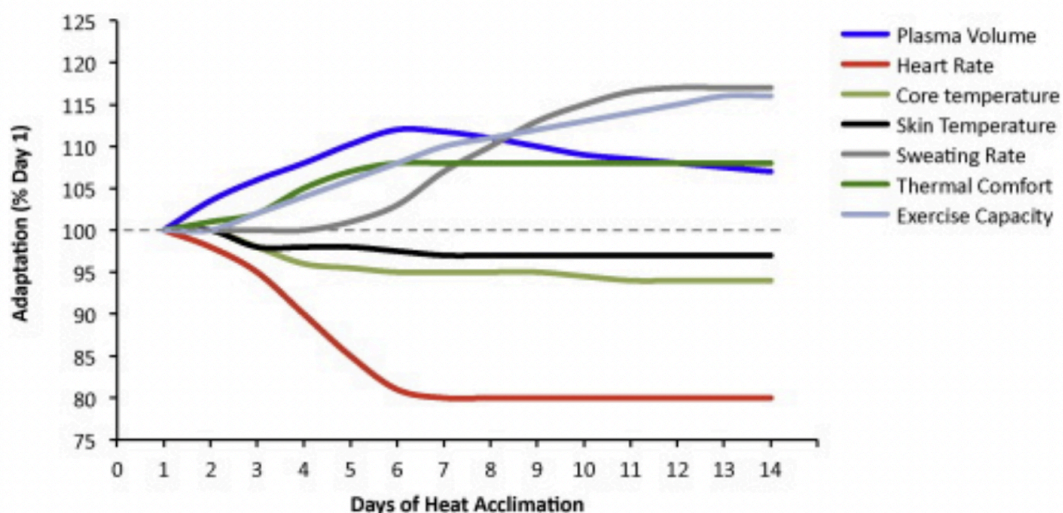


Heat stimuli:

To stimulate heat adaptations there is no need for long sessions. In general, 1-1h30' sessions would be enough to get to the targeted temperature and spend 40'-1h at that temperature. The adaptation is pretty quick after a few sessions but the loss of adaptation is also quick once we stop doing heat training, so the frequency (number of sessions per week) is important. The goal is to reach our targeted temperature - temperature threshold (individual internal temperature to stimulate, average around 38-38,5°C) then stay there for the time we need (40'-1h is enough for general heat training, that can be longer if we are acclimatizing for long races in the heat). If you can't see the internal temperature, you can monitor the sweat loss (weight before-after minus what you drink) and target to lose at least 600-800 ml of sweat/h. It's best to do it in a heat environment (either hot room, or with warm/waterproof clothes) to do the stimuli while keeping it in moderate metabolic intensity (~120-140 ppm).

Examples of training blocks and sessions:

Heat training block: 1-2 weeks 5-7 days/week heat training. After that, do reminder sessions 2-3 times a week.



Maintaining heat during the season: That can help to reduce the number of heat training days to around 3 days per week but done during most of the year around.

Long heat sessions: If competitions are done in hot environments, longer sessions might be interesting, and to train at similar humidity and temperature for durations similar to the race.

Altitude + heat: It might be good to combine with altitude training but not at the same time since altitude decreases plasma volume and t° increases, best suggested to do altitude block to improve Hb, red cells...then heat block to increase plasma volume.

It's not very relevant the way of heat training (dressing, heat chamber, external...) but it is more effective to do it actively (exercising) than passive (sauna, hot baths).

Heat acclimation:

Here the goal is not to get the general benefits of heat training only but to get prepared for an event at high temperatures and/or humidity.

For heat acclimatization is important to do in the same humidity as expected conditions, that can be done only the last weeks before the event for maximizing acclimatization.

During heat sessions is important to keep hydration, to increase sodium intake moderately (3–6g/day or 1500–3000 mg sodium/day during heat block) - Supports fluid retention → larger plasma volume → better cooling. Stay well-hydrated during and between sessions

Non active heat training:

Alternatives from best to worst:

- Dry Sauna: 30'
- Hot tub: 60-90'
- Sauna: 60-90'
- warm clothes: 60-90'

Important to let the system rest before the event, so no heat training the last 4 days to let the system recover and to lower its core temperature.

Pre-exercise cooling:

The goal the last days and during the race day is to decrease the body temperature to a minimum:

- Drink cold water, slushies,
- Use wet and cold clothes (vest, cap)
- Put fresh water or ice at the back of the neck, wrists.

Fluid ingestion:

- See nutrition – hydration

Cooling strategies:

The most effective in terms of a strategy's ability to favorably alter body core temperature are in this order:

1. Aerobic fitness
2. Heat acclimation
3. pre-exercise cooling (using cooling vests, getting in cold spaces...)
4. Fluid ingestion (fluid, cool fluid, cool aliments...)

During the race we can use ice in bandanas or backpacks, sponges, cold water, light clothing to limit sun radiation, cover head and eyes from radiation.

Fueling in the heat:

Heat affects gut function → increases risk of GI distress. Ensure gut training beforehand. Use lower osmolarity drinks, avoid overly concentrated products.

Useful links:

 <https://onlinelibrary.wiley.com/doi/10.1002/cph4.70017>

Heat training adaptations predictor:

<https://www.canberra.edu.au/research/centres/uc-rise/research/environmental-physiology/exercise-heat-acclimation-predictor>

 <https://hal-insep.archives-ouvertes.fr/hal-02569281/document>

 <https://pubmed.ncbi.nlm.nih.gov/12391114/>

Increase of plasma volume:

 <https://www.ncbi.nlm.nih.gov/books/NBK236240/>

Nutrition and heat training:

 <https://www.mdpi.com/2076-2607/12/6/1160>

Fluid ingestion:

 <https://pubmed.ncbi.nlm.nih.gov/30842739/>

 <https://pubmed.ncbi.nlm.nih.gov/20724560/>

 [A physiological perspective of the relevance of sweat biomarkers and their detection by wearable microfluidic technology: a Review](#)

 [The Importance of Maintaining and Improving a Healthy Gut Microbiota in Athletes as a Preventive Strategy to Improve Heat Tolerance and Acclimatization](#)

 <https://help.corebodytemp.com/>

Oxygen

Oxygen is used in a process called cellular respiration, which happens mainly in the mitochondria of our cells. Is key in endurance sports to oxidize the substrates. There are several systems involved in bringing oxygen from the air to our muscles:

1. Pulmonary system

- Role: Oxygen intake and CO₂ removal.
- Adaptations:
 - Increased lung ventilation efficiency.
 - Slightly increased lung volume in some elite athletes.
 - Better matching of ventilation-perfusion in the lungs.

2. Cardiovascular system

- Role: Transport of oxygen via blood.
- Adaptations:
 - Increased cardiac output (mainly stroke volume).
 - Increased blood volume and plasma volume.
 - Greater capillarization in muscle tissue.
 - Higher red blood cell mass (especially with altitude exposure).

3. Hematologic system

- Role: Oxygen carriage.
- Adaptations:
 - ↑ Hemoglobin concentration (especially with hypoxic training).
 - ↑ Erythropoietin (EPO) production → more RBCs.

4. Muscular & mitochondrial system

- Role: Oxygen extraction and usage.
- Adaptations:
 - ↑ Mitochondrial density and volume.
 - ↑ Oxidative enzyme activity (e.g. citrate synthase).
 - ↑ Myoglobin content (oxygen storage inside muscle).
 - ↑ Capillary density for better O₂ diffusion.

The adaptations are made at 4 levels:

1. Ventilation
2. Alveolar-capillary diffusion
3. Circulation
4. Diffusion from capillaries to mitochondria.

1. Ventilation:

Capacity to get air from outside - breathing capacity - volume and frequency - pulmonary system) In general there is no limiter if the athlete has no special problems.

Training improves ventilatory efficiency: better matching of ventilation/perfusion (V/Q), lower dead space ventilation, and stronger respiratory muscles. This helps maintain higher alveolar PO_2 during intense effort.

2. Alveolar-Capillary diffusion:

The capacity of oxygen diffusion from the lungs to the blood flow. Some people are born with a higher oxygen diffusion capacity, due to larger lung surface area, thinner alveolar-capillary membrane, higher hemoglobin levels or affinity and greater capillary density in the lungs. These traits are genetically influenced and define your baseline diffusion capacity (also known as DL_{O_2} or DL_{CO}).

With training we can increase capillarization, the capillary density around alveoli and in muscles increases.

While thickness of the alveolar membrane is less trainable, reducing inflammation, oxidative stress, and maintaining good lung health optimizes gas exchange.

3. Circulation:

Blood volume and total hemoglobin, so more O_2 can be carried and bound, helping maintain the diffusion gradient.

4. Diffusion from capillaries to mitochondria:

The diffusion of oxygen from capillaries to mitochondria is the final segment of the oxygen delivery chain, often called the "oxygen extraction". Improving this step means more oxygen gets into muscle cells and is efficiently used for ATP production during endurance activities.

Oxygen moves by passive diffusion from high to low partial pressure (PO_2):

1. Capillary Blood (High PO_2)
2. Interstitial Space
3. Muscle Cell Membrane (Sarcolemma)
4. Cytosol
5. Mitochondria (Low PO_2)

Oxygen follows a partial pressure gradient. The lower the PO_2 in mitochondria, the stronger the diffusion gradient.

Factor	Effect
Capillary Density	↑ Contact surface area for exchange
Myoglobin Content	Transports O_2 from sarcolemma to mitochondria
Mitochondrial Density	↓ Intracellular O_2 diffusion distance
Mitochondrial Respiration Rate	Creates lower PO_2 in mitochondria (↑ gradient)
Muscle Fiber Type	Type I (oxidative) fibers have higher capillarization and myoglobin
Interstitial Space Composition	Affects O_2 diffusivity

Systems involved

- Cardiovascular System: Delivers blood/ O_2 to the capillaries.
- Muscular Microcirculation: Capillary structure and perfusion control.
- Myoglobin System: Intracellular oxygen shuttle.
- Mitochondrial System: Consumes oxygen → maintains low PO_2 inside.

Training stimuli to improve the oxygen system:

1. Endurance training:

Leads to mitochondrial biogenesis, capillary growth, improved substrate oxidation.

Low intensity, high volume training ("Zone 2")

- Increases capillary density
- Stimulates angiogenesis (via VEGF)
- Enhances myoglobin expression
- Promotes mitochondrial biogenesis

So more oxygen can be extracted per unit of blood delivered.

Long intervals at LT2 or $\text{VO}_{2\text{max}}$

- Stimulates both peripheral and central adaptations.
- Increases oxygen demand → drives more oxygen extraction.
- Recruits Type I and IIa fibers → mixed adaptation.

2. Strength-Endurance / Blood Flow Restriction (BFR)

↑ Local hypoxia → upregulates HIF-1 α → ↑ VEGF → capillarization

Also increases oxidative enzyme expression

Example: Strength sessions or uphill running with local hypoxia (blood flow restriction) in the main muscles.

3. Heat Acclimation / Hyperthermic Training

Can increase plasma volume, improving perfusion and may enhance angiogenic responses with combined endurance load.

4. Hypoxia

Low O_2 tension → ↑ capillary and myoglobin adaptations. Stimulates mitochondrial oxidative capacity.

Simulated (hypoxic tents, masks) or real (living/training at altitude).

↑ EPO → ↑ RBC mass and hemoglobin.

Increases efficiency of oxygen usage at the muscle level.

5. Respiratory Muscle Training

Breathing exercises (resistance devices, breath-holds).

↑ strength and endurance of respiratory muscles (diaphragm, intercostals).

Can reduce perceived effort and respiratory fatigue.

Example: Strong expirations / inspirations until feeling dizzy (1') then holding in apnea lungs empty until exhaustion, then taking air 1 time and maintaining apnea (lungs full -until exhaustion) Repeat several times 3-5.

Molecular and physiological adaptations

Stimulus	Adaptation	Molecular Mechanisms
Endurance	↑ Mitochondria	↑ PGC-1 α → mitochondrial biogenesis
HIIT	↑ VO ₂ max	↑ VEGF → angiogenesis; ↑ mitochondrial enzymes
Hypoxia/Altitude	↑ RBC, Hb	↑ HIF-1 α → ↑ EPO production
Breathing	↓ Work of breathing	↑ Muscle strength and fatigue resistance

Protocols of Altitude / Hypoxic training:

Altitude vs intermittent altitude:

If goal is acclimatizing or creating benefits for lower altitude:

In altitude , blood count (hb, Epo)... so transport, the problem is that it costs more to keep intensity at high altitude. especially if they can't keep the same intensity.

Responder (altitude, load, less perfusion-intestine damage)

Need to do well individualisation (load, nutrition, intensity...)

Intermittent:

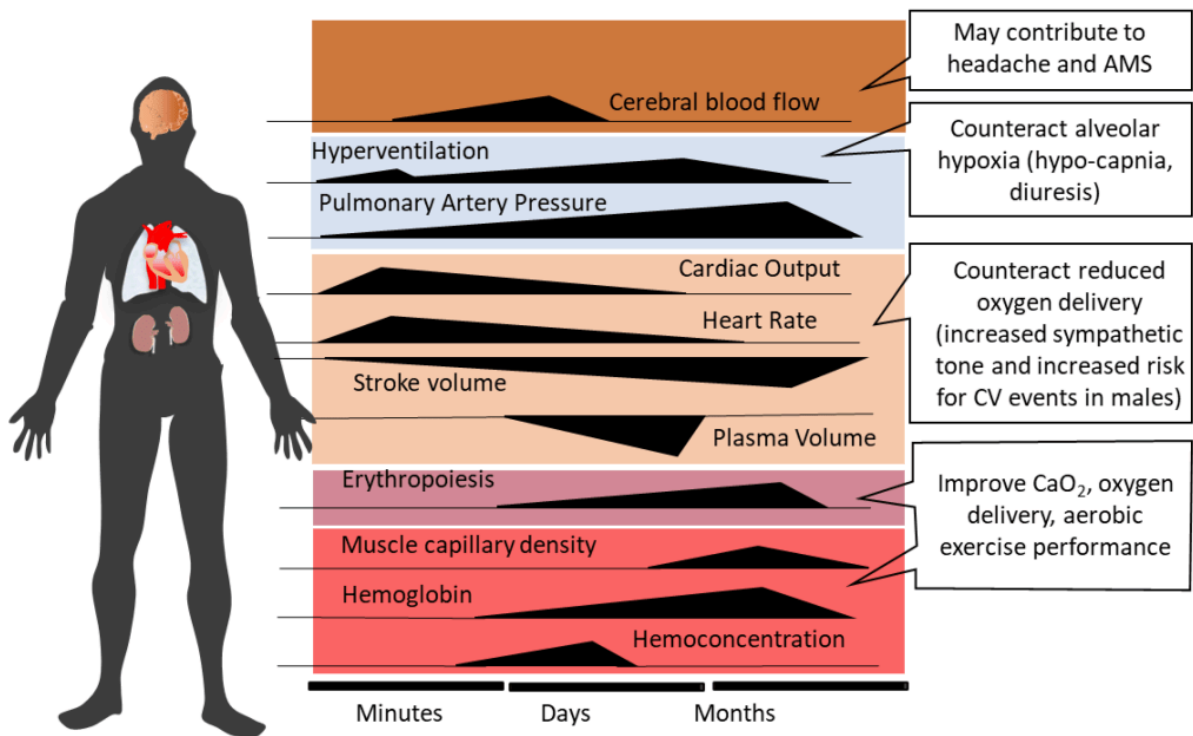
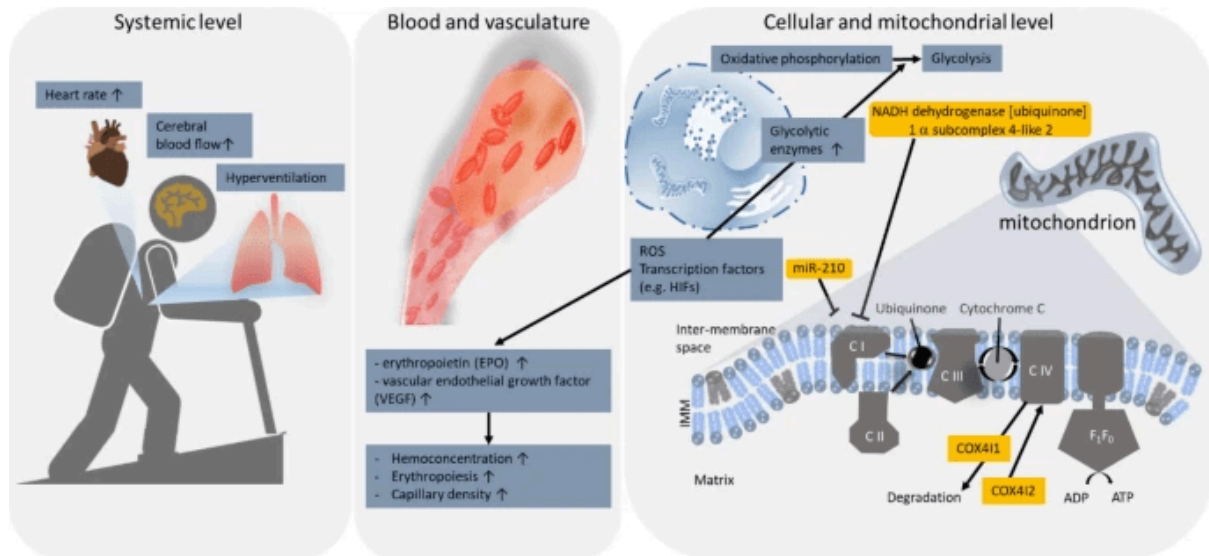
Not in transport (not enough time in altitude) but functions metabolic or extraction of O₂ muscle, reduction during stimuli of O₂ concentration.

Altitude:

From when we reach a higher altitude: We increase our ventilation. So consequently we increase the CO₂ production: Increases body acidity, this is regulated by the kidneys (regulation of O₂ and brain). So to not have kidney problems and stop this regulation of Co₂

is important to drink a lot (6L/ day) or don't lose water (breath warm- humid). If we stop peeing, it means that we have a kidney problem.

- 24h after reaching the altitude: We have an EPO production to increase the red cells, to increase the transport of O₂.
- 7 days: We can see an increase in the hematocrit and the effects of blood acclimatization
- 3 weeks / 400h: Hif 1 (diffusion O₂ to muscles) Enzyme and diffusion acclimatization. All our body can do in acclimatization is done.



First days in altitude go slow : Z1. Increase recovery. If very high altitudes even 1 day activity / 1 rest day.

If you haven't acclimatized the best is to arrive the day/ night before the race before the body makes responses to altitude. If your race is at 2000m, 3 to 5 days might be enough.

While being in altitude focus on:

- Easy paces at the beginning (until feeling acclimatized, Z1-Z2)
- Keep high hydration (drink more than usual to compensate liver work and often dryer air)
- If feeling unwell, get to a lower altitude quickly.

High intensity sessions at very high altitude

There can be some interest in high intensity sessions at very high altitudes (5000-6000m) for example 10x1', progressives, 15-20" sprints... where with a low mechanical impact we can reach some metabolic stimuli very high. Lactate will be much higher (easy to get to 15- 20 mm l/l in 3-4 sprints) Therefore it's interesting for glycogenesis stimulation + breathing + capillarization adaptations.

Molecular players in training adaptation:

Molecule	Role
VEGF (Vascular Endothelial Growth Factor)	Vascular endothelial growth factor. Stimulates capillary growth
HIF-1 α (Hypoxia-Inducible Factor)	Activates EPO, angiogenic and mitochondrial genes
PGC-1 α	Master regulator of mitochondrial biogenesis
Myoglobin	↑ with endurance and hypoxic training
Nitric Oxide (NO)	Regulates vasodilation and perfusion

Adaptations improving capillary to mitochondria O₂ Diffusion:

Adaptation	Mechanism	Training to Stimulate
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↑ Capillary density	↑ Surface area, ↓ diffusion distance	Zone 2, long intervals, BFR
↑ Myoglobin	↑ Intracellular O ₂ transport	Base training, hypoxia
↑ Mitochondrial density	↑ O ₂ usage, ↑ gradient	HIIT, Zone 2, altitude
↑ VEGF expression	Capillary growth	Hypoxia, BFR, heat
↑ HIF-1 α activation	Enhances O ₂ extraction mechanisms	Hypoxic/hard efforts

Useful links:

 <https://link.springer.com/article/10.1007/s42978-021-00150-0>

<https://www.frontiersin.org/journals/physiology/articles/10.3389/fphys.2021.735557/full>

<https://www.mdpi.com/1422-0067/24/2/1698>

Cognitive

At the same physiological capacities, the athletes with a higher pain tolerance will be able to perform better. At the end, what limits us, what makes us stop pushing is the central governor making us stop or slow down because it senses that it's entering a dangerous zone. But in life threatening situations we can go beyond those limits and push much more. Some of that is related to hormonal activation and release (Adrenalin, endorphins...) Similar to the ones we can face in competitions, and another is that in a fight-to-fight mode the central governor feels that it's more dangerous to slow down performance than to risk health or life due to pushing the systems.

The cognitive functions affects performance in:

- memory, encoding, storing, and retrieving information, shaping past experiences, and influencing future decisions
- attention, focus, filter distractions
- executive functions
- visual-spatial processing
- Knowledge
- Fear, attention, stress management.
- Pain management

How do we perceive and process the information?

The first step to process information is to perceive it. That is mostly done through our senses:

1. Sight (Vision) – perception of light, color, and shapes through the eyes.
2. Hearing (Audition) – perception of sound via vibrations detected by the ears.
3. Smell (Olfaction) – detection of airborne chemicals by receptors in the nose.
4. Taste (Gustation) – detection of flavors (sweet, salty, sour, bitter, umami) by taste buds on the tongue.
5. Touch (Somatosensation) – perception of pressure, vibration, temperature, and pain through the skin.
6. Proprioception – sense of body position and movement; knowing where your limbs are without looking.
7. Equilibrioception – sense of balance, controlled by the vestibular system in the inner ear.
8. Thermoception – perception of temperature (heat and cold).
9. Nociception – perception of pain.
10. Interoception – sensing internal sensations and bodily states like hunger, thirst, heartbeat, need to breathe, and bladder fullness.
11. Immunoception - Sensing our immune system.
12. Mitoception - sensing our cellular energy levels
13. Chronoception (debated) – sense of the passage of time, influenced by brain processes and rhythms.

Once we have perceived the information, we process that information in our brain, comparing it to our memory and knowledge:

1. Prefrontal cortex (PFC): Executive functions: planning, inhibition, working memory
Decides: *“Should I slow down?”*, *“Do I need fuel?”*
2. Anterior cingulate cortex (ACC): Monitors conflict or error: *“This pace feels harder than expected”*
3. Insular cortex: Processes interoceptive signals: thirst, hunger, pain, effort. Integrates internal state into decision-making.
4. Basal ganglia & limbic system: Involved in motivation and reward: *“Do I still want to keep pushing?”* Dopamine system tracks expected vs actual rewards.
5. Motor planning areas: Premotor and supplementary motor cortex decide on movement patterns: *step here, jump there, push harder.*

How cognitive fatigue affects performance:

Mental fatigue, caused by prolonged cognitive effort like concentration, decision-making, or emotional stress—has been shown to impair physical performance.

In training, high stress or mental fatigue, more risk of injury, inflammation, sleep deprivation, family, work...

The day before the race we often do nutrition, strategy the day before race, mental fatigue before race, life, travel...

1. Attention and Focus

- You need sustained attention for route-finding, pacing, and adapting to terrain.
- During fatigue or monotonous sections, sustained attention decreases, increasing the risk of mistakes or slowing down.

2. Executive Functions

- These include planning, decision-making, and inhibitory control.
- For example: resisting the urge to chase someone too early in a race, adjusting pace based on changing conditions.

3. Working Memory

- Keeps track of current pace, nutrition intake, route changes, or terrain features.

4. Emotional Regulation

- Helps deal with discomfort, frustration, or unexpected issues like weather or pain.
- Linked to motivation and perceived effort.

5. Perception of Effort

- Central governor theory and psychobiological models show that mental fatigue increases perception of effort, reducing performance even if muscles are still capable.
- Brain acts as a “performance limiter.”

Strategies to enhance cognitive functions for performance:

1. Cognitive Training

- Tools like dual-task training (running while solving math or memory tasks).

- Brain endurance training (BET): repeated mental tasks under physical fatigue.
- Apps (e.g., NeuroTracker, Lumosity) help train visual attention and working memory.

2. Mental Fatigue Exposure

- Train under slight mental fatigue to improve resilience (e.g., doing intervals after mental tasks).
- Cognitive control improves tolerance to perceived effort.
- After hard workouts or between intensities, for about 20-30' do some cognitive work:
Tasks: solving problems, focus: look for a target among distraction. Perception and situation, attention, rapid decision making, inhibiting automatic response. In a fatigue state to not lower training efficiency and to maximize cognitive load.

3. Mindfulness & Meditation

- Enhances awareness, emotional regulation, and focus.
- Reduces anxiety, helps with pain perception and pacing.

4. Nutrition & Hydration

- Maintain blood glucose for brain function.
- Tyrosine or caffeine can enhance alertness and reaction time.
- Proper hydration keeps neural transmission efficient.

5. Sleep and Recovery

- Crucial for memory consolidation, decision-making, and reaction time.
- Sleep banking before ultras can enhance focus and reduce errors.

6. Simulated Race Scenarios

- Practice decision-making under stress (route-finding, gear choice, nutrition).
- Includes emotional triggers and unexpected changes.

Uncertainty:

Train to be good at uncertainty, to adapt to situations (race with no perfect conditions – external and internal – sic...) to different terrains, length of effort (if someone attacks how long should I stay)

Train without information of data to get use on the feelings and to not rely completely on that

Train with no knowledge of duration, n° of repetitions...

During competitions:

The psychological and hormonal changes that occur during a race—often referred to as competition-induced stress response—can dramatically alter how athletes perceive effort (lower RPE) and enhance their ability to push harder than in training. This is a complex interplay of neuroendocrine activation, emotional arousal, and environmental factors.

Hormone/Neurotransmitter	Role in Race Context
Adrenaline (Epinephrine)	Increases HR, ventilation, blood glucose, muscle perfusion—fight or flight activator
Noradrenaline	Boosts alertness, mobilizes energy, sharpens focus
Cortisol	Increases glucose availability, suppresses pain and inflammation, supports endurance
Dopamine	Enhances motivation, reward prediction, motor drive
Endorphins	Reduce pain and discomfort (natural opioids), contributing to lower RPE
Endocannabinoids (e.g., AEA)	Elevate mood, reduce anxiety, and enhance well-being
Serotonin	Regulates mood and fatigue perception; modulated differently in race vs. training

1. Arousal and focus (Yerkes-Dodson law)

- Moderate stress raises cortical arousal, improving reaction time, focus, and motivation.
- The presence of competition boosts sympathetic drive, which sharpens attention and suppresses distractions.
- Capacity of zooming in (having high attention) and zooming out (lowering the attention level)

2. Motivation and purpose

- Races activate extrinsic motivators (audience, rivals, results) and intrinsic motivators (self-identity, goals).

- This creates a dopaminergic drive to overcome fatigue signals and push beyond comfort.

3. Perceived threat vs challenge

- In competition, the body interprets the stressor as a challenge, not a threat, leading to a performance-enhancing response (vs. fear-based shutdown).
- Appraisal theory suggests the perception of control and purpose modifies the physiological response and reduces perceived effort.

4. Lower RPE in races vs. training

- The brain recalibrates effort perception based on context and reward expectations.
- Signals of fatigue are downregulated in areas like the insula, anterior cingulate cortex, and prefrontal cortex during high-stakes situations

5. Greater motor output and drive

- Central governor model: In a race, the brain allows more motor unit recruitment before it limits output, sensing the reward outweighs the "risk."
-

6. Emotional regulation enhances output

- Athletes in a flow state suppress the default mode network (self-referential thoughts), allowing better focus and emotional regulation.

Factor	Effect in races
Hormonal response	Elevated catecholamines and cortisol boost energy and pain suppression
Neurotransmitters	More dopamine, endorphins, endocannabinoids enhance mood and motivation
Cognitive reappraisal	Race context seen as meaningful → reduced RPE
Sympathetic activation	Increased readiness, blood flow, ventilation
Psychological environment	Crowd, rivals, stakes increase effort and tolerance

Strategies to keep cognitive functions during competitions:

1. Visualization

- What it is: Mentally rehearsing the course, key moments (e.g. climbs, pain), and your desired emotional and physical reactions.
- Benefits:
 - Activates the same brain areas as physical performance.
 - Prepares for adversity and improves confidence.
- How to use: Visualize yourself staying calm, focused, and powerful through difficult moments. Include multisensory details (sights, sounds, fatigue).

2. Implementation Intentions (“If–Then” Planning)

- What it is: Making plans for specific challenges.
- Examples:
 - “If I feel my legs getting heavy on the last climb, then I’ll focus on my breathing rhythm.”
 - “If I feel overwhelmed, then I’ll remind myself I’ve trained for this.”
- Benefit: Reduces cognitive load by preparing automatic responses to stressors.

3. Goal Setting

- Use process goals (e.g. "keep cadence steady", "stay relaxed") rather than only outcome goals.
- Helps focus attention and conserve mental energy during long races.

During the Race

1. Self-Talk

- What it is: Internal dialogue to regulate effort, emotions, and attention.
- Two types:
 - Instructional: "Relax your shoulders", "Focus on stride".
 - Motivational: "You’re strong", "You can handle this".
- Best when pre-planned and practiced in training.

 *Blanchfield et al. (2014)* showed that trained self-talk increased time to exhaustion in cyclists.

2. Attention Control / Focus Shifting

- Alternate between:
 - Associative focus (on body, breathing, terrain).

- Dissociative focus (music, rhythm, scenery) during less critical sections to reduce mental fatigue.
- Helps prevent cognitive overload and mental fatigue buildup.

3. Breath Control / Mindfulness-in-Action

- Using breath to reset attention or manage emotions.
- Helps calm the sympathetic system (fight-or-flight), preserving prefrontal control and decision-making.

4. Chunking

- Mentally break the race into manageable segments (aid stations, climbs, time blocks).
- Reduces perceived task duration and mental load.
- Can be paired with mini-goals: “Just run well until the next ridge.”

5. Positive Reframing

- Actively reappraise discomfort: “This pain means I’m pushing well,” or “This is when I grow.”
- Keeps emotional fatigue in check.

Lifestyle constraints:

Motivation:

Quality of motivation, the why: Why are you running, racing, doing that particular race or project?

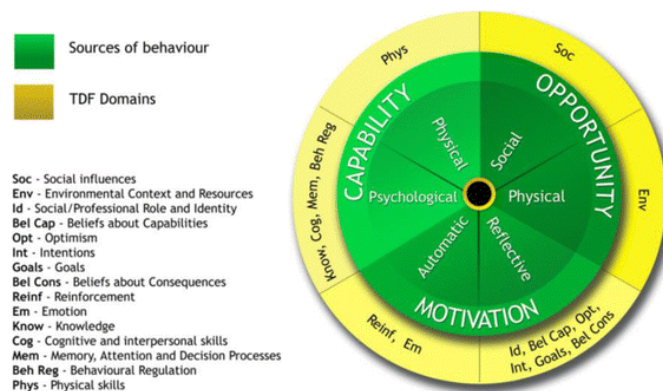
Quantity of motivation: How much effort you are willing to put to achieve something (pain tolerance) value of reward, need of reward, behavior regarding achieving the reward (thoughts you will achieve it or not, how you interpret and how you react on them)

The motivation is important on the race day but mostly during the training process to keep the drive and the consistency.

The further away is the reward (the race) the hardest is to keep the motivation, therefore the “why”, the focus on the process and enjoyment on the activity and secondary goals have more importance.

External motivation: It can be useful for short periods, but if you need a lot of them to keep training you need to rethink why you’re doing sport.

To train or execute any action we need the capacity (physical capacity and knowledge of how to perform the task), the opportunity (the conditions, resources to do it) and the motivation (something that makes us do it).



CAPABILITY

Physical Capability

Developing and refining technical and biomechanical running/climbing/skiing techniques.

Psychological Capability

Knowledge: Understanding training principles, nutrition, recovery, pacing.

Cognitive and Interpersonal Skills: Self-monitoring, decision-making under fatigue, race strategy.

Memory, Attention, Decision Processes: Staying attentive during long efforts, making critical decisions (e.g., navigation, hydration).

Behavioral Regulation: Planning and adjusting training, managing effort across races and seasons.

2. OPPORTUNITY

Physical Opportunity

Environmental Context and Resources: Access to trails, training infrastructure, weather conditions, gear, time.

Social Opportunity: Coaching, training partners, community support.

3. MOTIVATION

Reflective Motivation

Identity: Seeing oneself as an athlete, belonging to an identity.

Beliefs about Capabilities: Confidence in self capacities. Knowing that one will or can win. Creating momentum: When a strike of winning or good results for oneself or the team, partners enforces the belief.

Goals: Setting training targets, race objectives, personal projects.

Intentions: Committing to a race or long-term project.

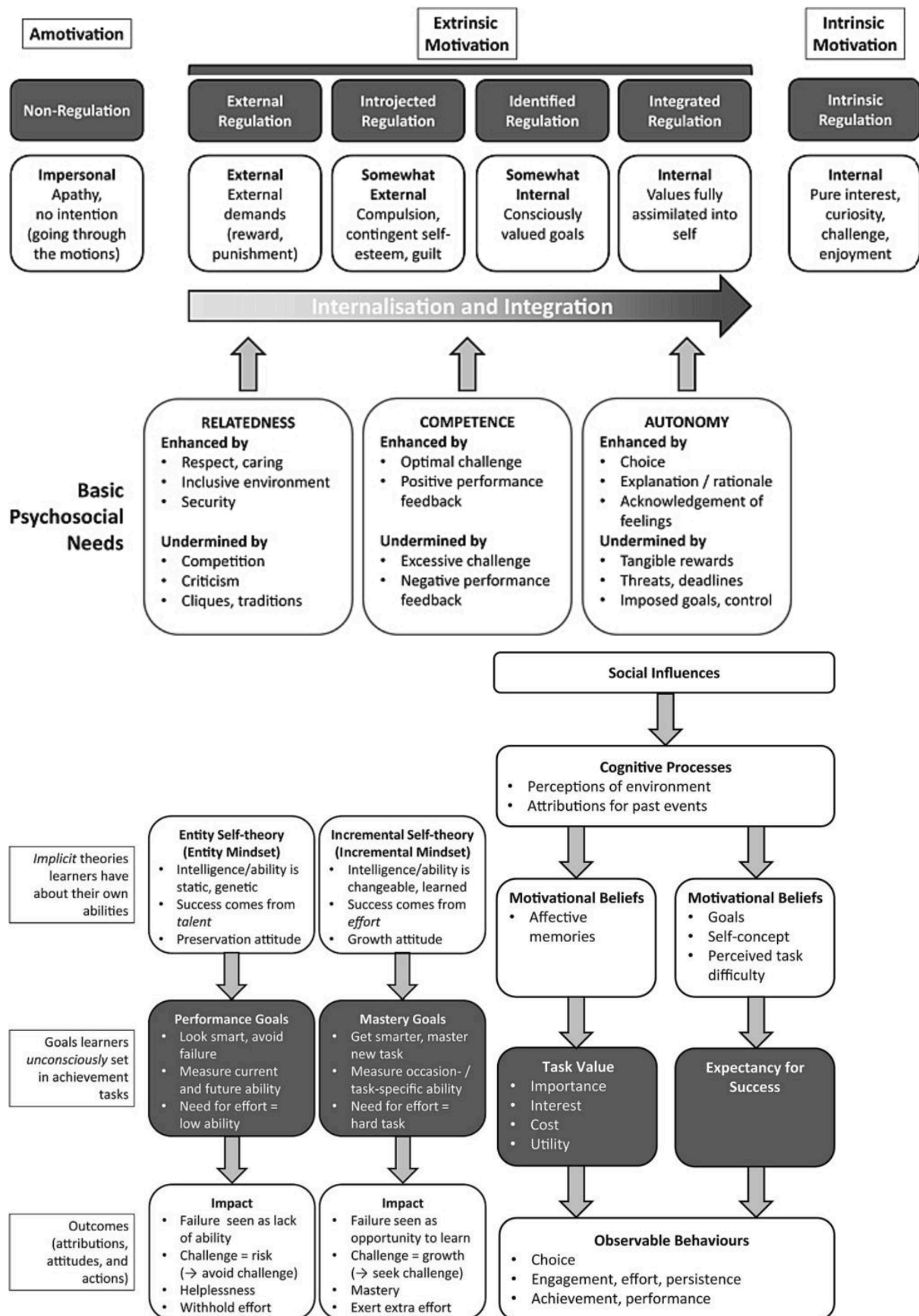
Beliefs about Consequences: Belief in health benefits, reputation, meaning, ethics...

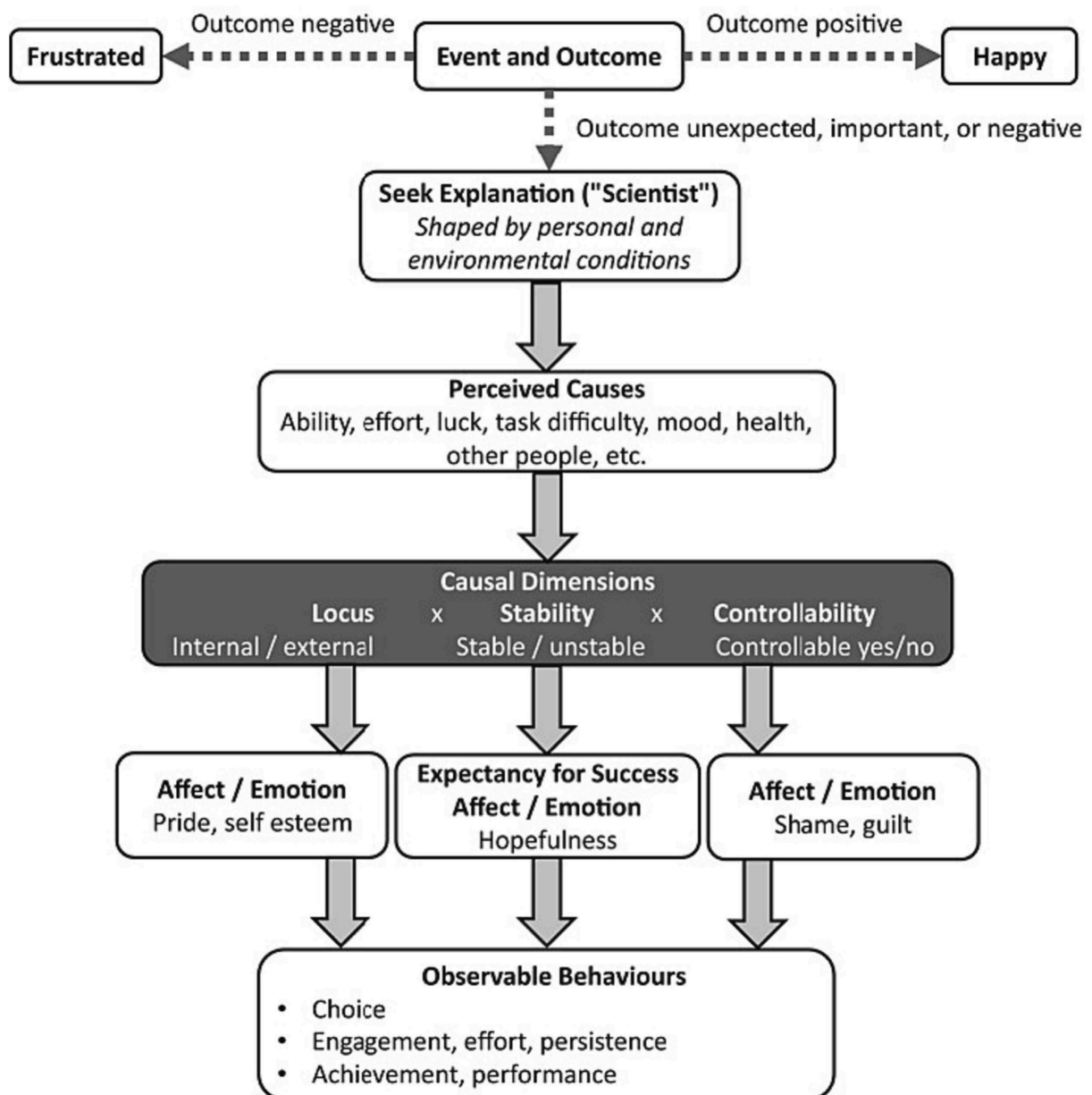
Optimism: Confidence in outcome, in that performance and learning will improve with training.

Automatic Motivation

Reinforcement: Enjoyment from nature, post-run endorphins, external rewards.

Emotion: Emotional connection to landscapes, flow states, to success, hormonal responses...





Overconnectivity & the Strava effect

In today's hyper-connected world, athletes are more visible than ever. Social media platforms like Instagram and training apps like Strava create an environment where every run, workout, or race can be tracked, shared, and compared. This can generate a subtle (or not so subtle) pressure to always perform, to always show up, and to constantly "do more."

This phenomenon often leads to FOMO, not just on races, but on the perceived progress of others. You scroll through your feed and see another athlete doing back-to-back long runs or a personal best, and you start questioning your own routine, even if it's well-structured and adapted to your goals.

This "Strava effect" can erode internal motivation, replacing it with external validation. We might stop listening to our bodies and start listening to what the algorithm favors. While these tools can provide community and inspiration, it's crucial to stay grounded in your own process, resisting the pressure to mimic what works for others.

Community and culture in training

There's often a search for the "best" training method, but history and observation show something else: athletes from different parts of the world succeed with very different training systems. Why? Because culture and community shape training habits.

In many cases, athletes don't follow a "scientifically" optimized plan—they follow what the people around them do. And surprisingly often, that works. It creates rhythm, consistency, and belonging. The sense of being part of something—a group, a tradition, a local style—can fuel motivation and bring emotional stability, which are often more important than the specifics of training zones or interval prescriptions.

Ultimately, performance thrives when we feel accepted and supported in the culture we live in. Feeling you belong in your training environment brings not only happiness, but also resilience.

Training camps

Training camps are a double-edged sword. On one side, they offer a unique opportunity: to increase focus, to adapt to a specific environment (altitude, heat, trails...), and to temporarily remove everyday distractions. Athletes often train better not just because of the location, but because they don't have to cook, clean, or manage family life, therefore recovery becomes easier. Camps can also offer a boost through group dynamics: training with others creates momentum, shared suffering, and joy in shared goals.

But on the other side, traveling to a camp can involve long journeys, flights, crowded airports, time zone shifts, and sleep disturbances. These are all stressors that may counteract the benefits of the camp itself. There's also a risk of getting sick from virus in those crowded spaces during the travels.

If your home setup already includes good terrain, proper recovery, and a sustainable routine, leaving it might not always be worth the risk. Sometimes the best training camp is the one you create at home.

Training alone vs. with others

Training alone gives you full control: you can stick to your pace, your plan, and listen closely to your sensations. It's ideal for developing discipline and precision. But motivation can fade more easily when you're facing wind, rain, or fatigue. Without external cues, the chance of undertraining, or skipping altogether, increases.

Training with partners, on the other hand, brings social motivation. Having someone next to you, even in silence, can make the session feel lighter. You're more likely to get out the door and finish the workout. But here, the risk is overstepping your plan, doing too much to keep up, or too little to stay together.

The balance

Training is just one form of stress. Life—work, family, social commitments, sleep, travel—is full of others. All these stressors interact. The most successful athletes are not those who train the most, but those who find a training load that is sustainable with their lifestyle.

That means shaping your training around your life's rhythm, not just cramming it into gaps. If weekends are full of family time, then maybe those are lighter days. If work is intense, maybe a lower volume week is better than forcing a high-load plan. The body doesn't separate stress from training and stress from life—it reacts to the total.

Long-term consistency and mental freshness come from this awareness. You're not weak for doing less, you're smart for doing what's sustainable.

Circadian rhythms

Circadian rhythms govern nearly everything in the body—from hormone release to core temperature to recovery speed. Disrupted rhythms (late nights, inconsistent meal times, artificial light exposure) can quietly degrade performance and health.

Keeping regular sleep-wake cycles, eating at consistent times, and managing light exposure (especially morning sunlight and avoiding bright screens late at night) help align your biological clock. This alignment supports better hormonal balance, immune function, mood, and energy, making training more effective without adding more volume or intensity.

Useful links:

 <https://onlinelibrary.wiley.com/doi/abs/10.1111/psyp.13343>

 Podcast Samuele Marcora about RPE

<https://podcasts.apple.com/no/podcast/inside-exercise/id1631457776?i=1000599122536>

 <https://www.sciencedirect.com/science/article/abs/pii/S0303264723002459>

 Cost of sleep deprivation: <https://pubmed.ncbi.nlm.nih.gov/12683469/>

 Mitoeption: https://osf.io/preprints/osf/ekj54_v1

 Non-conscious visual cues related to affect and action alter perception of effort and endurance performance <https://pubmed.ncbi.nlm.nih.gov/25566014/>

 Temporal discounting further away is the reward the more discount. (in behavior – reward)
<https://podcasts.apple.com/no/podcast/supporting-champions/id1330373242?i=1000658726445>

 Durability and psychological:
https://journals.lww.com/acsm-msse/fulltext/2023/01000/the_relationship_between_physiological.14.aspx

 <https://pmc.ncbi.nlm.nih.gov/articles/PMC9916354/>

Nutrition

We need to eat to survive and to produce energy. Our bodies require “fuel” to produce energy and different substrates and raw materials to maintain homeostasis, repair tissues, support growth and various physical functions.

The Basal metabolic rate (BMR) is the energy we need to survive, for the function of our organs and to repair our cells:

Building blocks for growth and repair:

Proteins and lipids.

Amino Acids from dietary proteins are used to synthesize structural proteins, enzymes and signaling molecules.

In intense training we might need around 1,5-2g/kg of protein intake for those functions.

Specially when getting older when muscle degradation is higher.

Lipids, especially phospholipids, are essential for building and maintaining cell membranes.

Metabolic regulation and biochemical reactions:

Vitamins and minerals.

Vitamins act as cofactors / coenzymes in enzymatic reactions, such as energy metabolism and DNA synthesis.

Minerals are involved in enzyme function, oxygen transport and nerve signaling.

Structural integrity:

Fats and proteins

Essential fatty acids (Omega 3, 6) are used to produce cell membranes and signaling molecules.

Proteins are critical for maintaining the extracellular matrix that provides structural support to tissues.

Immune function:

Proteins, vitamins (C, D, E...) and trace elements (zinc, selenium...)

Proteins provide the building blocks for antibodies and immune cells.

Vitamins and minerals support the antioxidant system, reducing oxidative stress and enhancing immune defenses.

Maintaining homeostasis:

Water, electrolytes, salts.

Electrolytes (sodium, potassium, chloride...) maintain fluid balance, nerve transmission and muscle contraction.

Water is vital for transporting nutrients, regulating temperature and facilitating biochemical reactions.

Brain function:

The brain uses ~20% of resting energy expenditure, almost exclusively oxidizing glucose, though in long fasting or keto states it also uses ketones. Brain activity during intense cognitive tasks will increase its energy use.

Mechanical energy production:

Carbohydrates, fats and proteins.

We need energy to contract the muscles when we exercise. The energy is provided in the form of ATP (adenosine triphosphate). Since the ATP stored in our body is very small, we need to be able to generate ATP quickly. Those systems are called the energy metabolism pathways.

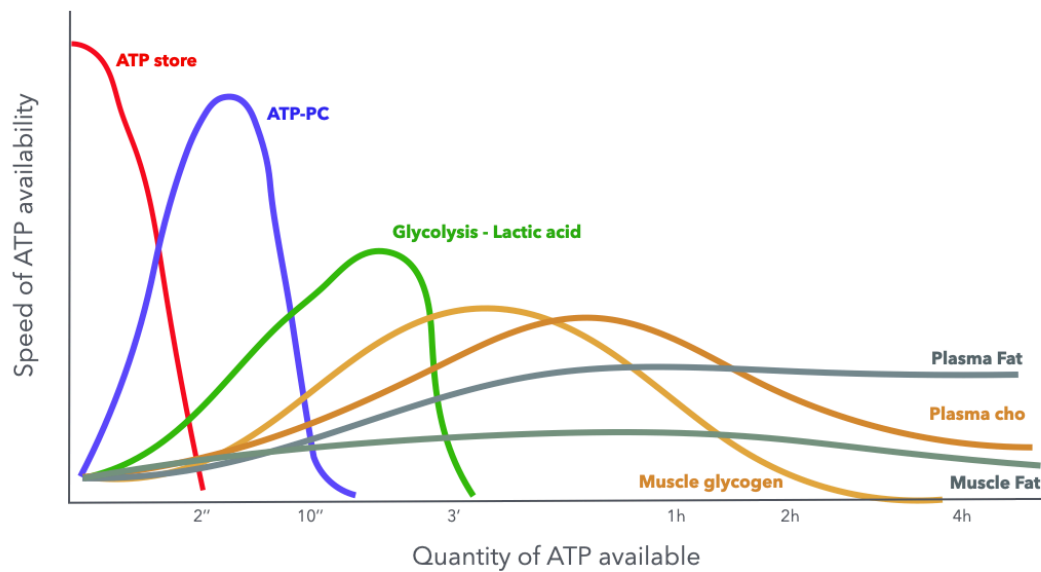
The 4 more important are:

- Phosphocreatine (PCr)
- Glycolysis and lactic acid

Mitochondrial respiration (requires oxygen)

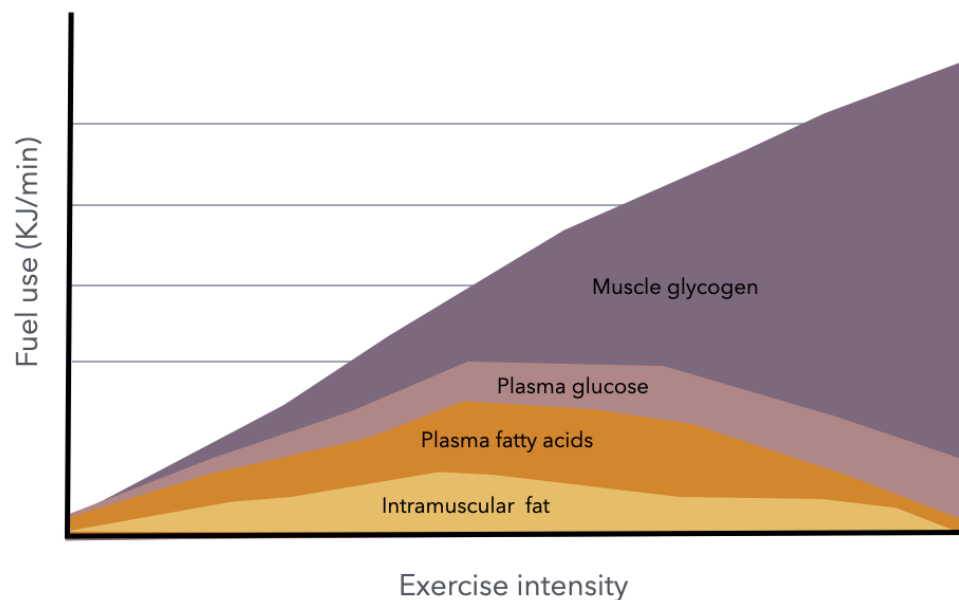
- Carbohydrate oxidation (muscle glycogen and plasma glucose)
- Fat oxidation (muscle triglycerides and plasma Fatty acids)

Those pathways vary in the speed and capacity they can generate ATP.

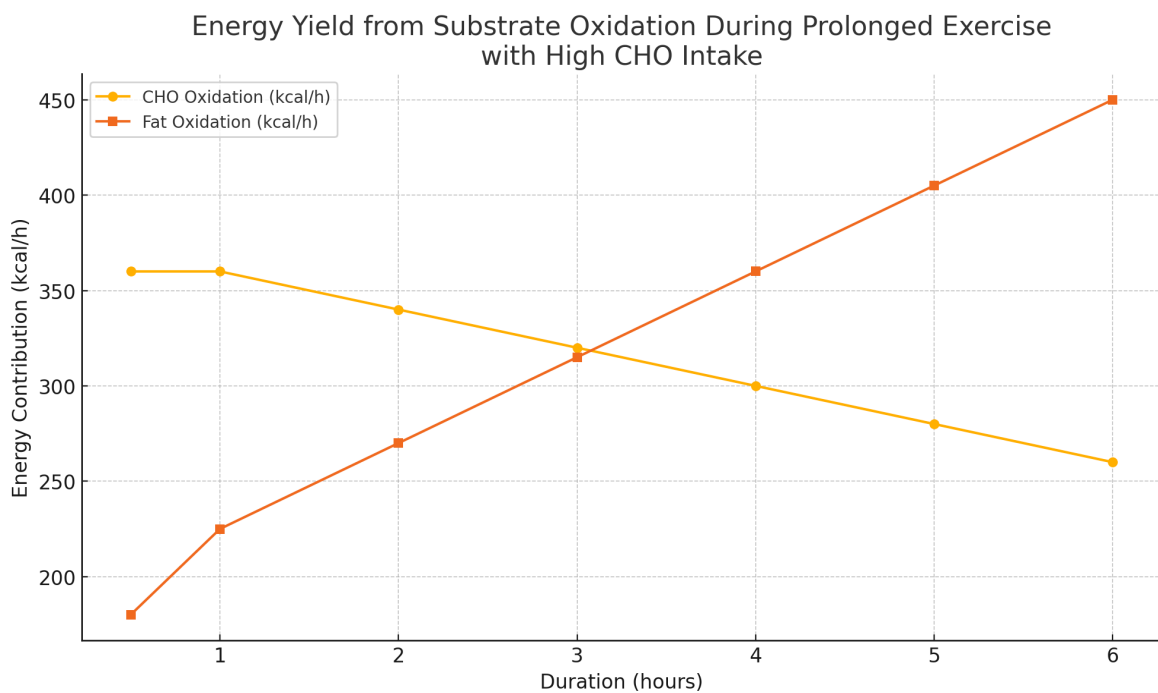
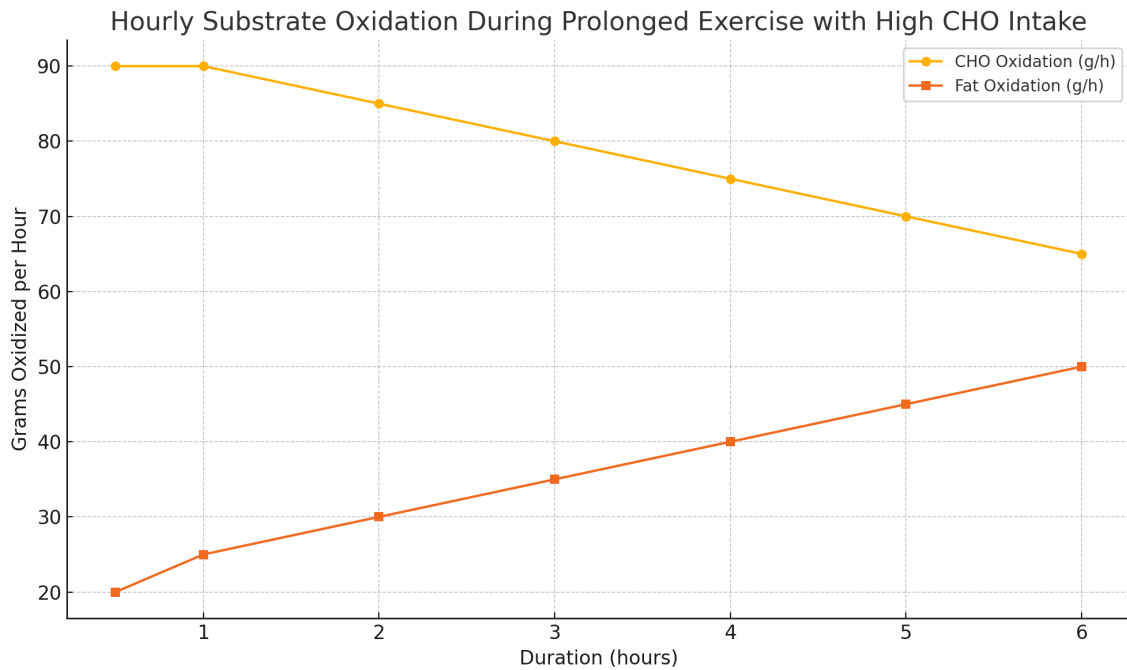


- All systems at one: When we exercise, all the systems are used at the same time. The more intense we exercise and therefore our energy needs increase, we use all the systems in a more extensive way.
- Muscle glycogen can deliver energy via glycolysis and via oxidation.

Fuel use at different intensities



Substrate utilisation at duration in same intensity:



Graph showing cumulative grams of carbohydrate and fat oxidized during prolonged submaximal exercise (e.g., 50–65% $\dot{V}O_{2max}$) with high carbohydrate intake (90–120 g/h)

- CHO oxidation stays high (~85–90 g/h) but gradually declines due to partial glycogen depletion and metabolic shifts.
- Fat oxidation increases slightly despite high CHO intake, though it remains lower than in low-CHO or fasted states due to insulin's suppressive effect on lipolysis.

Carbohydrates

- Oxidized mainly via glycolysis and aerobic metabolism.
- Yield ~4 kcal per gram when fully oxidized to CO₂ and water.
- Slight variations exist depending on the type (glucose, glycogen, etc.), but 4 kcal/g is the standard.

Fats

- Oxidized through beta-oxidation and the TCA cycle.
- Yield ~9 kcal per gram because fats are more reduced (i.e., they have more hydrogen atoms available for ATP-generating reactions).

Metric	CHO (Glucose)	Fat (Palmitate)
Energy (kcal/g)	~4 kcal	~9 kcal
Approx. ATP Yield per g	~30–32 ATP/glucose → ~15–16 ATP/g	~106 ATP/palmitate → ~9 ATP/g
O ₂ needed (L/g)	~0.83 L	~2.0 L
ATP per O ₂	~2.7–2.8	~2.3–2.4
Main Pathway	Glycolysis + TCA cycle + OxPhos	β-oxidation + TCA cycle + OxPhos

A significant portion of energy (within BMR and TEF) becomes heat:

- From biochemical inefficiency in ATP production (oxidation)
- Up to 80% of energy used in metabolism is lost as heat. That means that 75–80% of energy we “eat” becomes heat, only ~20–25% becomes mechanical work.

To perform you need to be healthy, so nutrition is not only to “fuel” exercising but to provide nutrients to stay healthy, repair and have a good immune system.

To train a lot you need to spend a lot of energy, therefore you need to eat a lot. But you need to be able to absorb the substrate (digestive system healthy and capacity to absorb and metabolize it – quantity and quality of mitochondria). There is a difference between what you are able to eat, what you are able to absorb and what are you able to utilize (oxidize in case of fat, cho, and use in chemical processes in minerals, vitamins, etc.)

Energy budget: balance between the energy consumed and the energy expended.

- **Energy Intake:** This includes all the calories consumed through food and drinks. For optimal performance and recovery, the quality of calories (nutrient density) is as important as the quantity. A balanced diet that provides sufficient macronutrients (carbohydrates, proteins, and fats) and micronutrients (vitamins and minerals) is essential for fueling workouts and supporting the body's recovery processes.
- **Energy Expenditure:** This encompasses the energy expended through:
 - Basal Metabolic Rate (BMR): The amount of energy expended while at rest, necessary for maintaining vital bodily functions such as breathing, circulation, and cell production.
 - Physical Activity: Includes both structured exercise (e.g., training sessions, competitions) and unstructured activity (e.g., walking, daily tasks). The intensity, duration, and type of physical activity significantly affect the total energy expenditure.
 - Thermic Effect of Food (TEF): The energy required for digestion, absorption, and assimilation of nutrients.
 - Non-Exercise Activity Thermogenesis (NEAT): The energy expended for activities that are not considered deliberate physical exercise, like fidgeting or maintaining posture.

Timing of fueling

Create a demand before creating an intake. If you're unable to oxidize X amount of CHO or fat there's no use in eating more or training your gut, because that substrate you are able to eat and even digest would not be metabolized. Before that you need to create a demand - the capacity of oxidizing more substrate - growing your mitochondria so then you're able to use what you're absorbing.

Looking from an evolutionary perspective, we have been developing to be efficient on expending energy, since the cost of getting food was very expensive in energy expenditure. So with more training our genetics will prioritize saving and be more efficient with less energy, not to be able to spend more. Efficiency > expenditure.

Leaving a long time without ingestion will let the intestines and digestive system rest and avoid inflammation. Therefore having less meals per day and having longer fasting periods will be beneficial.

How much and what type of macro and micronutrients we are eating will depend on individual factors and our training structure. We need to take into account our Base metabolic rate, the forecast of exercise expenditure of the next sessions, the digestion expenditure (when we are eating we spend also energy to activate the digestive organs, so the more we eat we need to put some extra energy for the expenditure of the eating itself) and the EPOC - post exercise oxygen consumption and heart rate recovery.

CHO intake:

- The CHO metabolism during exercise and at rest is different. During exercise, the hormonal response makes a sympathoadrenal activation and insulin secretion doesn't increase, or decrease, compared to rest.
- The blood flow is increased, so there are more nutrients entering the active tissues (muscles) via perfusion. The muscle temperature increases so + Glu gradient
- Activation via AMPK pathway (contraction) of Glut4 transporters. Increased abundance (membrane) and activity. Enhanced glycolytic flux and enzymatic activity (Hexokinase, etc).

So eating CHO during exercise vs at rest is more beneficial.

We can't really eat as much as we burn but we can aim high if we want to go fast. If we eat carbs during the exercise we will burn them directly and we will have less of the negative effects of them (intestinal permeability, pre-diabetes...) than if we eat it a long time before or after exercise. That way we can also focus the off-training to get other nutrients (protein, fat, vitamins, nitrate...).

Doing intensity fasted, can be interesting to increase efficiency of cho utilization.

Fat intake:

- FatOx maximization: some days week (1-2) where working with low carb (dinner day before and breakfast low carb + training session no fueling)

Microbiota

Microbiota refers to the collection of microorganisms—bacteria, viruses, fungi, and archaea—that live in and on the human body, especially in the gut. These microorganisms are not just passive passengers; they are essential partners in our biology.

Mostly resides in the intestines, especially the colon, but also on the skin, in the mouth, lungs, and urogenital tract. Contains trillions of microorganisms, more than human cells

Those species are essential for our life, they help digestion and nutrient absorption breaking down fiber and complex carbs we can't digest, it Produces short-chain fatty acids (SCFAs) like butyrate, fuel for colon cells and synthesizes vitamins (e.g., K, B12, folate, biotin)

They regulate and develop our immune system, preventing growth of pathogens (bad bacteria) by competing for space and nutrients and influencing inflammation and autoimmune risk.

It protects and barrier integrity, maintaining the intestinal wall and preventing "leaky gut" and modulating gut-brain signaling and neurotransmitter production.

Brain and mental health (Gut–Brain axis). Produces neurotransmitters like serotonin, dopamine, GABA. Affects mood, stress response, and cognitive functions.

Metabolism and Weight Regulation, Helping regulate energy harvest from food and influences insulin sensitivity, fat storage, and appetite signals

Detoxification and drug metabolism, Breaks down toxins and xenobiotics and affects how we metabolize medications.

Those bacteria eat fiber to fuel themselves, therefore the importance of eating fiber.

Prebiotics, fibers: Resistant starch (from cooked and cooled potatoes, rice...) AG, GOS,

Inulin pectin, raffinose beta-glucans, AX AXOS, polyphenols, FOS, XOS

Probiotics: B. longum B. bifidum L. plantarum L. reuteri S. thermophilus...

Species with evidence for endurance:

1. Modulating iron absorption

Iron is essential for hemoglobin and oxygen transport. Some gut bacteria improve iron bioavailability:

- Lactobacillus plantarum and Bifidobacterium spp. can:
 - Increase ferrous (Fe^{2+}) iron availability (the absorbable form).
 - Reduce gut inflammation, allowing better iron absorption.
 - Produce metabolites (e.g. SCFAs) that lower gut pH, improving solubility and absorption of iron.

This leads to higher hemoglobin levels, enhancing oxygen-carrying capacity.

2. Stimulating Erythropoiesis

Some gut bacteria influence EPO (erythropoietin) production through modulation of immune and metabolic pathways.

- For example, butyrate-producing bacteria (like *Faecalibacterium prausnitzii*):
 - Produce short-chain fatty acids (SCFAs) like butyrate.
 - Butyrate acts on HIF-1 α pathways, which also regulate EPO under hypoxic conditions.
 - Butyrate can also reduce inflammation in the bone marrow niche, supporting RBC synthesis.

3. Reducing inflammation

Chronic inflammation reduces RBC lifespan and impairs oxygen delivery. Some bacteria have anti-inflammatory properties:

- *Akkermansia muciniphila*:
 - Maintains gut barrier integrity.
 - Reduces systemic inflammation (e.g. via lowering LPS leakage into blood).
 - Better endothelial function → better capillary oxygen diffusion.

4. Improving Mitochondrial Function and Oxygen Utilization

Some gut-derived metabolites enhance mitochondrial efficiency, improving how oxygen is used in the muscle:

- SCFAs (especially butyrate) improve:
 - Mitochondrial biogenesis via PGC-1 α .
 - Fat oxidation, reducing reliance on glycolysis and lactate accumulation.
 - Efficiency of oxygen consumption (VO₂) at muscle level.

5. Blood Pressure and Blood Flow Regulation

Some bacteria influence nitric oxide (NO) production:

- Nitrate-reducing bacteria (e.g. *Veillonella*, *Prevotella*):
 - Convert dietary nitrate → nitrite → nitric oxide.
 - NO causes vasodilation, improving blood (and oxygen) delivery to muscles.

Microbiota Function	Mechanism	Impact on oxygen
Iron absorption	Lowers pH, reduces inflammation	↑ Hemoglobin
SCFA production	Butyrate → HIF-1 α , PGC-1 α	↑ RBCs, ↑ mitochondria
Anti-inflammatory	↓ Cytokines, protect endothelium	↑ O ₂ diffusion
Nitrate reduction	↑ NO → vasodilation	↑ O ₂ delivery
Gut barrier support	↓ Endotoxins	Stable systemic physiology

Species:

1. *Veillonella atypica*
 - Converts lactate (produced during exercise) into propionate, a short-chain fatty acid (SCFA) that can improve endurance by providing extra energy.
 - A 2019 study (Scheiman et al., *Nature Medicine*) found that this bacteria was more abundant in marathon runners.
2. *Akkermansia muciniphila*
 - Strengthens the gut lining, reduces systemic inflammation, and improves metabolic efficiency.
 - Associated with better glucose metabolism and fat utilization, key for long endurance efforts.
3. *Bifidobacterium longum*
 - Supports immune function and reduces gut permeability (“leaky gut”), which can be an issue in high training loads.
 - Some strains like *B. longum* 35624 have been linked to reduced inflammation and better GI tolerance during exercise.
4. *Lactobacillus plantarum* & *Lactobacillus rhamnosus*

- Aid in protein digestion and absorption, immune modulation, and reducing GI discomfort.
- May help decrease oxidative stress and speed up muscle recovery.

5. *Faecalibacterium prausnitzii*

- Produces butyrate, another SCFA that fuels colon cells, reduces inflammation, and supports recovery.
- Lactate shuttle

Bacteria (Species & Strain)	Role in Activity/Health/Performance	Food Sources / Supplements
Lactobacillus plantarum	Enhances muscle recovery, antioxidant properties	Fermented vegetables (e.g., sauerkraut, kimchi), supplements
Lactobacillus rhamnosus GG	Supports gut barrier function and reduces inflammation	Yogurt, kefir, supplements
Bifidobacterium longum	Modulates immune function, reduces gut permeability	Infant formulas, dairy products, supplements
Akkermansia muciniphila	Improves metabolic health and gut lining integrity	Not available in food; found in specialized supplements (under study)
Faecalibacterium prausnitzii	Anti-inflammatory effects, supports gut health	High-fiber diets (feeds this bacterium), prebiotic supplements
Veillonella atypica	Converts lactate to propionate, may enhance endurance	Not in food; colonizes active individuals post-exercise
Bifidobacterium breve	Supports gut-brain axis, reduces GI symptoms	Breast milk, fermented dairy, supplements
Lactobacillus casei Shirota	Enhances immune response and reduces exercise-induced GI discomfort	Yakult drink, fermented dairy products

Bacteria for altitude:

At high altitudes, the body faces challenges like low oxygen levels (hypoxia), cold temperatures, and increased ultraviolet radiation. These conditions can disrupt the gut microbiota, but certain bacteria have been found to aid in acclimatization:[PubMed Central](#)

- **Blautia A:** This genus becomes more prevalent in individuals exposed to high-altitude hypoxia. Studies indicate that *Blautia A* helps maintain intestinal health and supports adaptation to low-oxygen environments by reducing inflammation and strengthening the gut barrier. [Nature+3BioMed Central+3PubMed+3](#)

- Firmicutes and Bacteroidetes: These phyla are dominant in the gut microbiota of high-altitude mammals and humans. A higher Firmicutes-to-Bacteroidetes ratio is associated with enhanced energy extraction from food, which is vital for maintaining body temperature and energy balance in cold, oxygen-poor environments. [PubMed Central](#)
- Prevotella copri: This species is abundant in both Han and Tibetan populations living at high altitudes, suggesting a role in long-term acclimatization.

High CHO intake training

Interest in increasing cho intake in competitions (90-120gr/h) but work in training sporadically (specially for elite athletes with a good mitochondria: need less gut training) to avoid permanent damage: intestinal permeability, diabetes...

First to see if already able to handle high amounts of CHO (with better mitochondria function they will have a high absorption of CHO without any gut training) and then see if it's necessary a gut training, a reduced version or not at all.

Supplements

I believe there should be no need for supplements. We should prioritize building a diet that ensures the needs we have in macro and micronutrients. In some cases, supplements might be useful to reinforce some missing in the diet or in some specific moments in training or racing. Also it's very important to think about the origin and traceability of the supplements, especially if being an elite athlete subjected to anti doping controls, since supplements might be contaminated and may not contain what you expect or y contain what you don't expect.

Vitamin D, specially in northern countries with low sun exposure.

Iron if training a lot and doing altitude and if ferritin <30 ng/mL; must be diagnosed—don't supplement blindly

Probiotics: *Lactobacillus*, *Bifidobacterium* if microbiota bacteria is low in species diversity or total number.

Omega 3: have ant-inflammatory properties. Modulate cytokine production and reduce exercise-induced inflammation.

Caffeine

Caffeine is a stimulant that possesses well-established benefits for athletic performance across endurance-based situations, and short-term, supramaximal and/or repeated sprint tasks. increased endorphin release; improved neuromuscular function; improved vigilance and alertness; reduced the perception of exertion during exercise

3-6mg/kg 60 min before exercise.

During exercise better consumed with CHO, 3-6 mg/kg every 2 to 6 hours (depending person and caffeine source)



<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7739593/>

Creatine

Supplementation increases muscle creatine stores, augmenting the rate of PCr resynthesis, thereby enhancing short-term, high-intensity exercise capacity and the ability to perform repeated bouts of high-intensity effort.

Loading phase: ~20 g/day (divided into four equal daily doses), for 5–7 days

Maintenance phase: 3–5 g/day (single dose) for the duration of the supplementation period

Note: concurrent consumption with a mixed protein/CHO source (~50 g of protein and CHO) may enhance muscle creatine uptake via insulin stimulation.

Bicarbonate

Found evidence in able to hold higher lactate levels (lowering ph) in short distance races, or sprints after long efforts. maybe accelerate recovery (base/acid balance in blood) and

PGC-1alpha

a dose (0,24gr/kg) approx every 6h.

In ultra experience says that it improves neuro capacity (control of muscles) muscle fatigue and lowers PH

The effects decrease after 300min.

Due to the risk of GI problems, it is better to take with CHO.

<https://www.mysportscience.com/post/sodium-bicarbonate-cheap-and-effective>

Nitrates:

In many green leaves and beetroot. (beetroot juice)

Nitrate augments exercise performance via an enhanced function of type II muscle fibres; a reduced ATP cost of muscle force production; an increased efficiency of mitochondrial respiration; an increased blood flow to the muscle; and a decrease in blood flow to VO2 heterogeneities

Do charge before training or races, day and hours before. Works best with Citrulline.

They take up to 12 mmol/day

High nitrate-containing foods include leafy green and root vegetables, including spinach, rocket salad, celery and beetroot.

Acute performance benefits are generally seen within 2–3 hours following an nitrate bolus of 5–9 mmol (310–560 mg).

Prolonged periods of nitrate intake (>3 days) also appear beneficial to performance and may be a positive strategy for highly trained athletes.

Beta-alanine

augments intracellular buffering capacity, having potential beneficial effects on sustained high-intensity exercise performance.

Daily consumption of ~4-6g/kg , ingested via a split-dose regimen (ie, 0.8–1.6 g every 3–4 hours) over an extended supplement time frame of 10–12 weeks.

Ketones:

Interesting for recovery or easy intensity (lactate will be absorbed before ketones so in high intensity it will not be useful)

Seems to have an individual response.

Ketones are more favorable with fat oxidation.

In high hypoxia: ketones in blood go up to 20% in non active tissues.

Exogen ketones: improves CO₂ pressure, SatO₂ in blood, lactate levels, Augments the concentration of O₂ with the hemoglobin and stimulates EPO secretion. Augments brain oxygenation.

Augments concentration

Should aim for 1.5-2 to 4 mmol/l in blood. Check your profile to see how much of ketones you need to reach these levels and how long it lasts (approx 20-30gr). It takes 30.45' to get to the maximum levels and about 3h to get back to basal levels.

Best to take:

1. Just after exercise
2. Just before sleep

During exercise (specially high intensity) It decreases performance, Problem of ketones: increases the acidosis! Must take with bicarbonate. Not to take before or during exercise except long events at low intensity where the positive effects will be only increase of concentration.

Nutrition for Heat Adaptation

Support plasma volume expansion

- Increase sodium intake moderately (3–6g/day or 1500–3000mg sodium/day during heat block)
 - Supports fluid retention → larger plasma volume → better cooling
- Stay well-hydrated during and between sessions
- Add broths, salted foods, electrolytes to meals

Antioxidants & anti-inflammatories (in moderation)

- Chronic heat stress increases oxidative stress → damage to gut and cells
- Polyphenols (berries, cocoa, green tea), omega-3s, and turmeric support recovery and reduce systemic inflammation
- Avoid high-dose supplements (like 1000mg vit C) during training blocks – they can blunt adaptations

Heat shock protein (HSP) support

- Heat training increases HSPs, which protect cells and improve adaptation
- Curcumin, green tea extract, and intermittent fasting may enhance HSP expression slightly
- Prioritize high-quality protein intake to support cellular recovery (1.6–2.0 g/kg/day)

Gut health & microbiota in heat

- Heat reduces blood flow to the gut → more permeability (“leaky gut”)
- Leads to GI distress, reduced absorption, immune stress, and worse performance
- A robust microbiota helps protect intestinal barrier, regulate inflammation, and improve nutrient absorption
- *Akkermansia muciniphila*: strengthens gut lining
- *Lactobacillus* & *Bifidobacterium* spp.: reduce inflammation, improve digestion and immunity
- *Veillonella atypica*: shown to enhance endurance by converting lactate to propionate
- How to Support Them:
 - Fermented foods (kefir, yogurt, sauerkraut, miso)
 - Prebiotics: fiber-rich foods (onion, garlic, oats, legumes, bananas)
 - Polyphenols: berries, cocoa, green tea
 - Consider probiotic supplements (clinically studied strains like *Lactobacillus rhamnosus* GG, *Bifidobacterium lactis*)

Race nutrition in heat:

Fuel Strategy

- Gut is more fragile in heat → train it!
- Use lower-osmolarity drinks ($\leq 6\%$ carbs), combine liquids + gels, and distribute intake (e.g., 30–40g every 20–30 min)
- In heat, malabsorption risk $\uparrow$ → focus on glucose-fructose mix, test during training
- Include small amounts of protein/fat in longer races to support gut and slow carb absorption (only if tested)

Electrolytes

- Match sodium loss: If 1.5g Na⁺/L sweat and 1L/h → aim ~1500mg/h sodium
- Use salt capsules or drink mix; spread through race (not just in water stops)
- Pair sodium with potassium (200–400mg/h) and small magnesium (~50mg/h if sweating heavily)

Cool the gut

- Cold drinks, ice slushy ingestion, or mint can lower perceived heat and protect gut
- Reduce caffeine or NSAIDs on race day if prone to GI issues
- Mouth rinse or drinking with menthol decrease RPE on heat but not core temperature. It can be interesting to do in the late or warmer sections of competition.

Race fueling plan:

The needs and nutrition for competitions are very individual. First we need to estimate the expenditure during the race and our oxidation capacity to make a plan.

In long races we will not be able to restore all what we spend, so we need to get as much as we can taking into account our oxidative rates and absorption rates to avoid GI issues.

In general, in races under 1h we don't need any fueling since our reserves are enough.

In races up to 4h, 30 to 60 gr/cho/h should be enough.

In longer races we will try to take more depending on expenditure - oxidation rates.

In ultras if GI problems or altitude (more difficult to absorb CHO) we can take fast CHO (gels) in key moments where metabolism is more demanded – faster or uphill- and slower CHO or fat in other periods (potato, rice, dates, nuts, avocado...) to maintain the CAL intake.

If the race has high demands, in the last hours of the race we can decrease intake (energy delivery from the use of butyrate, lactate as main source)

Race day:

DAY -1: Breakfast: fasting (fasted training increases insulin sensitivity making muscles more receptive to glucose)

Lunch and dinner: Carbs, but don't eat more than normal. Glycogen stores have a limit, around 500g in a trained athlete. Anything we overeat will simply accumulate as fat.

We avoid sweets, junk food and ultra-processed foods. Water intake: drink as much as your body asks for.

Make sure to hydrate at breakfast and before the test. If it is very hot, it would be advisable to increase salt intake with meals in the 24 hours prior to the test, seeking to retain fluids and start the test in a state of hyperhydration.

Useful links:



https://link.springer.com/article/10.1007/s00421-024-05437-y?utm_source=chatgpt.com#Fi_g1

Know more: Nutrient utilization in Humans: Metabolism Pathways

<https://www.nature.com/scitable/topicpage/nutrient-utilization-in-humans-metabolism-pathways-14234029/>

<https://www.nature.com/articles/s42255-020-0251-4>

Lactate pathway: <https://www.sciencedirect.com/science/article/pii/S2213231720300422>

Max expenditure: <https://www.science.org/doi/10.1126/sciadv.aaw0341>

Genetics: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6389634>

<https://podcasts.apple.com/no/podcast/inside-exercise/id1631457776?i=1000622814365>

<https://link.springer.com/article/10.1007/s00421-023-05331-z>

Gut and injuries:

https://www.researchgate.net/publication/374749719_What_if_gastrointestinal_complications_in_endurance_athletes_were_gut_injuries_in_response_to_a_high_consumption_of_ultra-processed_foods_Please_take_care_of_your_bugs_if_you_want_to_improve_endurance

Nutrition in altitude: <https://link.springer.com/article/10.1007/s40279-019-01159-w>

Fueling and nutrition for women: 

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8566643/>



<https://www.mysportscience.com/post/the-female-athlete-considerations-for-fuel-storage-and-utilization>

Gut microbiota and injuries:

https://www.researchgate.net/publication/374749719_What_if_gastrointestinal_complications_in_endurance_athletes_were_gut_injuries_in_response_to_a_high_consumption_of_ultra-processed_foods_Please_take_care_of_your_bugs_if_you_want_to_improve_endurance

Hydration

Hydration: dehydration brings chronic acidity, Kidney and liver problems do not detoxify well. Liver toxicity capacity: the liver's ability to detoxify.

Less than 2% of body mass loss for dehydration doesn't affect performance.

Do a Sweat test (measure how much liquid you lose every hour during exercising at X temperature and X intensity) then you can see how much you need to drink in the duration of a race to lose around 2% of body weight.

In a hot environment small to moderate levels of dehydration (3-4%) will lead to a decrease in VO2max of +10% with a corresponding drop in endurance capacity of more than 20%.

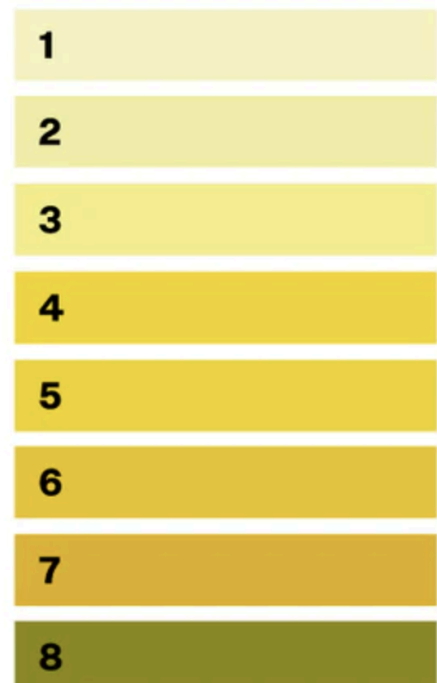
The main reasons dehydration has an adverse effect on exercise performance:

- Reduction in blood volume
- Decreased skin blood flow
- Decreased sweat rate
- Decreased heat dissipation
- Increased core temperature
- Increased rate of muscle glycogen use

The kidneys are a key player in this water balancing act while kidney excretion (urine) can provide clues to your hydration status. The kidneys will restore equilibrium in fluid balance within hours provided enough fluid is consumed during and after the ride. The kidneys achieve this rapid rate of rehydration via two hormones:

Aldosterone which targets sodium and potassium balance and Antidiuretic Hormone (ADH) which controls water balance.

Unless dehydration becomes extreme (exceeding 3% body weight) these two hormones work to maintain hydration within 1% of normal limits. Knowing this, it's possible to deduce a lot around hydration level from the properties of urine. As a simple guide: urine should be dilute and slightly opaque when you are hydrated (1 to 3), and when you are dehydrated urine should be darker in color.



The maximum amount of fluid that empties from the stomach during exercise is 0.8-1.2L/
Sweat sodium test to determine how much sodium we are losing on sweating (very individual). To determine the amount of sodium and electrolytes we should be taking in our races. That will change a bit during the season, normally we will be losing more in winter and less in summer.

In general it is important to be well hydrated (drink during training or be sure to drink well enough in the hours after to refill) to avoid renal problems and inflammation.

Gaining weight when you eat less due to Inflammation due to liquid accumulation in the muscles due to catabolism.

Useful links:



<https://us.humankinetics.com/blogs/excerpt/dehydration-and-its-effects-on-performance>

Periodization of stimuli

What do we need to maximize our performance on the goal?

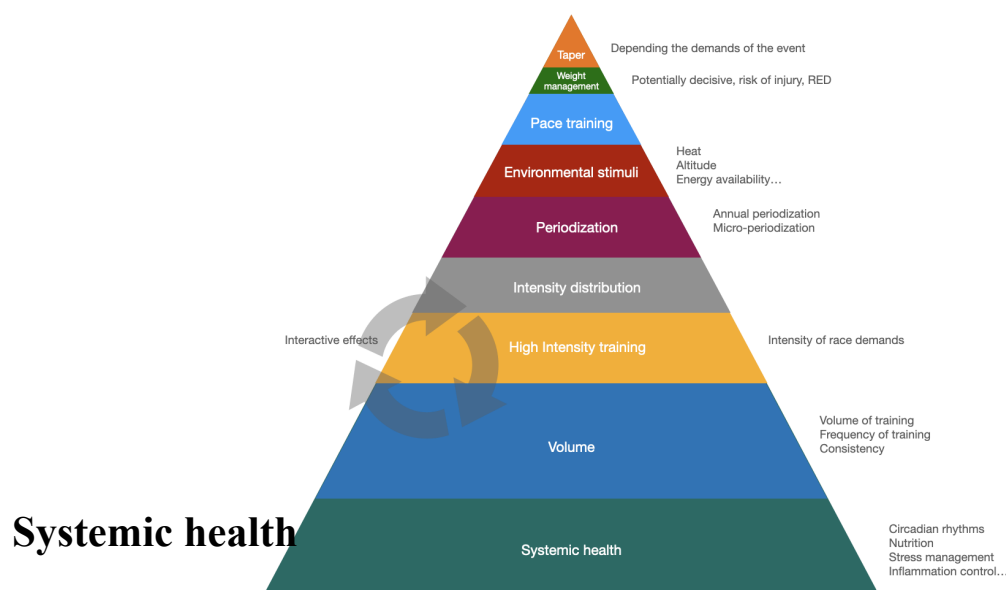
First goal will be to create a biological inertia, so our long term adaptations change our biology / epigenetics. Biological inertia is the cellular tendency to maintain their migration direction for some time even when a stimulus is reversed if the stimulus has been previously been in their environment for a long period.

For that it is key to keep stimulation over long periods (years / decades). The periodization is therefore less important in the short periods (if for some months or a year you don't stimulate something specific but you have created the inertia, the growth there is possible).

Secondly, and where periodization is important is on getting the systems as prepared as possible for the race requirements. That normally doesn't take so long, a few months maximum should be enough.

Performance is not a game of maximization but of optimization, therefore we need to see our availability (time we have, off-training load, training load capacity, access to facilities...) to select whose stimuli that will maximize our performance.

Pyramid of training needs: Focus on the base, and when acquired, get on the higher needs:



The Basis of systemic health explores genetic potential functions first. After we can bring specificity. If the nervous system / immune system is low we cannot.

- Reduction of inflammation / capacity of recovery
- Systemic inflammation is high then doesn't recover. Training brings inflammation. Sport brings inflammation, if your body works well you have more capacity to desinflammate, eliminate toxins, and metabolic function...
- Regulation of systemic functions: circadian rhythms, inflammation regulation, nervous system functions (tonification, motor efficiency, precision capacity – stress...) renal function, electrolytes function, oxygenation, metabolism, and purification (renal-hepatic) functions. If you don't have that you cannot accumulate training, injuries can come from extension of reactivity systemic – toxins accumulation, intestinal permeability, inflammation chronicle, not resting...
- Routine (hydration, rest, nutrition (quality of the aliments, more than quantity or frequency) base physical activity.

Signaling of the stimuli:

The signal that will make the body do adaptations. To make a strong stimuli we need to get fatigued (the higher the level the more we need to get fatigued to get a stimuli)

Do a workout and when start to get “worked” (we might start to feel tired on a particular way (muscle, metabolism, we need to put more focus, we need to concentrate or the movement is getting clumsier...)) then is when the stimuli is there, we are in a systemic state, then we do max ¼ more of workout (2-3 more reps if intervals for example) to make the stimuli but not more to not get overtrained.

What to train?

Before planning the season we need to see:

- Time / availability / resources we have
- Race profiling
- Gap assessment between athlete profile and race profile.

Then we make a plan prioritizing what are the most important elements we need to change.

The key is to keep stimuli all the time, for that the most important is to be consistent, much more than the quality of a single workout is the capacity of accumulating many workouts, for

that is important to have a good planification to enable a variety of different stimuli and a sustainable training load that permits us to train for long without getting inflamed (injured or overtrained).

Sessions

What is the goal of warming up?

Warm up helps to set the body ready for the efforts of the training or competition. That means the best basal levels to perform. And also to get mentally activated for the activity we will do.

Warm up until levels of Lactate lower, higher muscle O₂ sat. 15, 20' easy + small stimuli to the speed of the workout or race.

What about cooling down?

There's no scientific evidence that supports that cooling down will have any effect on recovery. (On The only evidence is that keep low intensity exercising in the minutes after a very high intensity effort is clearing lactate faster.

After a race with muscle damage, like most of the trail running races that involve downhill, there might be more damage to cool down running than to rest.

Single session vs double session:

- In running the volume will be limited by the mechanical charge – risk of injuries and muscle damage – so it can be interesting to break training into multiple shorter sessions (double session) specially when mechanical intense work (flat, downhill, track) to improve muscle and mechanical recovery between sessions.
- In non mechanically aggressive activities (cycling, skimo, xc skiing, swimming, mountaineering) the volume limit will be marked by the metabolic work, which is more than the mechanical therefore it might take more time to recover the metabolism from longer sessions and then is better to do only one session per day.

Base Period (duration: 2-3 month minimum)

The goal here will be to work your aerobic base:

Importance of low intensity training, unstructured training – z1 z2 with some intensity (depending the specificity of the athlete and competitions).

Here we look to create physiological adaptations, so our intensity will be measured with internal markers.

Athletes with bigger fat metabolism / Type I muscle fiber predominance will generally support big amounts of load but not much intensity needed to get in shape. If too much – inflammation – injury. Athletes with type II muscle fiber predominance might struggle more with high load but can do more intensity work without creating inflammation.

If this it is a long period it can be good to introduce specific stimuli every now and then to maintain (to do a B race, some specific sessions every few weeks)

Cross-training: In this period it is especially interesting to do some of the training on other activities, to change the muscular and mechanical impact, to avoid injuries and increase the low impact volume: Skimo, cycling, cross country skiing...

key sessions:

- Z2 Long progressive: Start at Z2 and every / uphill / 10, 20, 30, 40 min / augment the intensity progressively into Z3 (under threshold) the last 1 or 2 stages should be when you start to feel “worked”.
- Z1 to Z2 long sessions, low intensity long sessions to improve our aerobic capacity and lactate clearance.
- Z4 Threshold short intervals: doesn't need to be many, doesn't need to feel “worked” but just to remember the system to use the glycolysis (can be 2'-3' or 30"-30" reps at Z3-Z4) not many reps! all of them feeling fresh. The goal at this point of the season is just to activate and remind this metabolism, not to really work on it.

Specific Period: 6-12 weeks before the competitions.

Specificity period (depends athlete will take longer or shorter time to get in racing shape, know that to not overtrain)

Analyze the competition (profile, temperature, terrain) and your profile (strengths, weaknesses) to see what you should focus and not during the specific period:

- maintain strengths
- improve weaknesses you can improve
- forget about weaknesses you can't improve
- don't focus on improving qualities that you don't need for this race.

Increase progressively the training versus race simulation (systemic similarities of the race)

2 - 3 “fast” sessions per week are enough to create adaptations with compensation time. More might not be beneficial.

Tapering and shapering:

Tapering, if the training is sustainable, it shouldn't be more than 5-7 days of tapering (reduce volume by 40 to 60%, maintain stimuli) No neuromuscular work last 10 days (captures fatigue)

- maintain frequency of training
- reduce volume of the sessions
- keep intensity in reduced volume of it

Last days activation to get movement back and mental activation.

Shapering: weight management, sustainable loss for 1-2 weeks.

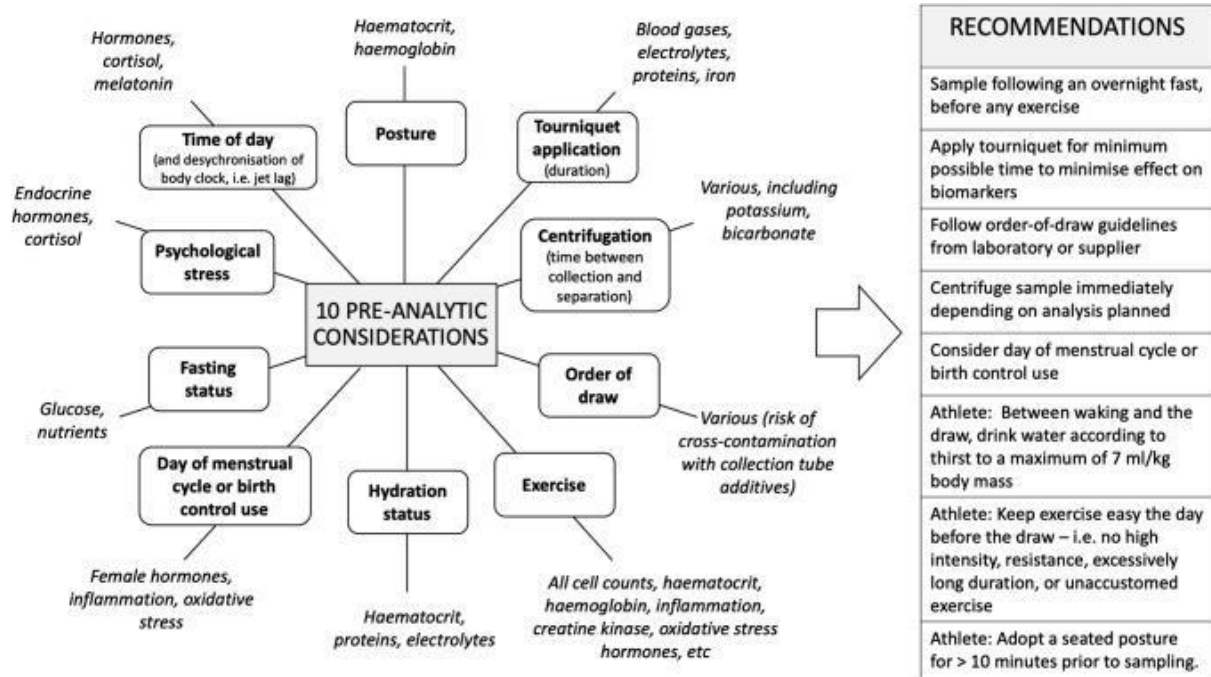
IF'S:

1. If in **good shape too early**: rest: take XX time (depending how early, if months a week, if less 1-3 days) to not get overtrained.
2. If you **feel too tired for days**. Underfueling, resting
3. **Sick**: don't try to overcompensate the sessions. take easy days (Z1) or rest and then get back to the training without trying to "recover" the lost sessions.
4. At Warm-up heart rate low, the lactate at a high speed in the lower range, heart rate was higher than normal: you can push
5. At Warm-up heart rate high, heart rate in session not possible to get high: stop and rest
6. **Feel tired, in a bad mood...** take 1-2 easy or rest days, do not try to recover the lost sessions after that.
7. If **feeling tired on a specific area**, cardio, muscle. Not try to do (for someone capable of running at X speed, running at x-5 speed don't give improvement) but work in other areas: temperature, altitude...(breathing exercises, strength...)
8. **If mechanical injury** (from broken bone to articulation mov reduction, pain that changes our biomechanics...) we can keep doing metabolic work on other sports if it's not painful (skimo, cycling, walking, swimming, rowing...). If we are not very used to this sport we might avoid high intensity and keep Z1-2 to avoid creating new mechanical problems.

Monitoring performance:

- Performance (field tests, lab tests, races)
- Training load in the different systems (metabolic load, heat load, oxygen, neuromuscular impact) special stressors (cognitive, hydration)
- Weight, body composition.
- Fatigue (mood, feedback during training, HRV...)

- Microbiota analysis
- Blood work:



Useful links:



<https://podcasts.apple.com/no/podcast/slow-medicine-revolution/id1678548062?i=1000651501670>



tapering:

<https://podcasts.apple.com/es/podcast/fast-talk/id1490521721?i=1000558964354>

Training volume: <https://physoc.onlinelibrary.wiley.com/doi/epdf/10.1113/JP277634>

Recovery

Recovery $\neq$ Adaptation (recovery is the time to get rested after a stressor, Adaptation is when the cumulation of stressors and recovery has created a biological adaptation)

Fatigue:

Metabolic Fatigue

Metabolic fatigue is associated with changes and depletion of the body's energy stores and accumulation of metabolic byproducts. During endurance training, such as long-distance running or cycling, the body primarily uses aerobic metabolism, which requires oxygen to generate energy from carbohydrates and fats.

Key points about metabolic fatigue:

- **Energy depletion:** Long or intense activities can deplete the glycogen (stored carbohydrates) in the muscles and liver, leading to fatigue.
- **Byproduct accumulation:** Prolonged exercise leads to an accumulation of lactate and hydrogen ions, which can interfere with muscle contraction and contribute to the feeling of fatigue.
- **Impact on performance:** Metabolic fatigue limits endurance performance by reducing the muscles' ability to generate energy efficiently.

Neuromuscular Fatigue

Neuromuscular fatigue refers to the decline in muscle function due to intense or prolonged activity. This type of fatigue is influenced by both the central nervous system (brain and spinal cord) and the peripheral components (nerves and muscles).

Key points about neuromuscular fatigue:

- **Muscle exhaustion:** Repeated muscle contractions can lead to a reduction in the force-generating capacity of the muscles.
- **Nerve signal efficiency:** Fatigue can reduce the efficiency of nerve signals that control muscle contraction, potentially due to alterations in the neurotransmitters and ion channels involved in signaling.
- **Recovery needs:** Neuromuscular fatigue requires adequate rest and nutrition for recovery, particularly protein for muscle repair.

Neurological Fatigue

Neurological fatigue or central fatigue is primarily related to the central nervous system. It involves a decrease in voluntary muscle activation during exercise and can be one of the most complex forms of fatigue to diagnose and manage.

Key points about neurological fatigue:

- **Mental tiredness:** Psychological factors such as motivation and mental stress play a significant role.
- **Central nervous system exhaustion:** Prolonged or intense exercise can lead to a decrease in the brain's ability to effectively recruit muscle fibers.
- **Signal reduction:** There might be a reduction in motor signals from the brain to the muscles, possibly due to changes in neurotransmitter levels.

Systemic fatigue:**Interaction and Training Implications**

In endurance training, these types of fatigue often interact. For example, metabolic byproducts can influence neuromuscular function, and both metabolic and neuromuscular fatigue can affect the central nervous system's ability to maintain exercise intensity.

Biomarkers: analytics for inflammatory process

monitor: Hrv, sleep h and basal pulse, mood, training/effort feeling

Acidity: get hydrated and lower PH: alkaline drink: water with moringa, aloe vera, lemon, bicarbonate...

Metabolic fatigue:

Basically focus on sleep, good training load (not too much and good recovery between the stimulus) Stress (work, social and couple relations...) environment, hydration, good nutrients and eat the calories you burn)

Short term HR development, variation considered is approximately 10bpm:

- Lower than normal during training: Indication of a tired metabolism.
- Higher than normal during training: Indicates good shape or sickness.
- Higher than normal during rest/sleep: Indicates sickness or high stress levels in general.

Recovery:

Increase calories intake (up to 20% more)

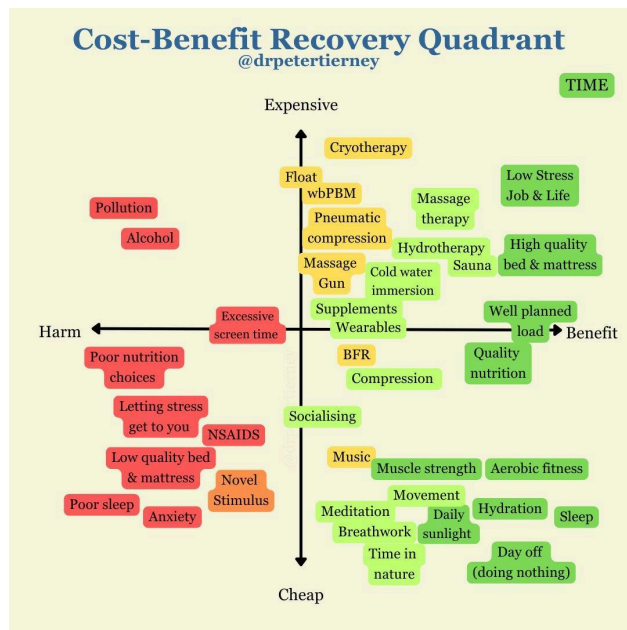
Protein (1,6-3gr/kg day) 20-30 gr x meal with 3gr leucine (1 egg 1,09gr)

Omega 6-3

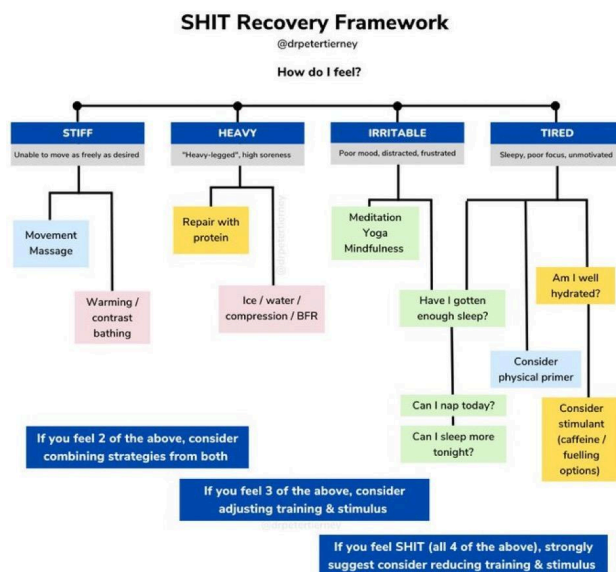
0,1gr/kg/day creatine

Vit D

Vit C



When you train you produce inflammations, those inflammations are necessary to create adaptations, so to cut them might feel good to be “recovered” faster but it’s cutting adaptations: don’t decrease temperature- ice – don’t take anti inflammatories. Pain is a great indicator to see when recovery is done.



Useful links:



Microbiota and injuries:

<https://www.aimspress.com/article/id/64e46d65ba35de4e5f973dc9>

sleep: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6879580/>

Problems during the race:

IF...	THEN...
Pissing a lot	Add salt to fueling (gels or drink), If this produces vomiting, just stop eating salt.
GI problems	Slow down and cool off so the blood flow can go to the stomach and digest. Eat when bouncing the less (walking better than running)
Blisters	1. Clean (if you have alcohol pad)2. Puncture the blister with needle, 3-4 holes (better sterile, or scissors but you can also sterilize bib pins) to let all fluid drain.3. Squeeze fluid out the blister4. Clean and dry5. Place tape or bandage, blister and surrounding skin – 3-4 layers. If the blister is open, put a non adhesive tissue in that part.
Muscle breakdown	Vary your stride (heel to forefoot) Bicarbonate or alkaline aliments.
Hot	Decrease core temperature as fast as possible (slow down, put cold things on you, drink cold...)
Cramp	Prick the muscle cramping with a needle (eg. from bib), drink or eat something bitter... That could make the cramp disappear in the moment (neural signal). For the long term, keep hydration-sodium, and manage intensity.

Training for Mountain Running

By: Scott Johnston

Posted: September 5, 2024

To be most effective, training for mountain running needs to be viewed differently than road or track running.

Mountain running and road running are, at best, distant cousins. Yes, both are “running,” like freestyle and butterfly are both “swimming” but require different techniques. Running is a form of locomotion that contains a flight phase, where both feet are off the ground at the same time, and only one of the runner's feet touches the ground at a time during the stride*. Both road and mountain running fit this definition. But that's where the similarity ends.

**whereas in walking, there is a time during the stride when both feet touch the ground.*

Training for mountain running and running on flat, smooth ground share only passing similarities. This article will explore the variety of training methods that mountain runners should employ.

When running on smooth, flat ground, be it road, track, or trail, every stride the runner takes will be virtually identical to all the others. This makes learning and perfecting the technique of running simpler to master and simpler to train for the repetitive movement.

Mountain running, which is almost always conducted on rough trails with significant and often frequent gradient changes, necessarily forces the runner to adapt the technique to the terrain. There will always be a great deal of technique variation during mountain running. Some flat sections of the trail may require a running technique identical to road running technique. However, the technical and fitness demands differ significantly for uphill sections, where “running” can look more like climbing a set of stairs, and runners are often forced to walk. On downhills, the runner must resist gravity lest the speed become so great that a fall is unavoidable. Resisting gravity in this way means that the impact forces the runner's legs must absorb are much greater than running on flat ground. The trail's roughness will greatly affect the runner's stride length, frequency, and height each foot must be lifted over obstacles.

Technique

Mastering any technique takes time and focus. Mastering four different techniques requires considerably more time and focus and will require prioritizing how that time is spent. Flat running, uphill running, downhill running, and uphill hiking are four different means of locomotion needed to be a mountain runner. From the standpoint of muscle recruitment patterns, each is its own “sport.” Each requires a different training method, presenting the mountain runner with a bigger challenge in allocating training time and energy. And we have not even mentioned the use of poles for steep hills where they can make a substantial difference.

Swimming faces a similar challenge. If a swimmer wants to be competitive in the individual medley event, they must train all four strokes, which, even though they are all “swimming,” are distinct techniques that each need to be trained in isolation.

Why all this talk about technique?

A good level of fitness is, of course, what will allow you to run a sufficient volume needed to handle long runs in the mountains. But good technique will allow you to cover that distance faster and with less energy.

Flat

Runners who are making a transition from the road to the mountains often bring with them good flat running technique and speed. This can be a significant advantage. One of the technique boxes has already been checked. These runners have developed both the technique and fitness to run substantially faster than they need to run in the mountains. That speed component is another helpful box to have checked. Mountain runners without a road

or track background often struggle with their pace on the flats. Running fast* on the flats requires its very own type of strength and muscular endurance. Tendons act like springs in that they can return much of the energy they are loaded with. This is why high-level runners appear to be running so effortlessly. They have learned how to extract more of this spring energy.

**Fast here means relative to the individual runner.*

Uphill

This is where most people's attention is focused when they think about mountain running, and for good reasons. Uphill running speed relies less on technique and more on fitness. Depending on the gradient and the trail's roughness, uphill running speeds can easily drop into the 20-30min/mile range. This means you may take 2-3 times as long to cover a mile uphill than on the flats or downhill.

Let's take a hypothetical 60km loop mountain course that has a mix of 25km on up hills, 25km on down hills, and 10km on flats. You could easily spend 60-75% of the race time going uphill, even though this course's distance is only 40% uphill. Simple arithmetic tells you that you can gain or lose the most time in the up hills. This is why uphill fitness gets so much attention from mountain runners. When Ruth Croft set the women's course record of 8:03 at the 2024 Transvulcania 100km race, she spent over 5 hours, or 66% of the time, going uphill.

Downhill

We can't talk about downhill running without speaking about the Impact forces of running because these heavily influence training for downhill running.

The force transmitted through the leg each time a runner's foot strikes the ground on a flat grass surface at about a 9min/mile pace will be about three times their body weight.

Running at the same 9min/mile pace on a rather typical 9% downhill gradient, the total (normal + parallel vectors) impact force will more than double what would be in the flats.

That force has to be absorbed primarily by the leg muscles and tendons each time your foot hits the ground. For most runners, this means 5-6 times their body weight when landing on each foot when running down a typical trail. Absorbing that impact force is done with eccentric* muscle contractions. At these high-force eccentric contractions, muscle damage is easily induced. Is it any wonder your quads get sore when running downhill? Training for this requires its own unique form of strength training that can be done on the trail or in a gym.

**Eccentric contractions are those where the muscle lengthens under force. These can produce higher forces than concentric contractions, where the muscle shortens under force.*

Walking Uphill

As the grade of the trail steepens, at some point, it will become more efficient to shift from running to walking. You've no doubt felt it when you reach this point going up a hill that steepens. This is important when you are on a long climb or are out for several hours. You don't have extra energy to waste; balancing the most economical mode with the fastest speed can be a critical decision.

At grades steeper than about 20%, running no longer includes the flight phase, where both feet are off the ground during the stride. So, even though it looks like running and has a higher stride rate than walking, it no longer fits the definition of running.

At what grade do runners make this transition to walking? This depends on several factors. These include: the length of the hill and how long the runner has been on the trail, leg length and strength, but most importantly, the runner's local muscular fatigue. While you can't change the length of your femur, fatigue resistance is highly trainable.

Of all the studies I can find on walking vs. running uphill, only a tiny handful, all conducted by an even smaller number of researchers, have used elite mountain runners and relatively short testing times. One study is mainly concerned with world-class Vertical Kilometer racers, race courses that typically average over 30% grade, and races that last 30 minutes. There is nothing wrong with investigating this, but we can't draw meaningful takeaways for runners doing far more common events that last many hours.

The duration of the individual climb and how many hours you have already been on the trail are hugely important to your walk/run decision. Fatigue plays the most critical role in long-distance running, especially when deciding when to walk and when to run. The more tired you are, the sooner you transition to walking as the grade steepens. That's why we can't just extrapolate from a study that looks at steady-state tests lasting a few minutes. These tests give accurate heart rate and VO2 data so that the relative energetic costs of running and walking can be compared for different grades and speeds. But do these tests reflect what happens after the runner has been on the course for many hours? If our experience coaching many mountain runners is any indication, these studies do not give meaningful information on multi-hour events.

Performance Limiters

You have limited time to train. More importantly, you have limited energy to devote to your training. You want to apportion your time and energy in the most effective means to yield the best returns on your time investment. To do that requires an analysis of the demands of mountain running.

Mountain running for durations longer than 3 hours will, of necessity, be conducted predominately at relatively low intensity. Mostly in Zones 1-2, with some short forays into Zone 3 and only tiny amounts of time in Zone 4, if at all. By using fats and carbohydrates as fuel, your aerobic metabolism will provide virtually 99% of your energy demands for running at these submaximal paces. The fatigue from running long distances in the mountains comes from the duration rather than the intensity of exercise. It should only be in the last hour of a 4-hour run that you begin to feel fatigued.

Below are three real-world examples of the training data from six months' worth of training by some of the top ultra-runners we coach. Notice the intensity distribution as measured by time in the zone. This information comes directly from their Training Peaks logs:

- **Runner #1:** Zone 1 = 40%, Zone 2 = 52%, Zone 3 = 8%, Zone 4 < 1%
- **Runner #2:** Zone 1 = 35%, Zone 2 = 54%, Zone 3 = 9%, Zone 4 < 2%
- **Runner #3:** Zone 1 = 45%, Zone 2 = 44%, Zone 3 = 7%, Zone 4 < 1%

Real-world examples of those same runners' typical intensity distribution in races of 100km or longer:

- **Runner #1:** Zone 1 = 6%, Zone 2 = 76%, Zone 3 = 18%
- **Runner #2:** Zone 1 = 18%, Zone 2 = 62%, Zone 3 = 19%
- **Runner #3:** does not wear a HR monitor when racing

These elite pros spend 90% of their training time below their aerobic threshold (AeT) or at the top of Zone 2. They do this because 80% of their race time is spent in those same intensity zones. If your training intensity distribution deviates a lot from theirs, this is the first place you should focus on to improve the effectiveness of your training.

We have seen that one of the best predictors of ultra-distance performance is one's speed at the aerobic threshold (AeT). That's because your AeT is an intensity you can sustain for hours. This does not mean you will not become fatigued after hours of AeT-intensity running.

Nor does it mean that training above your AeT is unimportant. Both of these points will be discussed in later sections of this article.

Fatigue Mechanisms

Fatigue of all forms manifests as a reduction in muscle contractile force. The result is that your stride length shortens, and you slow down. Fatigue mechanisms are lumped into two categories: Central and Peripheral.

Central fatigue is a neural effect that affects the brain's ability to recruit and fire the propelling muscles with as much force as when rested. This effect can be caused by the motor cortex or all the way down the nerve path to the individual motor units losing the ability to produce sufficient force to maintain the desired pace.

Peripheral fatigue is caused by changes within the muscle cells that reduce the muscles' contractile force. Glycogen depletion is the most well-known of these. However, metabolite accumulation and muscle pH reduction (acidity increase – the burn erroneously attributed to lactic acid) are other commonly understood factors in peripheral fatigue.

In this case, the distinction between central and peripheral is a somewhat artificial construct, as these two systems are inextricably linked and interdependent.

In any endurance event, training aims to increase the athlete's fatigue resistance. Fatigue resistance is a somewhat amorphous term and a combination of many training adaptations. As such, it can be hard to determine what factors contribute to increasing fatigue resistance.

However, studies have documented that the fatigue resistance of the main propelling muscles is not well correlated with VO2 max. In other words, having a high VO2 max does not guarantee fatigue-resistant legs. However, a higher fatigue resistance allows the runner to sustain a higher speed, utilizing a higher percentage of their VO2 max for longer.

This concept is closely related to the many terms that have been applied to the athlete's maximum sustainable speed: Anaerobic Threshold (AnT), Maximum Lactate Steady State (MLSS), Lactate Threshold (LT), Second Ventilatory threshold (VT2), Critical Power or Pace (CP), Functional Threshold Power (FTP) just to name some of the more common terms. Despite the bewildering array of names, they all stand for the maximum speed (or power) the athlete can sustain for 45 minutes to 1 hour. Metabolically, it is the fastest pace/highest power at which the lactate removal rate equals the lactate production rate. Hence, the MLSS term.

Famed running coach Renato Canova has stated that the most significant limitation to speed at this MLSS intensity is Local Muscular Fatigue. Increasing resistance to local muscular fatigue should be part of every endurance athlete's training strategy.

What causes you to slow down when your mountain run continues over many hours?

We have a lot of data for mountain runners in many races, including podium finishers in some of the most competitive races. One prominent trait that runners of all abilities have in common is that as local muscular fatigue builds late in long races, we see a significant drop in heart rate.

An example of a UTMB podium finisher:

On a 45-minute 16% climb 3 hours into the UTMB: avg HR=156. Avg pace 17min/mile

On a 1hr 18% climb 1/2 way through UTMB: avg HR=130. Avg pace ~20min/mile

On a 1hr 17% climb late in UTMB: avg HR=120. Avg pace ~20min/mile

Heart rate and oxygen consumption have a nearly linear relationship at sub-max efforts. This means that in the aerobic intensity realm (zones 1, 2, and 3), if you increase the heart rate X%, you will see a Y% rise in oxygen consumption (VO_2) throughout these intensity zones.

Because of this linear relationship, we can estimate the oxygen consumption of these athletes during races. At heart rates exhibited late in long races, even top athletes only use 65-70% of their maximum aerobic power ($\text{VO}_2 \text{ max}$). Their hearts have way more pumping capacity than their legs can utilize as local muscular fatigue builds late in the race. Their speed is not limited by how much blood/oxygen the heart can deliver. Their speed is limited by the muscles' ability to utilize that oxygen.

Because it takes less oxygen to run slower than faster, as their main propulsive muscles fatigue and the runner slows, the demand for more oxygen decreases, and the heart responds by lowering its output by reducing the heart rate. The phenomenon responsible for this decline in heart rate on oxygen consumption is Local Muscular Fatigue.

Therefore, improving the fatigue resistance of the main propelling muscles will allow the athlete to utilize a higher % of their $\text{VO}_2 \text{ max}$ and climb faster, especially later in the race.

NOTE: While it is true that these athletes are racing at a fraction of their maximum aerobic power, 65% of a big $\text{VO}_2 \text{ max}$ is a larger number than 65% of a small $\text{VO}_2 \text{ max}$. So, having a high $\text{VO}_2 \text{ max}$ is a benefit. However, it may be impossible for experienced runners who train consistently to increase their $\text{VO}_2 \text{ max}$ at all or only by less than 5%. In comparison, that same runner may have the possibility (with proper training) to improve their fatigue resistance by 20%.

Key Training

Regarding any multi-hour event, from our experience, and the conclusions we can draw from the above discussions, we have found this simple training approach: 1) The athlete's speed at their aerobic threshold and 2) Improved fatigue resistance of the main propelling muscles. These two points are listed in their relative order of importance

Speed at Aerobic Threshold

When athletes are aerobically deficient, they will not have the aerobic capacity to support running at anything but very slow speeds. Simply improving the aerobic capacity will dramatically increase those runners' speed at their Aerobic Threshold. Why is the aerobic threshold speed so important? That is the intensity at which you will run for more of your race. This can be accomplished in only one way: running a lot at an intensity commonly called Zone 2. We often see the aerobically deficient runner's speed at AeT increase 20-30% within 6-12 months of consistent Zone 2 training. Results will vary based on genetics, age, and training volume, which are the big three. This speed change comes about primarily due to these three adaptations in your muscles: 1) An increase in the mitochondrial mass. 2) An increase in the capillary density in the working muscles. 3) An increase in aerobic enzymes in those mitochondria. Rodent studies have shown a doubling in

mitochondrial mass in the working muscles after just a few months of training. During this aerobic base-building period, strides should be included in your normal Zone 2 runs and hill sprint workouts.

The well-endurance-trained athlete with a high AeT as measured by HR or pace (one whose AeT and AnT are within 10% of one another) will use a different approach. To improve their speed at the aerobic threshold, this runner needs more Zone 3 intensity in their training and must be cautious of doing too much Zone 2. Contrast the aerobically deficient runner mentioned above who can do a high volume of Zone 2 because their pace will be relatively slow (and less neuromuscularly taxing) with the aerobically strong runner whose AeT running pace can exceed 5min/mile, imposing a much higher neuro-muscular load that requires more recovery time or less volume. This faster pace will mean the stronger runner needs fewer Zone 2 runs and will fill in that time with Zone 1 runs. Refer back to the three pro runner examples to see that they do 35-45% of their training in Zone 1.

Fatigue Resistance

Once one's aerobic house is in order, improving local muscular fatigue resistance becomes the key to unlocking better performance in ultra-endurance events. As explained above, ultra-endurance events are conducted at sub-maximal intensities, primarily in Zones 2 and 3. Because of this, fatigue, typical for shorter endurance events, does not play much role in ultras. This means you will not be dealing with the accumulation of metabolic by-products or drop in blood pH associated with fatigue in shorter races like a Vertical Kilometer or a 10km road race. However, keeping up with carbohydrate use through race feeding can be a big challenge for many long-distance runs, resulting in peripheral fatigue.

While sports scientists are very interested in identifying the source of fatigue, either central or peripheral, coaches and athletes just want to improve performance. Several methods of improving fatigue resistance of the main propelling muscles exist.

Training Methods

At its simplest level, all endurance training is designed to increase fatigue resistance. It is a general rule of thumb that the athlete who does the most work in the least time during their training will get the greatest boost in fatigue resistance. Several caveats accompany this: the work must be specific to the event being trained for in intensity, duration, and modality. For example, swimming 10x100m repeats at a Zone 4 intensity means doing much hard work. Still, these will not significantly help improve the work capacity of an athlete training for a 100km mountain running race.

If there is one thing we know about all successful long-distance runners, it is that they run a lot. That has been the foundational element that all endurance sports share. The best athletes are almost always the ones who train the most.

Indeed, if you are new to long mountain running, you must slowly build up to getting a lot of miles along with elevation gain and loss into your legs; if you are targeting a certain distance that exceeds anything you have done before, you must build not only the endurance to cover that distance but also the confidence that comes from knowing that you will make it to the end.

Increasing training volume is a time-honored way of improving the athlete's fatigue resistance in all endurance sports. However, running those mega miles takes a toll on your body that can break some less gifted athletes. Newer or amateur runners with much more limited available training time and less tolerance to very high mileage programs can benefit from adding a different type of stimulus.

Likewise, for the experienced mountain runner with years of high-volume training history, piling on more distance each week will eventually lead to diminishing returns on the training investment and potential injury. Why is that? Those athletes are likely approaching their limits in terms of genetics and time and energy allocation that the same training stimulus will give. These runners must look for new stimuli if they want to continue to improve.

If you feel like you have ridden your current training horse as far as it will take you and are looking for some new ideas, keep reading.

The typical way to add this new stimulus and potentially reduce the training volume is to add high-intensity running to the program. Unquestionably, high-intensity running will increase fatigue resistance at sub-max speeds by increasing local muscular endurance. The downside to using training above the MLSS very often is the global fatigue that comes with it, limiting the overall training volume.

Instead of asking our runners to train using Zone 4 interval-style workouts, we use two proven, successful methods.

Muscular Endurance Workouts

These training sessions target those pesky main propelling muscles by overloading them with low resistance/high repetition "strength" training. This article goes into some depth on the subject. In brief, this process works by working those muscles at an unusually high resistance level for many repetitions. The number of reps per workout can range from dozens to a few thousand for the elite. The extra resistance forces the limitation of the athlete's speed and duration under load to be the local muscular fatigue. While these workouts have a profound training effect and local fatigue. However, they do not impose a heavy global fatigue load and, as such, do not require a reduction in overall training volume.

We rely heavily on a block of this type of training as a precursor to beginning a more traditional interval training program. By elevating fatigue resistance, the athlete can handle a higher volume of conventional interval training with an even better effect.

Threshold Training (Zone 3)

We are using conventional naming for this type of training. However, what we mean by it may differ from your understanding. There is controversy over the best approach for raising the MLSS/AnT speed. Some feel it is most effective to train slightly above the top of Zone 3 (what we term the AnT). Having experimented extensively with Zone 3 and 4 training, we find that using even lower Zone 4 intensity in the interval sessions has a much higher global fatigue load than training in mid-Zone three and requires significantly reduced training volume in the target intensity zone.

Controlling the intensity of this sort of training is paramount. We are trying to target a specific metabolic effect by approaching the metabolic MLSS without stepping over the line. We use blood lactate measurements to ensure we hit the correct intensity. However, it can be done by carefully identifying the AeT and AnT paces and HR and training in the middle of Zone 3.

For the high-level runner, this method allows them to include two-plus hours of work at this intensity in a week. In comparison, getting more than 30-40 minutes in Zone 4 weekly is challenging for even the most well-conditioned athlete and risks pushing the athlete into an overtrained state. Our experience has been that a higher volume of Zone 3 done this way is very effective in improving fatigue resistance and can be the ultimate preparation phase for the ultra-endurance runner. It can also serve as the next to final stage of training for those doing races shorter than 2 hours.

We'll leave you with these final words. None of these cool-sounding, sexy training ideas amount to a hill of beans unless you can apply them using the following approach.

There are three cardinal axioms that every good training philosophy must adhere to to be effective:

Consistency: Optimal gains are made by repeatedly applying the training stimulus regularly and frequently. Every workout makes a small deposit into your fitness bank account. Only with consistency will those small deposits add up to enough money in the fitness bank account to allow you to go on a wild spending spree during your event without bouncing checks.

Gradual Progression: Our bodies will respond to training if the training load is gradually increased over weeks to months. They do not respond well to sudden jumps in training load. This principle is often called progressive overload.

Modulation: The training load must not only gradually progress but also modulate between hard and easy. Lower loads must follow heavy loads so the body has time to recover and make the adaptations that result in increased fitness.

This video, the third installment of the **Evoke Endurance Book Club**, features author **Scott Johnston** discussing Chapter 3 of **Training for the Uphill Athlete**. The session covers the theoretical and practical framework for athletic training, focusing on how to build work capacity and properly apply training stress.

****Key Concepts & Training Philosophy****

* ****Defining Event Demands :**

****Scott*** emphasizes that the first step in any plan is understanding the specific demands of your sport—whether it be mountaineering, trail running, or climbing—and identifying the fundamental qualities (strength, endurance, speed, technique) required to meet them.

* ****Capacity vs. Utilization :**

* ****Capacity Training:**

Focuses on long-term growth and increasing the body's functional potential, often using non-specific exercise. It is essentially "putting money in the bank."

- * **Utilization Training:** Aims to improve near-term performance by mimicking the specific demands of the target event. This is "writing checks on that bank account." Overusing this leads to burnout or hitting a performance plateau.
- * **The Endurance Cascade :** Explains the physiological signaling pathway (**AMPK**) triggered by metabolic stress (such as glycogen depletion). This signaling stimulates protein synthesis, resulting in increased mitochondrial mass, which enhances aerobic performance.
- * **Intensity Zones:** **Scott** details the five-zone system, noting that while heart rate is an imperfect proxy, it remains the most practical tool for athletes training across varying terrains.
 - * **Zone 2** is highlighted as the cornerstone for building aerobic capacity.
 - * **Zone 5** is primarily for power and speed, using stored ATP/Creatine Phosphate.
- * **Training Principles:**
 - * **Continuity:** Avoid the "weekend warrior" trap of only exercising sporadically.
 - * **Modulation:** Balance hard effort with recovery to trigger super-compensation.
 - * **Individualization:** Acknowledge that athletes respond differently to the same stimulus due to training history, lifestyle stress, and genetics (**Renato Canova** quote, 42:49).

Recovery & Adaptation

- * **Super-compensation:** Training induces fatigue, but proper recovery leads to a higher level of fitness than the baseline.
- * **Recovery Tools :** The primary drivers of recovery are **sleep** and **nutrition**. While tools like massage or compression boots (**gizmos**) may help, they cannot replace the foundational importance of rest.

According to Scott, the three foundational principles for effective athletic training are:

- * **Continuity:** Athletes should avoid sporadic 'random exercise' and instead maintain frequent training stimuli to ensure the body adapts consistently rather than breaking down from shock.
- * **Gradualness:** The body responds best to a gradual overloading of training stress. Making huge, sudden leaps in intensity or volume can be counterproductive and may even lead to detraining.
- * **Modulation:** This involves balancing periods of hard effort with proper recovery. This 'hard-easy' approach allows the body to recover, adapt, and reach a state of super-compensation.

Scott Johnston explains that training induces fatigue, which is represented by a dip in an athlete's fitness level below their current baseline. During the subsequent recovery period, the body adapts to the training stress, eventually reaching a state known as **super-compensation**, where the athlete emerges fitter than they were before the workout.

He emphasizes that recovery is the period when actual improvement happens, as training sessions themselves initially make an athlete 'weaker' . He identifies several key drivers for effective recovery:

- **Sleep:** Cited as the most critical component for repair and adaptation.
- **Diet:** Essential for physical recovery.

- **Active Recovery:** Using light sessions, such as swimming, cycling, or walking, to keep blood moving and facilitate healing.
- **Manual Therapy:** Including self-massage or foam rolling.

Finally, he notes that while various market 'gizmos' (such as ice baths or compression boots) exist, their efficacy is less certain compared to these foundational recovery practices.

This video, featuring *Scott Johnston*, focuses on the critical art of **monitoring training** for mountain endurance athletes, emphasizing that true fitness gains occur during recovery, not just during the training sessions themselves.

Core Concepts of Training and Recovery

The Training Principle: The goal of endurance training is to create a controlled "embarrassment" to the body—a workload it cannot currently handle. By allowing for proper recovery, the body triggers **super-compensation**, leading to a higher baseline of fitness than before.

Frequent Small Overloads: Johnston argues against the "weekend warrior" approach of massive, infrequent sessions. The body responds best to frequent, smaller doses of overload, allowing for more consistent progress without necessitating days of total exhaustion.

Recovery Times: Recovery needs vary by intensity:

• **Zones 1-2 (Aerobic):** Typically require 12–36 hours for super-compensation.

• **Higher Intensity (Zones 4-5) & Strength:** Often require 72 hours for full recovery.

Understanding Fatigue Mechanisms

Johnston breaks down fatigue into several categories to help athletes identify what their body needs:

1. **Metabolic Fatigue:** Involves lactate accumulation (short-term high intensity) and glycogen depletion (longer-term energy stores).
2. **Central Nervous System (CNS) Fatigue:** A crucial factor in ultra-running. The brain reduces the power of nervous signals to the muscles as a protective mechanism. Studies, such as those at the *Ultra-Trail du Mont-Blanc*, suggest that voluntary power loss is often neural rather than muscular.
3. **The Central Governor Theory:** Proposed by *Tim Noakes*, this theory suggests the brain acts as a "governor," limiting performance to prevent catastrophic physical damage. Training may help "push" the limit of this governor.

Monitoring and Diagnosis

Subjective Perception: Above all data, how you feel is the most important metric. If you aren't feeling fitter on a week-to-week basis, you are likely overdoing it.

Practical Tests:

• **Grading Workouts (A-F):** A simple system to track how you feel daily. Consecutive "C" or "D" grades are major red flags.

* **Resting Heart Rate & Step Tests:** Monitoring your resting heart rate or heart rate recovery after a simple stair-stepping test can signal when the autonomic nervous system is stressed.

* **Overreaching vs. Overtraining:**

* **Overreaching** is a planned training tool followed by adequate rest.

* **Overtraining** is a medical condition. It involves a systemic breakdown, often leading to performance stagnation that can take months or even a year to recover from. He warns that it is a serious condition that can end athletic careers.

Periodization

Johnston explains that periodization is about transitioning from general to specific training:

* **Transition Period:** For general conditioning.

* **Base Period:** Focuses on increasing work capacity in fundamental qualities individually.

* **Specific Period:** Bringing those qualities together (e.g., combining strength, aerobic base, and muscular endurance for mountaineering) as you approach an event.

* **Competition Period:** Managing multiple events while mostly focusing on recovery rather than gains.

Johnston concludes by advising that in periods of rest or between seasons, athletes should focus on **fun activities** they previously had to sacrifice, rather than strictly sticking to a training log.

This video, led by *Scott Johnston* of *Evoke Endurance*, serves as a deep-dive into the second part of Chapter 5 from the book *Training for the Uphill Athlete*. The session focuses on the practical application of **strength, power, and endurance** training for mountain athletes.

Key Concepts Explained

1. Strength vs. Power & Neurological Adaptation

* **Strength** is a neurological effect; it involves training the brain to recruit a larger number of motor units to apply force.

* **Power** is defined as the *rate* at which you create force. Johnston emphasizes that simply lifting heavy weights slowly (like in a heavy deadlift) does not necessarily translate to the explosive power needed for mountain sports.

* **Specificity:** The speed of recruitment during training should ideally mimic the demands of the event (e.g., a single-leg box step-up is often more beneficial for a mountaineer than a standard two-legged squat).

2. Developing Power without Heavy Weights

* Johnston highlights that you can achieve high levels of strength and power through methods other than traditional weightlifting.

* **Hill Sprints:** Considered the 'no-brainer' for mountain athletes. They should be short (10–15 seconds), performed at maximum intensity, and ideally on steep terrain or stairs. Heart rate is an ineffective metric here because the effort is too short for the heart to respond.

- * **Plyometrics:** Should be used for *reactive strength* (e.g., jumping off a box and bouncing immediately). These require low repetitions and long recovery periods between sets to ensure full neurological recovery.

3. Recovery and Restorative Training

- * Recovery is individual and depends on the depth of fatigue.
- * **Active Recovery:** Extremely low-intensity aerobic work is often more restorative than total inactivity, as it helps clear metabolic waste and improves blood flow.

4. Aerobic Base and Training Intensity

- * Volume remains the key driver for aerobic base development.
- * Johnston shares anecdotes of legendary coaches like *Mihaly Iglói* and *Phil Maffetone* to illustrate that consistent, moderate-intensity training allows for steady, year-over-year gains in aerobic threshold.

5. High-Intensity Training: The Role of Zone 3

- * Johnston argues for the effectiveness of **Zone 3 training** for high-intensity work. Unlike Zone 4, which is extremely taxing on the central nervous system and takes longer to recover from, Zone 3 allows for higher overall training volume.
- * **Scientific Context:** There is evidence that excessive high-intensity (Zone 4+) training that causes high blood acidity can actually *down-regulate* the genetic expression responsible for increasing mitochondrial density, which is the foundation of aerobic fitness .

Conclusion

Johnston emphasizes that the 'best' training is what an athlete can consistently recover from and repeat. He advises against mimicking the training loads of elite professionals like *Kilian Jornet* or *Tom Evans* without considering the years of capacity building that preceded their current performance levels.

According to the presentation, Scott Johnston defines **muscular endurance** (ME) as the ability to create or expend a high percentage of one's maximum force for as long as possible. He highlights several key points regarding this concept:

- **Training Correlation:** Johnston states that muscular endurance is highly trainable and arguably contributes more to overall endurance performance than almost any other factor outside of fixed genetic variables.
- **Event Duration:** He explains that the importance of maximum strength versus muscular endurance changes based on the length of the event; in shorter, high-intensity events, the athlete must exert a higher proportion of their maximum strength, whereas, in multi-hour events, the required force is much lower, reducing the need for extreme maximum strength.
- **Sustainable Speed:** Johnston notes that muscular endurance is directly related to an athlete's sustainable speed, which is a key performance determinant alongside anaerobic threshold and movement economy.

This video, presented by *Scott Johnston* of *Evoke Endurance*, provides a comprehensive look at Chapters 7 and 8 of *Training for the Uphill Athlete*. The discussion focuses on the

distinction between general and specific strength, how to assess these attributes, and how to program training for endurance athletes.

Key Concepts in Strength Training

- * **Absolute Strength vs. Athletic Performance:** Johnston emphasizes that for ultra-endurance sports, absolute strength (like max deadlifts) is less critical than the ability to move quickly and efficiently. Training for maximal force often requires slow movement, which is not sport-specific for runners or climbers.
- * **The Importance of Single-Leg Stability:** Most endurance activities involve single-leg movement. Because stable, two-legged lifts allow for higher loads by reducing the need for stabilizer muscle activation, they are often less effective for mountain athletes. Developing single-leg stability is the primary goal to prevent injury and improve power.
- * **The Valgus Knee Issue:** Weak hip stabilizers often lead to a "valgus knee" (where the knees cave inward under load). This misalignment increases stress on the *lateral condyle* of the femur and can lead to chronic injury. Strengthening the hip stabilizers is the recommended mitigation strategy.

Assessment and Progression

Johnston outlines three levels of assessment and training:

- * **Level 1 (Activation):** Uses simple exercises like banded air squats and side-lying leg raises to help the brain learn to engage the hip stabilizers.
- * **Level 2 (Stability):** Incorporates movements like box step-ups, front lunges, and single-leg Romanian deadlifts to challenge stability under higher control.
- * **Level 3 (Strength):** Once activation and stability are mastered, athletes introduce resistance (80-90% of max) to build raw strength in these functional patterns.

Core and Upper Body

- * **Core Strength:** The focus is on the *transverse abdominis* to stabilize the spine and hips. Key exercises include planks (for the transverse abdominis), windshield wipers (for obliques), and hanging leg raises (for hip flexors).
- * **Upper Body:** Strength is assessed through push-ups, dips, and pull-ups. For those unable to perform full repetitions, he suggests regressions like scapular retractions and incline pull-ups.

Specific Strength: Power and Muscular Endurance

- * **Power Training:** Power is developed through short, high-intensity efforts like hill sprints or stair sprints (10-12 seconds). Johnston notes these should not be exhausting; the workout ends when power output drops.
- * **Muscular Endurance:** This training targets the "Frontier fibers"—those used at higher speeds that lack aerobic conditioning. It involves high-repetition, sub-maximal intensity work (often with heavy packs on steep terrain or stair machines) to specifically improve local muscular fatigue resistance.

Programming Differences

* **Runners vs. Mountaineers:** While runners often prioritize muscular endurance in the base period to support later interval training, mountaineers typically focus on general maximum strength (like weighted step-ups) early on, as their sport directly mimics carrying heavy loads uphill.

Scott Johnston emphasizes that **muscular endurance** is arguably the most specific and important form of strength training for endurance athletes. He provides several key insights regarding its role, theory, and practical application:

* **Defining the Limitation:** He characterizes muscular endurance as a **local or peripheral fatigue**, distinct from the global fatigue caused by heavy breathing. It is the specific sensation of "dead legs" where you want to maintain speed but the muscles physically cannot sustain the effort.

* **The "Frontier Fiber" Model:** Johnston introduces his theoretical framework for why this training is effective. He explains that as an athlete increases speed, the brain recruits a wider pool of motor units. The muscle fibers recruited at higher intensities (those closest to your current limit) are "Frontier fibers." These fibers are often less aerobically adapted and fatigue quickly. The goal of muscular endurance training is to repeatedly engage these fibers so they become more **aerobically efficient** and fatigue-resistant.

* **Training Methods:**

* **Gym-based:** He references specific workouts detailed in his books and available as free PDFs on the *Evoke Endurance* website.

* **Overload Hiking:** A primary method is hiking steeply uphill (30–50% gradient) while carrying a heavy pack. He notes that doing this on a stair machine in a gym is often more effective than finding a suitable outdoor hill, despite being less stimulating, because it guarantees a consistent, steep grade for maximum local muscle engagement.

* **Periodization:** He notes that training frequency should rarely exceed **once per week**. For runners, this work is typically introduced in the **base period** to build local fatigue resistance before transitioning to higher-intensity interval training, ensuring that their legs don't become the limiting factor during those more intense sessions later in the cycle.

In this second part of his post-UTMB series, coach *Scott Johnston* of *Evoke Endurance* explains the core training methodologies used by his athletes, *Tom Evans* and *Ruth Croft*. Johnston argues against an over-reliance on traditional VO2 max training, instead focusing on building **fatigue resistance** through specific physiological adaptations.

Key Concepts

* **Muscular Endurance (ME):** Borrowing from the work of *Tudor Bompa* and *Yuri Verkhoshansky*, Johnston defines ME as the ability to perform many repetitions of a sport-specific movement under resistance. The goal is to target **"frontier motor units"**—those fibers right at the athlete's endurance limit (anaerobic threshold)—without incurring the massive global fatigue associated with high-intensity interval training. Methods include:

* **Weighted uphill hiking/running:** Adding 8-15% of body weight to perform sport-specific uphill efforts.

* **Stair climbing:** Using weighted packs or stair machines for slow-turnover resistance.

* **Gym-based protocols:** Specific strength endurance routines for athletes without access to suitable terrain.

* **Anti-Glycolytic Training:** This approach aims to maximize performance while minimizing reliance on the glycolytic (anaerobic) metabolic pathway. Because glycolysis produces metabolites that eventually force an athlete to slow down, Johnston suggests that excessive high-intensity training can actually reduce an athlete's aerobic capacity. Anti-glycolytic methods include:

* **Short speed work:** Incorporating 10-15 second hill sprints or flat strides with full recovery to recruit powerful motor units without stimulating the glycolytic system .

* **Extensive training:** Steady-state work conducted between the first (aerobic) and second (anaerobic) thresholds to improve the efficiency of the aerobic system .

* **The Frontier Fiber Model:** Johnston explains that training must be specific to the fiber types recruited at the intensity of the event. By targeting the fibers at the "frontier" of the endurance limit, athletes can improve their speed at the anaerobic threshold, allowing for late-race surges without early depletion.

Updated Thinking on Aerobic Assessment for Mountain Athletes

By: Scott Johnston Posted: October 26, 2022

Until recently, there were limited options for an endurance athlete to get a handle on his or her metabolic response to exercise at varying intensities. There are several good options, but I'm going to focus on the ones we usually default to for determining the two thresholds that represent important metabolic shifts. Knowing these two points will set you up well for controlling the intensity of your training. Be aware that the first test requires serious technology, such as an expensive heart rate monitor/GPS watch. For the no-tech test, skip ahead to the Nose Breathing section. But the second test is as simple as can be. It's just demanding.

Testing Aerobic Threshold (AeT)

Heart Rate Drift

After wearable heart rate monitors became widely available in the early 80s, astute coaches and athletes noticed that on runs over about 30-45 minutes, the runner's heart rate might rise while they maintained the same pace or even if they slowed the pace. Coaches called this "heart rate drift", but it was only understood qualitatively. With the advent of the newer generation of heart rate monitors combined with a GPS in the same watch, the analysis of the "drift" moved from being a qualitative oddity to becoming a quantifiable, valuable, and actionable piece of data.

If while holding the same pace, your heart rate climbs more than 5% during an hour of continuous exercise, OR if, while keeping your heart rate steady, your pace slows more than

5% during an hour of continuous exercise, the chances are excellent that you began that hour at a heart rate that was above your aerobic threshold (AeT) or the top of Zone 2.

From a metabolic standpoint, heart rate drift means your energy requirements exceed your aerobic capacity. As you know from this article, when your energy demands exceed your aerobic capacity, your anaerobic system must make up the difference. It is this anaerobic contribution to the overall energy requirement that is causing your heart rate to slowly drift up during that hour.

We have had extensive experience administering, interpreting, and evaluating HR drift tests for hundreds of athletes over six years. The HR drift test correlates very closely to the expensive laboratory-administered gas exchange test (GET) we'll discuss shortly when establishing your AeT. Besides the money savings, it has the benefit of being able to be re-tested whenever you desire. More on that when we get into the testing in the next section. This has become our go-to test, even for our elite athletes, to establish the top of Z2.

How to conduct this test

Our first choice is to do this test on a treadmill or a StairMaster (these look like short escalators) for mountaineers. This allows closer control and fewer variables to deal with. Runners should do this running on a 2% grade. Those who can't run should or will do most of their aerobic base training hiking and should set the treadmill to 15% and hike. Mountaineers can use either a treadmill set to 15% or a stair machine.

NOTE: If you're using one of these machines, do not hang on to the handrails. You will not have bars when you go outside, and hanging on makes it too easy.

Indoors HR Drift test on a treadmill or stair machine: Warm up gradually for at least 10 minutes, never exceeding a nose breathing pace.

During the warm-up, you can fiddle with the pace (while nose breathing) to establish a starting pace where your HR stabilizes in a range of plus-minus 2-3 beats around a number for at least 3 minutes. If your HR continues to climb during these 3 minutes, the pace is too fast. Slow it down till your HR stabilizes for 3-4 minutes. It is essential to let your HR stabilize at a pace before starting the test. After you find that stable HR pace, **DO NOT TOUCH THE SPEED CONTROL FOR THE REST OF THE TEST.** This is the most common mistake folks make.

If, say, 10 minutes into the test, your HR is ten beats higher than the start and continuing to climb, the starting pace/HR was too high, and you will need to redo the test. Slowing the treadmill speed down during the test invalidates the result. It is probably best to come back another day to do the test again at a slower pace. If all is good, though, continue this test for 60 minutes. Some gyms limit treadmill time to 60min, so you may need to do all this fiddling around during the warm-up, stop the machine, start your watch over again, and record a new session so you can go an hour.

By holding the treadmill speed constant, the only variable you need to consider is HR. Using a Training Peaks premium account, upload this workout, click the analyze button (upper right

of the workout window), and see a graph like this one from a real HR drift test one of our athletes did. The red line is the HR data, the only data you need to look at.

Using your cursor, select the first half of the session. In this graph, you would start that selection by dragging the cursor from about the 5min mark where the HR settled down till a bit past 30 minutes. It is not necessary to select the halves of the test exactly. Plus/minus a few minutes is fine.

You can disregard the Pa: Hr because the pace and distance are probably inaccurate because the GPS does not work indoors. But you will see that the average HR was 128 for this first half of the test. Doing the same thing for the second half will give you the average HR for the second half.

In this test, there was virtually no upward drift in HR. That tells us that the starting HR (about 127 in this case) was well under this athlete's aerobic threshold. Because this test showed no drift, I would normally suggest the athlete redo the test with a starting HR of about 135.

Outside HR Drift Test for Running

Use a Training Peaks premium account to analyze your heart rate drift by clicking on the analyze button in the data displayed in the upper right corner. You will see **Pa: Hr**. That's the HR drift. This test must be conducted on either a flat course like a running track or a gently rolling loop course with less than 100ft (30m) elevation change per mile (1.6km). **DO NOT USE AN OUT-AND-BACK COURSE.** Inevitably one way will be more uphill than the other, and the results will skew the test. It is impossible to do this test hiking on a natural gradient outdoors. You will not be able to find a consistent grade that continues for one hour.

Warm up with the same protocol as above. Find a comfortable nose breathing pace and start the test when you establish that. Upload the data to your Training Peaks premium account and click the "analyze" button. A screen like the one below will show, and **Pa: Hr** is the number you are looking at because your GPS works outdoors.

In this case, an outside running HR drift test of 3.6% indicates that he was within his aerobic capacity. Notice that over 10 miles, this run only gained and lost around 100ft.

Nose Breathing Test and Intensity Monitoring

This is the ultimate low-tech method of determining your AeT. Not only that, but it also has the best real-time feedback on your metabolic state outside of a lab..... IF YOU PAY CLOSE ATTENTION.

Exercise scientists call the rate and depth of breathing ventilation. With specialized equipment, they can measure how much air your lungs are pumping in and out each minute and the percentage of oxygen and CO2 in each breath.

You may have heard the terms "first and second ventilatory thresholds (VT1 and VT2)". That same specialized lab equipment can pinpoint these points which occur at important metabolic events. We, laymen, can also get close to locating these points by carefully monitoring our ventilation (rate and depth of breathing). The first metabolic threshold you

cross (VT1, often called the aerobic threshold) corresponds very closely for almost all of us with the upper limit of nose breathing or where you can speak in complete sentences.

VT2 corresponds to another noticeable jump in ventilation where you can only manage one or two words. Below this, breathing will be steady and deep but not labored. Above this point, you are in a state of hyperventilation where your body is trying desperately to blow off excess CO₂. In the old days, we called this point break-away breathing.

For the perceptive athlete, there is a noticeable jump in ventilation at both these thresholds. Since they occur at significant metabolic points, it behooves you to notice them. That way, when you are out for a run, hike, or skin and you've forgotten your HR monitor, you will be able to determine your metabolic state. Are you in Zone 2 or pushing well into Z3 because you are forced to breathe through your mouth?

Be aware that your AeT and AnT heart rates will vary daily, depending on your recovery state. So by blindly adhering to heart rate, you may be overreaching by trying to hit your AeT HR when you are tired.

Whereas ventilation is a direct result of the metabolism fueling your exercise, paying attention to your breathing during your training will alert you to when your AeT is a bit lower on a day when you are tired. It is like having that fancy lab testing equipment with you during each run.

Gas Exchange Test

I used to call this the gold standard: A gas exchange test done in a laboratory and usually costing \$200-\$300. The best use for this test is to determine where you are getting the calories that propel you: Are they from fat or carbohydrates? This is best done with a metabolic efficiency test (MET), which only some testing labs are familiar with. Most labs will offer to have you do a maxVO₂ test. The protocols for these two tests are quite different. For the MET, you should be in a fasted state. For the maxVO₂, you must have eaten some carbohydrates within 30-60 minutes. There are other differences that I won't go into here. If all you want to do is determine your aerobic threshold (AeT, top of Zone 2), this test is overkill. This test is for the uber-curious or the elite athlete who plans to get tested frequently at the same lab.

Blood Lactate Test

Long ago, this was my go-to test for determining AeT and checking if the skiers I was coaching were staying under AeT during long, easy workouts. But I needed to be there in person to test. You can test yourself. But this test is, again, for the curious who plan on frequent testing to control the progression of their training. It's a bit overkill for most folks, and these lactate meters are a few hundred dollars. The test strips are also expensive. I have written an in-depth "how to" lactate test article that I will link to soon.

Over the past few decades, I have moved from basing training zones entirely on blood lactate tests or laboratory Gas Exchange Tests to the more straightforward but high-tech HR drift test, which still requires expensive equipment and a Training Peaks account. This is still the test we administer to all of our coached athletes. We confidently recommend it to

anyone. However, there is something incredibly alluring about the no-tech nose breathing/ventilation test. It may tune you into your body even more than paying attention to your heart rate. Suppose you are going to experiment with the nose breathing test. In that case, we recommend that you do so with a heart rate monitor and learn to correlate these two physiological markers to each other and, most importantly, to how you feel.

Testing Anaerobic Threshold (AnT)

I like to use this very simple, but in my view, the best test for determining the anaerobic threshold. The results of this test are the simplest to interpret, easiest to implement, and, therefore, most useful by a coach or athlete. This threshold, by any of its several names, is your endurance limit. The value you determine using this test, whether using heart rate, running pace, or power, will be an actual measure of the maximum work rate you can sustain for an extended length of time. Your AnT will likely vary under different conditions/events. A test running on the flats is probably not going to yield the same average heart rate as an uphill hike with a pack. These two will most certainly yield different average paces.

If this doesn't seem "sciency" enough for you, consider an example:

You execute your own "Vertical Kilometer" (VK) time trial on a steep local hill with a heavy pack. It is not necessary that it actually gain 1km in elevation but say it takes you 43 minutes going as hard as you can go, without blowing up, to gain 285m. Your average heart rate for the test was 168bpm, and your average rate of climb was 285m/43min or about 6.6m/min. These numbers represent the limit of your endurance on this hill with this weight pack.

This is as actionable as a piece of data can be and much more useful to a coach or athlete in terms of programming than a blood lactate test running on a treadmill would have been for determining AnT.

The test can be as short as 20 minutes for someone just starting out with this kind of training or up to an hour for a world-class athlete. All you are going to do is run, ski, cycle or hike steeply uphill as fast as you can for the allotted time. It will feel like a race or time trial. If you are careful not to start too hard and blow up 5 minutes into the test, your average heart rate, pace, or power is going to be the maximum you can sustain for that period. Bingo. It is that simple.

I prefer this to any other test because it is actually measuring your performance. It is not measuring a proxy for performance, such as interpreting the RER from a gas exchange test or trying to suss out where the blood lactate curves make that bend upward.

Warning: This test is hard and should only be conducted when you are well-rested if the results are going to be meaningful. Doing this test tired will still feel very hard, but the average heart rate and pace will be lower than in a rested state.

What to do with this information: The 10% Rule

Once you have conducted these two tests, you will have a decent snapshot of how your metabolism responds to exercise. That's a powerful tool and can guide your training if you pay attention. By comparing the metabolic responses to exercise for an athlete before and after six months of the type of training we promote, we will be able to see how you can make the best use of the test info you just completed. While Bob is not his real name, this example is not fictitious. We see this kind of improvement in athletes we coach and hear from athletes training on their own.

On week one: Bob is 31 years old and has a busy work and family life. He has recently read *Training for the Uphill Athlete* and begun to train using the methods we promote. He has a high work capacity from playing ball sports through college and participating in 2-3 Crossfit workouts a week for the past three years. In the past, most of his running workouts have been done at a pace that was as fast as he could maintain for the time or distance he could squeeze in. If he had 30 minutes, he'd run as hard as he could for 30 minutes. If he had an hour, he'd run as hard as he could for that hour. He had the intensity needle pinned to the red zone for almost all his runs. The results of his heart rate drift test showed his aerobic threshold to be 136bpm at 11:30 min/mile (7:08 min/km), which he found to be a very easy effort. It was so easy that he was convinced that either he did the test wrong or we must be smoking something to suggest that training at this intensity could accomplish any training effect. After making sure that the test was correctly done, his Evoke coach explained to him that this result was, in fact, a good measure of his aerobic capacity. This was all the energy his aerobic system was capable of producing right now before the anaerobic or glycolytic metabolic system had to begin to provide more than about 50% of his energy needs.

Next came the anaerobic threshold test, an all-out 30-minute run where his average heart rate was 168 and his average pace was 9:20 minutes/mile (5:37min/km). The spread between his aerobic and anaerobic thresholds is 23.5% (in terms of heart rate; 169/136) and 23% in terms of his pace (11:30/9:20). Because of all the high-intensity training he had done in the past few years, his anaerobic system has a high capacity.

In setting his zones, we used his heart rate drift test to set the top of his Zone 2 and his anaerobic threshold test to set the top of his Zone 3. When we did that, we saw that his purely aerobic zones 1 and 2 were squeezed below 136 at what he felt was a silly slow speed. He had a very broad Zone 3 because his muscles required a heavy dose of glucose to provide the energy he needed. His anaerobic system was the predominant fuel source, even at slow speeds. To better understand why this is a problem for an endurance athlete and an ever bigger one for ultra-long events like many of us engage in, please read this article

<https://evokeendurance.com/endurance-its-evolution-psychology-meaning-physiology-and-how-to-improve-yours/>

The best course of action for him is to concentrate his training efforts on elevating his aerobic threshold to bring his aerobic and anaerobic metabolisms into a better balance for his endurance goals. He is suffering from a condition we see all too often: Aerobic Deficiency Syndrome.

Six months later, after doing almost all of his aerobic training in Zone 2 and having retested his aerobic and anaerobic threshold every 4-6 weeks or whenever he or his coach noticed an increase in his running pace at his aerobic threshold, his metabolism has begun to transform into that of an endurance athlete. His heart rate drift test now gives his aerobic threshold as 155 bpm and 9:45 min/mile (6:15 min/km). His anaerobic threshold has not moved because his training has been focused on fixing his aerobic deficit. There have been no workouts yet that provide a training stimulus to his anaerobic system. But now his aerobic and anaerobic thresholds are only 8% apart in terms of heart rate and 4.5% in terms of pace. His Zone 3 has shrunk to where it is only 14 bpm wide compared to 33 bpm wide when he started training this way.

The performance implications of this are profound. His aerobic capacity has increased dramatically. He can now comfortably sustain a running pace that is within 5% of the pace that took a maximum effort six months ago. His aerobic system is now able to fuel a much higher speed before having to call in the reserves, which are the anaerobic cavalry. Now his aerobic deficiency has been dealt a death blow, the gloves have come off, and the more interesting performance-related training is set to begin. His coach will begin programming more Zone 3 and 4 workouts which will elevate his anaerobic threshold heart rate and pace.

The good news for Bob is that he has done all this in just six months. And this is just the tip of the iceberg in terms of improving his endurance. While his aerobic threshold heart rate may not move up much more, his pace at that heart rate can continue to increase for several more years under this kind of training approach. Along with more intensity training, his anaerobic threshold pace will also move up substantially. This AnT pace is the best correlation with endurance performance.

Bob is a runner, but the same concepts, principles, methods, and results work with all endurance athletes.

10% Rule

This is not some formula derived from scientific studies. It comes from the observations of many elite athletes by their coaches. Elites have a very high aerobic threshold and a very small Zone 3. The difference between being in a full aerobic metabolic state (in Zone 2) and going as fast as they can sustain for several minutes at a time (Zone 4) is only a few beats per minute and a handful of seconds per mile or kilometer. When the spread between the aerobic and anaerobic threshold drops to 10% or less, the athlete is considered to have a well-developed aerobic base and should begin to introduce more high-intensity training and reduce the time spent in Zone 2 because the pace/effort is too demanding muscularly.

NOTE: Before any of you math nerds out there correct my percentage calculations, I know that this is not the proper way of calculating percentage differences. However, the difference is tiny and immaterial, and the simplicity of the above method makes it much quicker and easier to use.

Summary

The simplicity and free cost of these tests allow you to repeat them anytime you choose. At Evoke, our coaches have used these tests with hundreds of athletes over the past seven years.

As a nice gauge of fitness, I will sometimes have my athletes conduct a couple of time trials at the beginning of our training together and repeat the tests a few months later. One test is done at the AeT heart rate to gauge the progress in aerobic capacity. And one is done at the AnT heart rate to test their endurance. While you can use different courses for the AeT and AnT tests, be sure to repeat the tests on the same course(s) under similar conditions so you are comparing apples to apples.